

BASE24-atm
LINK (LIS5) INTERCHANGE
INTERFACE MANUAL
RELEASE 6.0



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PREFACE

Throughout this manual, the term LINK describes the network operated by LINK Interchange Limited for the transmission of transactions.

The BASE24 LINK (LIS5) Interchange Interface is a BASE24 product that provides an interface between a BASE24 acquirer or issuer system and LINK. The term LINK (LIS5) Interchange Interface or LINK (LIS5) Interface is used to describe the BASE24 process that interfaces with the LINK switch.

Intended Audience

This document is primarily intended for staff responsible for the configuration, operation or maintenance of the BASE24 LINK (LIS5) Interchange Interface. Users are expected to be familiar with the principles of BASE24 configuration, its operation and certain functions. Users should consult the relevant LINK, HP NonStop (formerly Tandem), or BASE24 documentation where appropriate.

This document can be used as an aid to training but readers are advised to attend the relevant training courses.

Software

This manual documents standard processing for BASE24 LINK (LIS5) Interchange Interface Release 6.0.

Custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Document Location

This document is located on the SW60LINK subvolume in a file called LINKMNWD. It may be copied as needed by ACI customers who have purchased ACI's BASE24 LINK (LIS5) Interchange Interface.



Location of Information

A list of section headings and related page references are given in the *Contents* page. The headings are function oriented.

Section 1, “Introduction,” provides an introduction to the purpose and functionality of the BASE24 LINK (LIS5) Interchange Interface, including the LINK network.

Section 2, “Processing Overview,” provides an overview of the functionality of the BASE24 LINK (LIS5) Interchange Interface.

Section 3, “Configuration and Files Maintenance,” describes the configuration requirements specific to the BASE24 LINK (LIS5) Interchange Interface.

Section 4, “Startup and Control,” documents the network control commands supported by the BASE24 LINK (LIS5) Interchange Interface and describes the network management message flow.

Section 5, “Transaction Handling,” describes the different transaction message flows between the LINK network and the BASE24 LINK (LIS5) Interchange Interface and provides further information on transaction processing.

Section 6, “Cutover and Reporting,” documents the procedures for settlement with LINK and describes the configuration of standard BASE24 interchange reporting.

Section 7, “EMV Support,” describes the support within the BASE24 LINK (LIS5) Interchange Interface for EMV transaction processing.

Appendix A, “Response Code Translation,” presents tables describing the translation of response codes between BASE24-atm and the LINK network.

Appendix B, “File Formats,” defines the data structures used by the BASE24 LINK (LIS5) Interchange Interface.

Appendix C, “Log Messages,” describes the messages logged by the BASE24 LINK (LIS5) Interchange Interface and provides guidance on their causes and the appropriate action to take.

Appendix D, “Glossary,” explains the definitions of the terms used in this manual.

Appendix E, “TSS Migration,” describes the steps required to configure the BASE24 LINK (LIS5) Interchange Interface for use with TSS.

Related Documentation

BASE24 Documentation

- *NET24-XPNET Network Control Operations Guide*
- *NET24-XPNET Network Control Command Reference Manual*
- *BASE24 User Security Manual*
- *BASE24 Files Maintenance Manual*
- *BASE24 Logical Network Configuration File Manual*
- *BASE24 Files and Record Structures Reference Manual*
- *BASE24 Tokens Manual*
- *BASE24 Transaction Security Manual*
- *BASE24 Transaction Security Services Processing Guide*
- *BASE24-atm Settlement and Reporting Manual*
- *BASE24-atm Transaction Processing Manual*
- *BASE24-atm EMV Support Manual*

LINK Documentation

- *LINK Switch Service Interchange Standard 5 (LIS5), 2008-1 “What’s New”*
- *LINK Switch Service Interchange Standard 5 (LIS5), 2008-1, Version 1.0*
- *LINK Switch Service Interchange Standard – LIS5 Security Standard*
- *Network Management Message Flows Document, Version 2.0*

Other Documentation

- *ISO Codes for the Representation of Currencies and Funds, ISO 4217-1981*
- *ISO Standard for Bank Card Originated Messages, ISO 8583 - 1987*
- *American National Standards Institute (ANSI) Standard X9.8 for PIN Management and Security*
- *American National Standards Institute (ANSI) Data Encryption Standard (DES) X9.9*
- *Europay/MasterCard/Visa (EMV): ICC Specifications for Payment Systems, version 3.1.1, dated 31st May 1998*

Document Control

Version	Date	Major Changes
1	01/02	LINK (LIS5) Interchange Interface Release 6.0, plus Dynamic Key Management and EMV Support
2	06/02	Non-Currency Dispense Support and other Release 6.0 Enhancements
3	07/02	Mobile Phone Top-Up Support
4	11/02	LIS5 Release 2003-1 Mandates
5	06/03	Offline PIN Management and LINK Deposit Support
6	02/04	LIS5 Release 2004-1 Mandates
7	08/05	LIS5 Release 2005-1 Mandates
8	03/06	Track-2 Data Suppression Support
9	01/07	LIS5 Release 2007-1 Mandates
10	06/07	Foreign Currency Dispense and Switch Terminal File Support
11	11/07	Manual updates to ensure compatibility with Release 6.0 version 8
12	03/08	LIS5 Release 2008-1 Mandates
13	11/08	Clarification of “Definition of a Unique STAN” functionality.

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1. INTRODUCTION

LINK is the national ATM switch for the U.K. The BASE24 LINK (LIS5) Interchange Interface connects a BASE24-atm network with the LINK Interchange.

1.1 BASE24 SYSTEM ARCHITECTURE

The BASE24 LINK (LIS5) Interchange Interface is a BASE24 switch product that provides an interface between a BASE24-atm acquirer or issuer system and the LINK Switch.

The BASE24 LINK (LIS5) Interchange Interface is compliant with the LINK Switch Service Interchange Standard (LIS5), up to and including Release 2007-1.

1.1.1 System Overview

The following figure provides an example of a BASE24 LINK (LIS5) Interchange Interface configuration.

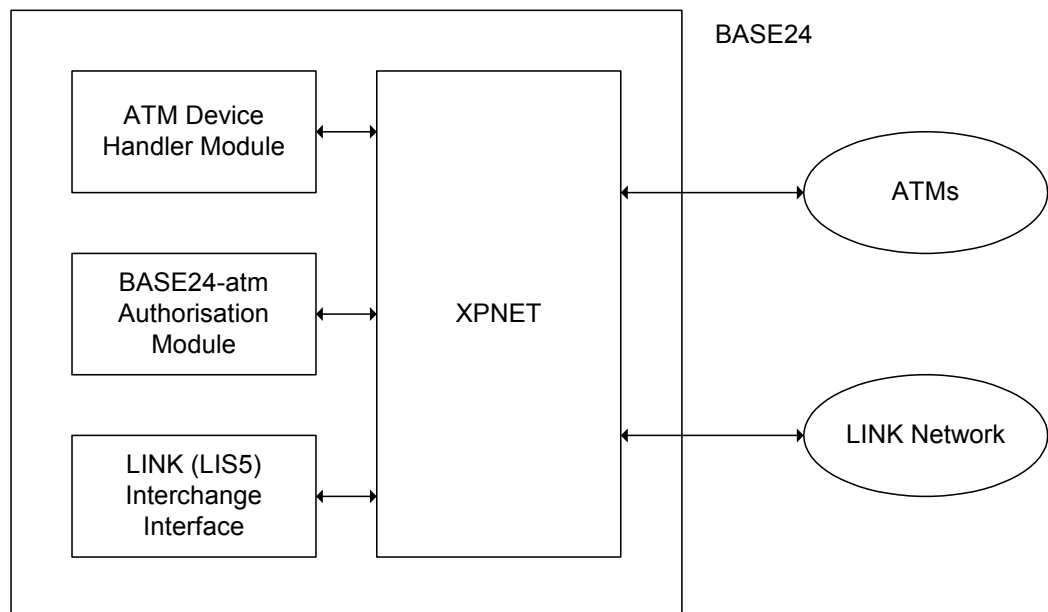


Figure 1-1 BASE24 System Configuration

1.1.2 Issuer And Acquirer Support

In the financial industry, the terms *issuer* and *acquirer* define the parties on opposite ends of the electronic message exchanges required to authorise and process transactions.

- *Issuers* are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorising cardholder transactions and keeping track of cardholder accounts.
- *Acquirers* are the parties initiating transactions on behalf of the card acceptors.

The BASE24 LINK (LIS5) Interchange Interface can function in the role of an issuer, an acquirer, or both. When the BASE24 LINK (LIS5) Interchange Interface sends financial transactions to LINK (LIS5) for authorisation, the BASE24 LINK (LIS5) Interchange Interface is representing the transaction acquirer, even though the transaction may not have originated at a BASE24-controlled terminal. When financial transactions are being sent to the BASE24 LINK (LIS5) Interchange Interface for authorisation from LINK, the BASE24 LINK (LIS5) Interchange Interface is representing the card issuer, even though BASE24 may only forward the transaction to an issuing system for authorisation.

1.1.3 Module Relationships

The BASE24 LINK (LIS5) Interchange Interface consists of two source code modules:

- Base module
- ATM module

The Base module contains procedures that are not specific to ATM transaction processing, e.g., Network Management Message formatting.

The ATM module contains procedures that are specific to ATM transaction processing, e.g., authorisation message formatting.

The BASE module and ATM module include a number of procedure calls ('hooks') where additional processing is required for EMV support. These hooks invoke either procedures that contain the logic required to support EMV transactions (for EMV customers), or dummy procedures (for non-EMV customers). Both the EMV procedures and the dummy EMV procedures are contained in separate source files.

1.2 LINK (LIS5) SYSTEM ARCHITECTURE

LINK (LIS5) is a telecommunications network operated by LINK Interchange Limited. The network allows communications between LINK Members for transaction authorisation. The network also acts as a “gateway” between LINK Members and the international card schemes, i.e., MasterCard Europe and Visa.

1.2.1 LIS5 Standard

The LIS5 Standard defines a set of messages that LINK Member Institutions will use to communicate with the LINK Switch.

The Standard details the format and content of the messages, the message flows and the associated processing requirements for LINK Members and the LINK Switch.

The LIS5 Standard supports the broad categories of functions defined below.

- The processing of ATM-based transactions. This is defined to be:
 - domestic ATM sharing within LINK
 - ATM reciprocity with international networks (specifically MasterCard Europe and VISA)
- The processing of cash advance transactions, i.e., cash disbursements performed on credit cards at Point of Service (POS) terminals. This is defined to be:

reciprocity with international networks (specifically MasterCard Europe and VISA)

- The processing of other POS transactions. This is defined to be:

reciprocity with international POS networks, specifically VISA.

Only POS transactions sponsored by magnetic stripe cards are currently supported (i.e., non-EMV).

1.2.2 Network Overview

The following figure provides an overview of the LINK network configuration.

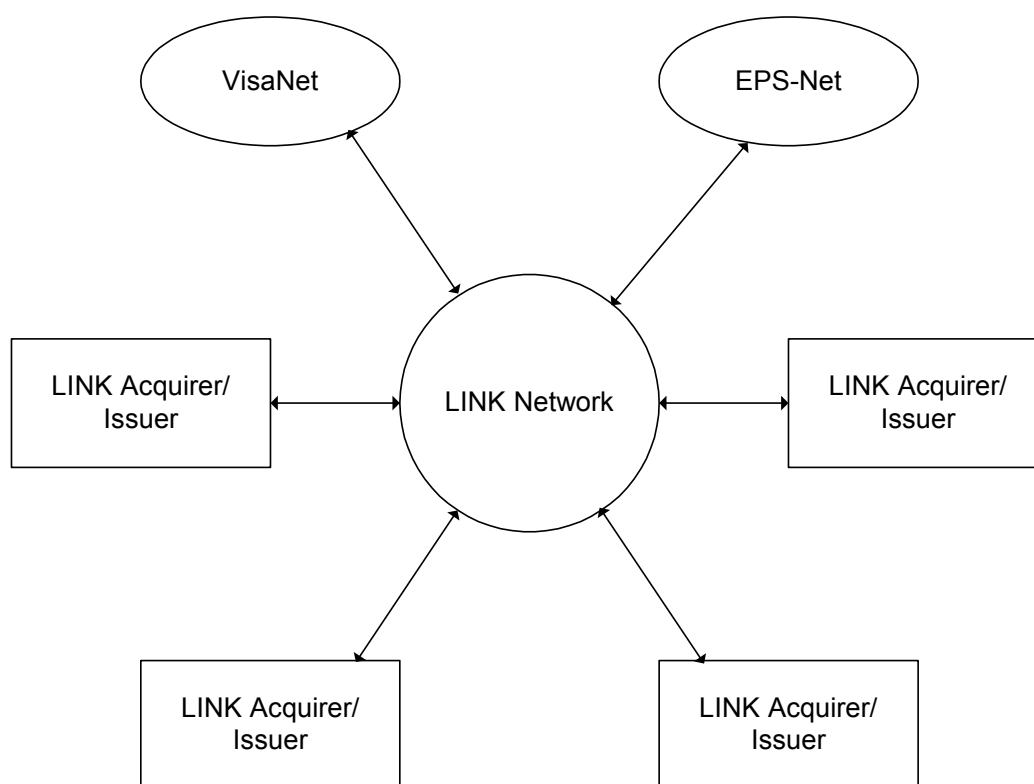


Figure 1-2 LINK Network Configuration

1.2.3 Authorisation and Settlement

Authorisation of transactions is performed using online authorisation messages through the LINK network, as defined in the LIS5 Standard. An authorisation request is sent from the acquirer to the issuer, and the issuer returns an authorisation response to the acquirer. The LIS5 Standard is based on the *ISO Standard for Bank Card Originated Messages, ISO 8583 - 1987*.

Settlement of transactions is performed using data logged during transaction processing.

1.3 MESSAGE PROCESSING

For a full description of the online interface between a member's host system and the LINK network, please refer to the *LINK Switch Service Interchange Standard (LIS5)*.

1.3.1 Message Types

The LINK network supports the following message types:

- 0100 Authorisation/Balance Enquiry/PIN Management request
- 0110 Authorisation/Balance Enquiry/PIN Management response
- 0200 Financial Transaction request
- 0210 Financial Transaction response
- 0420/0421 Reversal advice/Reversal advice repeat
- 0430 Reversal advice acknowledgment
- 0620 Administration advice
- 0624/0625 PIN Management failure advice/PIN Management failure advice repeat
- 0634 PIN Management failure acknowledgement
- 0800 Network management request
- 0810 Network management response

For all transactions passed between LINK members, requests for cash withdrawals are sent as 0200 messages, and balance enquiries and PIN management requests are sent as 0100 messages.

For transactions passed between LINK members and MasterCard Europe, all requests (including cash advances performed on credit cards) are sent through the LINK network as 0200 messages.

POS transaction requests passed from LINK members to VISA are sent through the LINK network as 0200 messages. All other transaction requests passed between LINK members and VISA (including cash advances performed on credit cards) are sent through the LINK network as 0100 messages.

1.3.2 Transaction Types

The LINK network supports the following transaction types:

- Purchase of Goods and Services
- Cash Withdrawal
- Quasi Cash
- E-Cash Load
- Mobile Phone Top-Up (also known as “pre-pay”)
- Balance Enquiry
- PIN Management
- Cash Deposits
- Cheque Deposits
- Cash & Cheque Deposits

1.3.3 Message Flows

The LIS5 Standard explicitly defines all interchanges of messages that are to be supported. The following section defines general principles that apply to all these message flows.

1.3.3.1 Types of Message Flow

Any party in the LINK network may send Administration Advices and Network Management Request messages (except PIN Management failures) to any other party to which they are directly connected. Since the LINK Switch forms the central hub of the network, either the originator or receiver of these messages must be the LINK Switch.

For all remaining supported messages, the circumstances under which, and by whom, they may be sent are defined by the role of the party as Acquirer, LINK Switch or Issuer.

1.3.3.2 Sessions

An institution may only send and receive messages whilst it is logged on to the LINK Switch. Only key exchange and logon related messages may be sent/received whilst not in session.

A session is initiated by the successful completion of the Institution Logon Functional Flow, and terminated by the Institution Logoff Functional Flow.

1.3.3.3 Logging

Details of most functional message flows are required to be recorded by Institutions and the LINK Switch for settlement and charging purposes.

1.3.3.4 Message Routing

The LINK Switch performs message routing based on the Primary Account Number (PAN).

1.3.3.5 Time-out

All responses to Advice and Request messages are required to be returned to the originator within a fixed period of time. Failure to respond within a time-out period gives rise to an exception condition.

1.3.3.6 Message Integrity

The following fields are used to match transaction requests to a reply and an original transaction to a reversal:

- Transaction Time (ISO field P-12)
- Retrieval Reference Number (ISO field P-37)
- Card Acceptor Terminal Id (ISO field P-42)

The combination of these data elements provides a unique reference in any given 24-hour period.

Matching can be improved by also including the Transaction Date (ISO field P-13), Systems Trace Audit Number (ISO field P-11) and the PAN (ISO field P-2) in the list of fields used for matching.

No matching fields are amended during the lifecycle of the message flow concerned.

The Systems Trace Audit Number is used to match other requests and advices to a reply.

1.3.3.7 “Must Deliver” Messages

Some of the message flows are defined under the LIS5 Standard as "must deliver", that is delivery of the request or advice to the intended recipient must be repeated until a response message is received.

If no response is received to a Must Deliver message within the specified time-out period, the message is resent. The sender repeats this process up to 5 times in total, after which the message is stored pending re-establishment of communications with the intended receiver.

1.3.3.8 Reversal and PIN Management Failure Processing

LIS5 reversals can be generated only after receiving an approved transaction response (i.e., on receipt of a response rather than timeout).

For PIN Management requests, PIN Management failures can be generated without receiving an approved transaction response (i.e., on timeout rather than receipt of a response).

1.3.3.9 Invalid Messages

A message will be invalid if it has an invalid message type identifier, a field format error such that the message cannot be parsed, or a mandatory data element is absent.

If an invalid message is received, the recipient must respond with an Administration Advice to the originator of the message.

1.3.4 Transaction Security

The LIS5 security procedures are based around ensuring that the Personal Identification Number (PIN), entered by the cardholder, does not appear in the clear during the life of the transaction.

The PIN management and security procedures are based on the *ANSI Standard X9.8 for PIN Management and Security*.

Message authentication is not currently implemented across the LINK network.

1.3.4.1 PIN Processing

The LIS5 Standard supports the entry of PINs of up to 12 digits. A minimum length of 4 is required.

Once entered, the PIN is held in encrypted form except when within tamper-proof physically secure devices. The PIN block encryption functions are based on the ***ANSI Data Encryption Standard (DES) X9.9*** algorithm.

Before the PIN can be encrypted, it needs to be formatted as a 64-bit block containing 16 hexadecimal characters. The PIN block format supported in the LINK network is ISO Format 0.

The LINK network supports PIN blocks encrypted using Triple DES.

To enhance the security of the network, the Issuer and the Acquirer use separate keys for encrypting the PIN blocks, thus creating discrete security zones.

The LINK Switch is responsible for encrypted PIN Block translation between Institution security zones.

The PIN and encrypted PIN Block are never logged (apart from temporarily as part of the whole message for diagnostic purposes).

1.3.4.2 Key Management

The LINK network supports the dynamic exchange of keys used for PIN block encryption.

An Acquirer Working Key (AWK) is used for encrypting the PIN block during message transmissions between an Acquirer and LINK. LINK is responsible for the generation of new Acquirer Working Keys, and sends the new Acquirer Working Key to the acquirer. The acquirer cannot generate new AWKs, but may request a change.

An Issuer Working Key (IWK) is used for encrypting the PIN block during message transmissions between LINK and an Issuer. The Issuer is responsible for the generation of new Issuer Working Keys and sends the new Issuer Working Key to LINK. LINK cannot generate new IWKs, but may request a change.

Issuer and Acquirer Working Keys must be generated independently.

Full details of the key change protocol are given in the ***LINK Switch Service Interchange Standard - LIS5 Security Standard***.

1.3.5 Acquirer Surcharging and Issuer Pre-Notification

The LIS5 Standard provides Acquirers with the ability to surcharge transactions. Acquirers may also provide cardholders with an “incentive”, i.e., a negative surcharge, to use their ATMs.

An Acquirer is not permitted to surcharge a transaction for a designated LINK Non-Debit card that charges an issuer fee. For a designated LINK Non-Debit card, the Acquirer may provide an incentive.

The LIS5 Standard provides Issuers with the ability to inform Acquirers of an issuer charge or an issuer rebate. Issuers may indicate an issuer charge only if the Acquirer is not applying a surcharge. If a surcharge is being applied, the Issuer may provide a rebate.

Issuer fees in respect of non-debit card transactions need not be specified, but a generic warning that such fees will be applied must be given at the time of the transaction, with the opportunity to cancel or proceed. Where the Issuer of a non-debit card returns the amount of the fee, this must be notified to the customer at the time of the transaction, with the opportunity to cancel or proceed.

All fees in respect of debit card transactions must be pre-notified to the customer at the time of the transaction with the opportunity to cancel or proceed. The customer receipt (for both debit card and non-debit card transactions) must show all known fees, positive and negative.

1.3.6 EMV Support

The LINK (LIS5) network supports the processing of ATM transactions performed on EMV-compliant chip cards. The LIS5 Standard accommodates the transmission of EMV-specific transaction data in tertiary data fields within the ISO 8583 message format.

1.4 IMPLEMENTATION CONSIDERATIONS

This section documents environmental issues to be considered when implementing the LINK (LIS5) Interchange Interface.

1.4.1 Software Dependencies

The LINK (LIS5) Interchange Interface is compatible with XPNET version 3.0 and version 3.1.

The product is a Switch 6.0 interface, and therefore operates in a Release 4.0, a Release 5.0, a Release 5.1, a Release 5.3, or a Release 6.0 BASE24 system.

The BASE24 subvolume, SW60SWPM, contains the source file libraries required to support Transactional Interface Monitoring within an interchange interface. This subvolume will be required so that the LINK (LIS5) Interchange Interface can support T.I.M.

To provide full support for the encryption of PIN blocks using Triple DES, the Transaction Security Services (TSS) must be installed and configured. Appendix E, “TSS Migration”, describes the steps required to configure the BASE24 LINK (LIS5) Interchange Interface for use with TSS.

1.4.2 Disk File Requirements

Each LINK (LIS5) Interchange Interface process writes records to an Interchange Log File (ILF) and a Store-and-Forward File (SAF). Sufficient disk space must be available for LINK (LIS5) Interchange Interface processes to write the required information to each of these files.

1.4.3 Security Requirements

The minimum device and firmware requirements for HSMs in order to support Triple DES in the interface to the LINK switch are as follows:

- Thales e-security (formerly Racal) HSMs must be RG7000 with Revision 1.05/5.05 firmware or greater.
- Atalla HSMs must be an A9000 or A10000 with Revision 2.8.4 firmware.

For customers who use Atalla HSMs, it is not possible to comply fully with LINK's dynamic key exchange requirements without using the Transaction Security Service. This is because the Atalla HSM returns only 4-digit Key Check Values (KCVs) in the key generate and key translate commands used by BASE24 (without the TSS), instead of the 6-digit KCVs required by LINK.

1.4.4 Acquirer Surcharging and Issuer Pre-Notification

To support acquirer surcharging and issuer pre-notification, the BASE24 Surcharge product must be licensed and installed.

If directly-connected ATMs are supported, the ATM Device Handlers must comply with LINK's acquirer surcharging and issuer pre-notification requirements. The following BASE24-atm Device Handlers support this functionality:

- BASE24 Interbold 10XX/478X Device Handler
- BASE24 Interbold 10XX/478X Device Handler with Dial-up Extension
- BASE24 NCR 5XXX Device Handler
- BASE24 NCR 5XXX Device Handler with Dial-up Extension
- BASE24 Triton Device Handler (Dial-up).

Within BASE24, the support provided for LINK's requirements is known as Surcharge Interactive Response Notification.

1.4.5 Non-Currency Dispense

To support the acquiring of Quasi Cash transactions, the BASE24-atm NCD product must be licensed and installed.

If directly-connected ATMs are supported, the ATM Device Handlers must support NCD processing. The following BASE24-atm Device Handlers support the NCD functionality:

- BASE24 Interbold 10XX/478X Device Handler
- BASE24 Interbold 10XX/478X Device Handler with Dial-up Extension
- BASE24 NCR 5XXX Device Handler
- BASE24 NCR 5XXX Device Handler with Dial-up Extension
- BASE24 Triton Device Handler (Dial-up).

Details on how to configure this feature can be found in the individual device support manuals.

1.4.6 EMV Support

To support EMV processing, the BASE24-atm EMV product must be licensed and installed.

2. PROCESSING OVERVIEW

The LINK (LIS5) Interchange Interface process is responsible for the online message transfer between BASE24 and the LINK network.

This process performs the following functions:

- Manages communications between BASE24 and the LINK network.

The LINK (LIS5) Interchange Interface sends and receives network management messages to determine the status of the connection between BASE24 and the LINK network.

- Facilitates message exchange between BASE24 and the LINK network.

When BASE24 receives a transaction from the LINK network, the LINK (LIS5) Interchange Interface translates the message into the BASE24-atm internal message format and sends the message to a BASE24-atm Authorisation process. When BASE24 receives a transaction for the LINK network to authorise, the LINK (LIS5) Interchange Interface translates the message from the internal message format to the external message format and forwards the message to the LINK network. LINK (LIS5) supports communication with the LINK network using the message formats defined in the *LINK Switch Service Interchange Standard (LIS5)*. These formats are based on the ISO 8583 (1987) message standard.

- Provides store-and-forward capabilities.

When the connection between BASE24 and the LINK network is down, the LINK (LIS5) Interchange Interface stores certain messages intended for the LINK network in its Store-and-Forward File (SAF). When the connection is restored, these messages are retrieved from the file and sent to the LINK network.

- Facilitates settlement between BASE24 and the LINK network.

The LINK (LIS5) Interchange Interface produces a set of interchange reports to assist in reconciling transactions between BASE24 and the LINK network. These reports allow the BASE24 participants to reconcile the settlement positions of the LINK network and BASE24.

2.1 FILE USAGE

The LINK (LIS5) Interchange Interface process uses a number of BASE24 files. These files, which can be divided into control files and transaction files, are identified below.

2.1.1 Control Files

Control files contain configuration information and parameters that determine how the LINK (LIS5) Interchange Interface process operates. The control files that contain configuration information for the LINK (LIS5) Interchange Interface include:

- Network Environment File (NEF)
- Logical Network Configuration File (LCONF)
- Interchange Configuration File (ICF)
- Key File (KEYF or KEY6)
- Switch Terminal File (STF)
- Log Message Configuration File (LMCF)
- Token File (TKN)

For information on how these files are used to configure the LINK (LIS5) Interchange Interface process, refer to section 3.

For information on the switch-specific ICF data definition, refer to Appendix B.

2.1.2 Transaction Files

Transaction files are used to store information about individual transactions processed through the LINK (LIS5) Interchange Interface. The transaction files used by the LINK (LIS5) Interchange Interface include:

- Interchange Log File (ILF)
- Store-and-Forward File (SAF)

The ILF and SAF DDLs can be accessed using DDLDOC. For information on DDLDOC, refer to the *BASE24 Files and Record Structures Reference Manual*.

2.2 NETWORK MANAGEMENT

The BASE24 LINK (LIS5) Interchange Interface connects a BASE24-atm system with the LINK network. The communications between the networks are managed by ACI's XPNET software and application-level network management messages.

For further information on network management processing in the LINK (LIS5) Interchange Interface process, refer to section 4.

2.2.1 XPNET

Stations provide the means by which communications takes place between BASE24 and the LINK network. All messages sent to the LINK network are actually sent to the stations defined for the LINK network. Likewise, all messages received from the LINK network are actually received from stations.

All protocol-level functions between BASE24 and the stations are handled by the XPNET process using the communications control software, provided with the HP NonStop (formally Tandem) hardware. The LINK (LIS5) Interchange Interface process sends and receives messages through the XPNET process. It is the responsibility of the XPNET process to maintain the communications connections with the stations according to the protocols under which the stations are defined.

Configuration of an XPNET system includes defining attributes for each XPNET node and all of its associated processes, links, lines, stations, and devices. All objects, including the XPNET node itself, can be added, changed, and deleted online without stopping the XPNET system. Configuration information relating to each XPNET node is held in the node's Network Environment File (NEF). There is one NEF for each XPNET node in the system. Each NEF is managed by the XPNET process in that XPNET node.

No two XPNET systems are configured exactly alike. In addition to being composed of different applications, processes, devices, and stations, there are a number of settings that depend entirely on how you want your system to work.

For complete information on the syntax of the commands, attributes, and directives used to configure an XPNET system, see the *NET24-XPNET Network Control Command Reference Manual*.

2.2.2 Base Message Types

The LINK (LIS5) Interchange Interface module supports network management and administration message types, known collectively as BASE messages.

The following table indicates the base message types supported by the LINK (LIS5) Interchange Interface.

Type	Description
0620	Administration advice
0800	Network management request
0810	Network management response

Administration Advice messages are used when invalid message are received. If either the LINK (LIS5) Interchange Interface module or the LINK Switch receives an invalid message (e.g., the message cannot be identified or a mandatory data item is absent), the recipient of the invalid message formats and sends a 0620 Administration Advice message to the sender.

Network Management Request/Response messages are used to support the following functions:

- Echo Test
- Logon
- Logoff
- Business Day Cutover
- Dynamic Key Exchange.

2.3 TRANSACTION PROCESSING

The BASE24 LINK (LIS5) Interchange Interface connects a BASE24-atm network with the LINK network. It allows a LINK cardholder to initiate a transaction at a terminal belonging to a network not operated by the card issuer.

For further information on transaction processing in the LINK (LIS5) Interchange Interface process, refer to section 5.

2.3.1 Message Formats

This section defines supported message types, the external and internal message formats, and the translation between the two message formats.

2.3.1.1 Supported Message Types

The LINK (LIS5) Interchange Interface module supports both acquirer and issuer processing for ATM transactions.

The following table indicates the transaction message types supported by the LINK (LIS5) Interchange Interface.

Type	Description	Acquirer/Issuer Processing
0100	Account balance request	Acquirer & Issuer
0110	Account balance response	Acquirer & Issuer
0100	PIN management request	Acquirer & Issuer
0110	PIN management response	Acquirer & Issuer
0200	Financial transaction request	Acquirer & Issuer
0210	Financial transaction response	Acquirer & Issuer
0420	Acquirer reversal advice	Acquirer & Issuer
0421	Acquirer reversal advice repeat	Acquirer & Issuer
0430	Acquirer reversal advice response	Acquirer & Issuer
0624	PIN management failure advice	Acquirer & Issuer
0625	PIN management failure advice repeat	Acquirer & Issuer
0634	PIN management failure advice response	Acquirer & Issuer

The LINK (LIS5) Interchange Interface module does not support “gateway” functions, that is:

- transactions originating from the Visa and MasterCard networks;
- transactions destined for the Visa and MasterCard networks.

2.3.1.2 External Message Format

The external message format is used whenever information is routed between the LINK (LIS5) Interchange Interface process and the LINK network. The external message format is defined in the *LINK Switch Service Interchange Standard (LIS5)*.

2.3.1.3 Internal Message Format

The internal message format is used whenever information is routed between the LINK (LIS5) Interchange Interface process and the rest of BASE24. The internal message is commonly referred to as the BASE24-atm Standard Internal Message (STM).

For more information on the STM, see the *BASE24-atm Transaction Processing Manual*.

2.3.1.4 Message Translation

The LINK (LIS5) Interchange Interface process translates messages it receives from the LINK network into the internal message format, then sends the internal message to the BASE24-atm Authorisation process. When the message has been processed, the authorizing entity sends an internal response to the LINK (LIS5) Interchange Interface process. Using information from the internal message, the LINK (LIS5) Interchange Interface process formats an external response message and sends it to the LINK network.

The LINK (LIS5) Interchange Interface process translates the internal message it receives from BASE24-atm into the external message format, then sends the external message to the LINK network for authorisation. When the message has been processed, LINK sends an external response to the LINK (LIS5) Interchange Interface process. Using information from the external message, the LINK (LIS5) Interchange Interface process formats an internal response message and sends it to the BASE24-atm Authorisation process.

2.3.2 LINK Network Security

The transaction acquirer passes the PIN data to the card issuer in ISO field P-52. The BASE24 LINK (LIS5) Interchange Interface supports the transmission of PIN Blocks encrypted using Triple DES only.

When sending a request message to LINK, the encrypted PIN Block is retrieved from the internal message. The PIN Block is translated to encryption under the Acquirer Working Key (AWK). It is this encrypted value that is carried in P-52.

When receiving a request message from LINK, the PIN Block is retrieved from P-52, encrypted under the Issuer Working Key (IWK).

Both Acquirer and Issuer Working Keys are dynamically exchanged between the institution and the LINK network by using network management messages.

2.3.3 Supported Transaction Types

This section defines supported transaction types, and the processing of these transactions by the LINK (LIS5) Interchange Interface.

2.3.3.1 Transactions Received from the LINK Network

The LINK (LIS5) Interchange Interface may receive the following transaction types from the LINK network for authorisation:

- Balance Enquiry
- Cash Withdrawal
- Quasi Cash
- E-Cash Load
- Mobile Phone Top-Up
- PIN Management
- Cash Deposits
- Cheque Deposits
- Cash and Cheque Deposits

When a Quasi Cash transaction is received from the LINK network, the message is interpreted as a Non-Currency Dispense (NCD) transaction. These transactions can be identified by a processing code of '11xx00' (where 'xx' is any valid 'from' account type) in the generic switch token added to the internal message.

When an E-Cash Load transaction is received from the LINK network, the processing code is mapped to a BASE24-atm transaction code of '10' (cash withdrawal). These transactions can be identified by a processing code of '13xx60' and '13xx61' (where 'xx' is any valid 'from' account type) in the generic switch token added to the internal message.

When a funds authorisation for a Mobile Phone Top-Up transaction, a Foreign Currency Dispense (FCD) transaction, or a Non-Cash financial transaction is received from the LINK network, the processing code is mapped to a BASE24-atm transaction code of '10' (cash withdrawal). These transactions can be identified by the processing code in the generic switch token added to the internal message. The processing code will be '00xxyy', where 'xx' is any valid 'from' account type, and 'yy' is '80' (Mobile Phone Top-Up transaction), '81' (FCD transaction), or '82' (Non-Cash financial transaction).

When a PIN Management transaction is received from the LINK network, the processing code is mapped to a BASE24 transaction code as a PIN change ('81') or EMV PIN unblock ('82').

When any Deposit transaction is received from the LINK network, the processing code is mapped to a BASE24 generic deposit transaction code of '20'. These deposit transactions may not contain a PIN.

The LINK (LIS5) Interchange Interface supports transactions received from the LINK network where the card details are manually-entered and/or a PIN is not used, e.g., at a Post Office counter.

2.3.3.2 Transactions Sent to the LINK Network

The LINK (LIS5) Interchange Interface forwards the following BASE24-atm transactions to the LINK network for authorisation:

- Balance Enquiry
- Cash Withdrawal
- Quasi Cash
- Mobile Phone Top-Up
- PIN Management
- Cash Deposits
- Cheque Deposits
- Cash and Cheque Deposits

BASE24 does not generate an E-Cash Load processing code in request messages to the LINK Interchange.

Other BASE24-atm transaction types may be routed to an alternate authorising destination, (e.g., the ATM Authorisation process).

The LINK (LIS5) Interchange Interface does not support transactions destined for the LINK network where the card details are manually-entered and/or a PIN is not used (deposits, balance enquiries, withdrawals).

2.3.4 Acquirer Surcharging and Issuer Pre-Notification

LINK has defined Field P-28, Amount, Transaction Fee, to be used for the surcharge/incentive amount in financial request, reversal and repeat reversal message types. The field is conditional and is only used when a surcharge is applied.

When BASE24 is acting as the acquirer, the surcharge amount (if present) is set up in Field P-28. When BASE24 is acting as the issuer, the surcharge amount is retrieved from Field P-28 (if present).

LINK has defined Field P-30, Amount, Transaction Processing Fee, to be used for the pre-notification of issuer charges. This field is transmitted in financial transaction response messages from issuers, and contains the 'issuer charge or issuer rebate' to be displayed on the acquirer's ATM screen. Field P-30 may be absent in financial transaction responses for non-debit card transactions, indicating an "unknown" issuer fee.

When BASE24 is acting as the acquirer, the issuer/fee rebate amount is retrieved from Field P-30 (if present). When BASE24 is acting as the issuer, the issuer/fee rebate amount (if present) is set up in Field P-30.

A reversal reason code of '9' identifies reversals resulting from customers rejecting either the acquirer surcharge or the issuer charge.

Within BASE24, the support provided for LINK's requirements is known as Surcharge Interactive Response Notification.

2.3.5 Transaction Logging

The LINK (LIS5) Interchange Interface logs the following types of messages to the Interchange Log File (ILF):

- Requests containing error conditions
- Responses
- Reversals
- PIN Management failures
- Force posts

Whenever a reversal occurs, the LINK (LIS5) Interchange Interface attempts to update the ILF record already logged for the transaction.

2.3.6 Store-and-Forward Function

A store-and-forward function is incorporated into the LINK (LIS5) Interchange Interface process to collect transaction messages during periods when a LINK network is unavailable. When the LINK network is unavailable, messages required as notification to the LINK network (“must deliver” messages) are stored in the Store-and-Forward File (SAF). These messages are then transmitted later when communications are restored.

2.3.7 Triple DES Support

The BASE24 LINK (LIS5) Interchange Interface supports the transmission of PIN Blocks encrypted using Triple DES only (Single DES is no longer supported).

Triple DES will be supported fully only in Release 6.0 (the full solution), using the KEY File (KEYF). To facilitate a phased migration, BASE24 Switch 6.0 interchange interfaces can be used in a 5.x environment to provide limited Triple DES support (the interim solution), using the KEY6 File (KEY6).

2.3.7.1 Triple DES “Full” Solution

With the advent of Triple DES, ACI has introduced a new architecture for managing transaction security. In BASE24 Release 6.0, all cryptographic functions, such as PIN translation, can be managed through the new *BASE24 Transaction Security Services*. A full Triple DES environment is based on BASE24 6.0 using the Transaction Security Services.

In the traditional BASE24 environment, the application processes (device handlers, authorisation modules and interface processes) manage their own cryptographic services. This includes storing and retrieving keys, and making calls to a Hardware Security Module (HSM). In an environment using Transaction Security Services (TSS), the applications call the TSS process rather than the HSM. TSS manages the storing and retrieving of keys and making calls to the HSM.

When using BASE24 Transaction Security Services, the application processes use “key locators” in place of actual keys. Key locators serve as tags, or indexes, representing a key. The actual keys are stored in new key files managed exclusively by Transaction Security Services. The TSS uses the key locators to retrieve the actual keys, before formatting and sending commands to the HSM.

The full solution impacts the LINK (LIS5) Interchange Interface because of the new configuration requirements. The Key File (KEYF) stores key locators to the TSS database instead of the actual PIN block encryption keys themselves.

Further details of the TSS are given in the ***BASE24 Transaction Security Services Processing Guide***. Appendix E, “TSS Migration”, describes the steps required to configure the BASE24 LINK (LIS5) Interchange Interface for use with TSS.

2.3.7.2 Triple DES “Interim” Solution

Transaction Security Services is not supported prior to Release 6.0, however, the same version of BASE24 interchange interfaces can run in all releases of BASE24. To enable Release 6.0 interchange interfaces to run in all releases of BASE24, modifications have been made to interchange interfaces to support Triple DES when the interface is running in a BASE24 Release 5.x network. The interchange interfaces must support translating a PIN block in transactions originating at the interchange from Triple DES encryption to Single DES encryption. The interchange interface must support translating a PIN block in transactions originating in BASE24 or a host from Single DES encryption to Triple DES encryption. To accomplish this, interchange interfaces must call utilities added to support double length PIN encryption keys in PIN block translate calls to an HSM. A new parameter indicates when the special PIN block translation will occur.

This interim solution impacts the LINK (LIS5) Interchange Interface in the following ways:

- Support is required for the new KEY6 file, replacing the Key File (KEYF). This new file stores double length PIN block encryption keys instead of the existing single length PIN block encryption keys.
- Support is required for translating PIN blocks encrypted using Single DES (received from the BASE24-atm Authorisation process) to encryption using Triple DES (for transmission to LINK).
- Support is required for translating PIN blocks encrypted using Triple DES (received from the LINK network) to encryption using Single DES (for transmission to the BASE24-atm Authorisation process).

An additional call to the HSM (to convert a PIN Block from Triple DES encryption to Single DES encryption) will be required to support PIN Blocks received from LINK if the Triple DES “interim” solution is implemented.

2.4 INTERCHANGE REPORTING

The LINK (LIS5) Interchange Interface produces two reports: SETL01 and STAT09. The purpose of these reports is to facilitate settlement between BASE24 and the LINK network.

For information on these reports, refer to section 6.

2.5 CHIP-BASED DEBIT/CREDIT SUPPORT

LINK supports chip-based debit and credit functionality, based on the EMV/MasterCard/Visa (EMV) Standards: *ICC Specifications for Payment Systems, version 3.1.1, dated 31st May 1998*.

For authorisation systems such as BASE24, this involves supporting additional data elements in the LIS5 message and supporting additional values in existing data fields.

EMV support is provided by a separately licensable module, which is built on top of the core LINK (LIS5) Interchange Interface modules.

For information on the EMV extension module, refer to section 7.

Full details of EMV support in BASE24 are provided in the *BASE24-atm EMV Support Manual*.

2.6 UNSUPPORTED FUNCTIONALITY

Support for the following optional LIS5 functions may be added in a future release of the LINK (LIS5) Interchange Interface:

- Acquiring Foreign Currency Dispense (FCD) transactions.
- “Gateway” transactions.
- “MoniLink” service, i.e., where banking services are accessed via a mobile phone.
- Online Key Verification messages for validating a new Zone Master Key (ZMK).
- Format-2 Interchange Log Files (ILFs).
- The BASE24 Release 6.0 ‘Transactions Allowed’ enhancement.

In addition, BASE24-atm may be enhanced to allow issuers to configure issuer rebates, and to allow issuers greater flexibility in the configuration of issuer fees.

3. CONFIGURATION AND FILES MAINTENANCE

This section describes the configuration files that must be set up or verified in order that the LINK (LIS5) Interchange Interface process can be configured into the BASE24 network.

This section also provides details of the BASE24 files maintenance screens and field descriptions that have been enhanced to support the BASE24 LINK (LIS5) Interchange Interface module. Function keys that operate differently from the standard function keys are also documented.

For full details of the screen layouts, standard BASE24 function keys and online help screens, refer to the *BASE24 Base Files Maintenance Manual*.

3.1 XPNET PROCESS

This section provides information on the physical communications used to connect a BASE24 system with the LINK network.

A single BASE24 LINK (LIS5) Interchange Interface process can support a maximum of ten stations with the LINK network.

This section provides a brief explanation of the way in which BASE24 lines, stations and subdevices are defined for the LINK (LIS5) Interchange Interface process.

See the *NET24-XPNET Network Control Operations Guide* for descriptions of the XPNET network control functions. This manual provides descriptions of the various functions used in configuring, monitoring, and controlling an XPNET system.

See the *NET24-XPNET Network Control Command Reference Manual* for descriptions of the XPNET network control commands. This manual provides comprehensive command descriptions of all network control commands used to monitor and control an XPNET system.

3.1.1 Terminology of Lines, Stations and Subdevices

All BASE24 communication to external devices takes place using logical entities known as stations and lines. For asynchronous and byte synchronous protocols, a line is synonymous with a physical communications line and there can be one or more stations per line, depending upon the protocol. For X.25, however, a line is equivalent to an X.25 virtual circuit and only one station can be defined per line.

3.1.2 Network Management

This section presents sample BASE24 process, station, line, and device objects definitions for the LINK (LIS5) Interchange Interface.

3.1.2.1 Process Definition

One process definition must exist for each LINK (LIS5) Interchange Interface process. An example of a process definition for the LINK (LIS5) Interchange Interface process is given below.

```
SET PROCESS PROGRAM $DATA.SW60OBJ.LINK
SET PROCESS CPU 1
SET PROCESS BCPU 0
SET PROCESS LIBRARY $DATA.XPNET.SKELB
SET PROCESS PRIORITY 149
SET PROCESS PPD $P1E1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 20
SET PROCESS QMI 85
SET PROCESS QMT 150
SET PROCESS SWAP $DATA
ADD PROCESS \SYS1.P1A^NODE.P1A^LINK1
```

3.1.2.2 Device Declarations

The BASE24 Network definition must contain one device declaration for each generic station type.

An example of a device declaration for an X.25 device is given below.

```
SET DEVICE RECVLEN 4095
SET DEVICE XMITLEN 4095
SET DEVICE ADD DESTSTRING 1, P1A^DPROC, 2, %HA0
SET DEVICE ADD DESTSTRING 2, P1A^EPROC, 2, %HA1
SET DEVICE ADD PROTOCOLSTRING 1, ITI, 10, %HF1
SET DEVICE ADD PROTOCOLSTRING 2, ATALLA, 10, %HF2
SET DEVICE ADD PROTOCOLSTRING 3, DATAFONO, 10, "?"
ADD DEVICE \SYS1.P1A^NODE.D1A^X251
```

3.1.2.3 Station Declarations

There must be a station declaration for each station in the network. For the X.25 protocol, there must be one station declaration for each switched virtual circuit used by the LINK (LIS5) Interchange Interface process.

An example of an X.25 station declaration is given below.

```
RESET STATION
SET STATION DEVICENAME \SYS1.P1A^NODE.D1A^X251
SET STATION LINENAME \SYS1.P1A^NODE.L1A^X251
SET STATION DESTINATION \SYS1.P1A^NODE.P1A^DPROC
SET STATION UC RATE X25STARATE
SET STATION UC STATE X25STASTATE
SET STATION UC QUEUE X25STAQUEUE
SET STATION CLASS GROUP1
ADD STATION \SYS1.P1A^NODE.S1AX251
```

3.1.2.4 Line Declarations

There must be one line declaration for each communication line used by the LINK (LIS5) Interchange Interface process. For the X.25 protocol, there must be one line declaration for each X.25 switched virtual circuit used by the LINK (LIS5) Interchange Interface process.

An example of an X.25 line declaration is given below.

```
RESET LINE
SET LINE PORT $AX10
SET LINE PROTOCOL X.25
SET LINE ETE NONE
SET LINE WFCALL ON
SET LINE DUPLEX FULL
SET LINE ATIMER 1
SET LINE VERDTE OFF
SET LINE UC RATE X25LINRATE
SET LINE UC STATE X25LINSTATE
SET LINE UC QUEUE X25LINQUEUE
SET LINE CLASS GROUP1
ADD LINE \SYS1.P1A^NODE.L1A^X251
```

3.2 LOGICAL NETWORK CONFIGURATION FILE (LCONF)

The Logical Network Configuration File (LCONF) is the central location of file information and parameters for the BASE24 Logical Network; one LCONF exists for each logical network.

The LCONF contains assigns that link the physical names of objects in the logical network to names used in the BASE24 code and parameters (params) that define processing values at the logical network level.

The LCONF assigns and params listed in this subsection are required by the LINK (LIS5) Interchange Interface process to initialise and process correctly.

For more details, refer to the *BASE24 Logical Network Configuration File Manual*.

3.2.1 LCONF Assigns

Unless otherwise stated, the assigns specified below are optional.

Assign	Description
ICF (Required)	The name of the Interchange Configuration File (ICF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
ILF (Required)	The name of the Interchange Log File (ILF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides. The ILF assign must be associated with the symbolic name of the Interface process. A template file name is required as the LINK (LIS5) Interchange Interface process creates the ILF periodically.
KEYF (Required if the TSS has been implemented)	The name of the Key File required for the Triple DES “full” solution. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
KEY6 (Required if the TSS has not been implemented)	The name of the KEY File required for the Triple DES “interim” solution. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
LMCF	The name of the Log Message Configuration File (LMCF) that is used by all BASE24 processes. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
RPT	The name of the BASE24 reports file that contains BASE24 and Interchange reports. A template file is required as the LINK (LIS5) Interchange Interface process creates the Interchange reports periodically.
RPT-ERROR-FILE	The name of the file to which the Interchange report programs route the report error log messages.

Assign	Description
SAF (Required)	The name of the Store-and-Forward File. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides. The SAF assign must be associated with the symbolic name of the LINK (LIS5) Interchange Interface process. A template file is required because the LINK (LIS5) Interchange Interface process creates the SAF periodically.
STF	The name of the Switch Terminal File (STF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
SW-RPT01-PROG-NAME	The name of the executable object code that generates the Interchange Clearings Statement Report (RP01SWI). The symbolic name of the LINK (LIS5) Interchange Interface process name is required for the assign. The format is: <i>\$volume.subvolume.RP01SWI</i>
SW-RPT09-PROG-NAME	The name of the executable object code that generates the Interchange Transaction Detail Report (RP09SWI). The symbolic name of the LINK (LIS5) Interchange Interface process name is required for the assign. The format is: <i>\$volume.subvolume.RP09SWI</i>
SW-RPT-OBEY-FILE	The name of the LINK (LIS5) Interchange Interface report obey file. The symbolic name of the LINK (LIS5) Interchange Interface process is required for the assign. The format is: <i>\$volume.subvolume.Swyymmdd</i>
TKN (Required)	The name of the Token File (TKN). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.

For assigns required for Hardware Security Module (HSM) use, refer to the ***BASE24 Transaction Security Manual***.

3.2.2 LCONF Parameters

Unless otherwise stated, the parameters specified below are optional.

Parameter	Description
ACCT-NUM-TYP	A value indicating whether account numbers are stored in binary or decimal form. The binary form allows the account number to be up to 28 digits and the decimal form allows an account number of 19 digits. This parameter is required for report processing. Valid values are: B = Binary format D = Decimal format (default)
ATM-LN-CUTOVER (Required)	The time of day the entire BASE24-atm Logical Network is cut over. The 24-hour clock format is used. Valid values are 00:00–23:59.
BASE24-REL	A code indicating whether the BASE24 logical network is currently running release 3.4/4.0 or release 5.x or release 6.0. Valid values are '40', '50' and '60' (default).
CO-OWNED- TERMINAL-BRANCH- ID	This optional parameter specifies the branch identifier used within the BASE24-atm system for non-branch co-owned terminals. Valid values are 4 characters containing alphanumeric characters and/or asterisks. Non-branch, co-owned ATMs are a subset of remote ATMs, and so the value of this parameter should ensure that a matching BRCH-ID in the STM also matches the LCONF parameter REMOTE-TERMINAL-BRANCH-ID.
DATE-DISPLAY-TYPE	The format of the month and day portion of the dates on reports. The parameter does not affect dates that include the year (i.e., dates in the format YYMMDD). Valid values are: 1 = DDMM 2 = MMDD (default)

Parameter	Description
DEFAULT-ISSUER-FEE	This optional parameter specifies the default issuer fee (specified in fractional currency units) to be used to populate field 30 in 0210 response messages to the LINK Interchange. Valid values are 1 to 8 numeric digits (note that a negative value is not permitted). For Default Issuer Fee processing to be supported, this parameter must exist with the ISSUER-FEE-ACCT-TYPES parameter.
DES-TRIPLE-SINGLE	This optional parameter indicates if Triple DES support in the interface is provided using the “interim” solution, i.e., with the KEY6 (Transaction Security Services not used). Valid values are: Y = Yes, KEY6 is used N = No, KEYF is used (default)
ILF-NOTIFY-INTERVAL	The number of writes made to the ILF between reading and calculating the percentage file full. Once the file reaches the percent full specified in the ILF-NOTIFY-PERCENTAGE param, a message is logged indicating that the file is becoming full. Valid values are 10–32,766. The default value is 50.
ILF-NOTIFY-PERCENTAGE	The percentage of file full at which the LINK (LIS5) Interchange Interface process logs a message indicating that the ILF is becoming full. Valid values are 10–99. The default value is 80.
INCR-LINK-TRACE-NUM-BY	This optional parameter allows the Systems Trace Audit Number (STAN) sent to LINK to be unique across multiple LINK (LIS5) Interchange Interface processes. The value must be equivalent to the number of LINK (LIS5) Interchange Interface processes configured within the BASE24 system. Valid values are 1 to 99. The default value is 1. Refer to section 5.3.2 for further information on how this parameter is used.

Parameter	Description
ISSUER-FEE-ACCT-TYPES	This optional parameter specifies the account types to which the default issuer fee may be applied. Valid values are up to 20 sets of two-digit numeric values, each delimited by a semi-colon (;). For Default Issuer Fee processing to be supported, this parameter must exist with the DEFAULT-ISSUER-FEE parameter.
LINK-ACQUIRERS-MIGRATED	This optional parameter is used by issuers to specify whether all LINK acquirers have migrated to the latest release of the LIS5 Standard. Valid values are: Y = All acquirers have migrated N = Not all acquirers have migrated (default)
LINK-ISSUERS-MIGRATED	This optional parameter is used by acquirers to specify whether all LINK issuers have migrated to the latest release of the LIS5 Standard. Valid values are: Y = All issuers have migrated N = Not all issuers have migrated (default)
LINK-PRE-PAY	This optional parameter is used to enable processing specific to the LINK Switch. Valid values are: Y = LINK Switch N = Normal issuer or acquirer (default)
LINK-START-TRACE-NUM	This optional parameter allows the Systems Trace Audit Number (STAN) sent to LINK to be unique across multiple LINK (LIS5) Interchange Interface processes. The value must be unique for each LINK (LIS5) Interchange Interface process configured within the BASE24 system. Valid values are 1 to 99. The default value is 1. Refer to section 5.3.2 for further information on how this parameter is used.

Parameter	Description
MAX-STF-RECORDS	The maximum number of records belonging to the LINK (LIS5) Interchange in the Switch Terminal File (STF). This parameter is used to determine the amount of extended memory required by the interface. If this parameter is set to 0, enough extended memory for all records in the STF is allocated. Valid values are 0 to 32766. Default value is 0.
REMOTE-TERMINAL-BRANCH-ID	This optional parameter specifies the branch identifier used within the BASE24-atm system for remote terminals. Valid values are 4 characters containing alphanumeric characters and/or asterisks.
SAF-NOTIFY-INTERVAL	The number of writes made to the SAF between reading and calculating the percentage file full. Once the file reaches the percent full specified in the SAF-NOTIFY-PERCENTAGE parameter, a message is logged indicating that the file is becoming full. Valid values are 10–32,766. The default value is 50.
SAF-NOTIFY-PERCENTAGE	The percentage of file full at which the LINK (LIS5) Interchange Interface process logs a message indicating that the SAF is becoming full. Valid values are 10–99. The default value is 80.
SW-ILF-FILE-RETENTION	The number of days an ILF remains on the system, after which the LINK (LIS5) Interchange Interface process purges the ILF from the system. Valid values are 3–99. The default value is 7.
SW-RPT-STARTUP	The time the LINK (LIS5) Interchange Interface process runs the report obey file. The 24-hour clock format is used. Valid values are 00:00–23:59.
SWI-ACQ-DISCR-DATA-SUPPRS	A value indicating whether discretionary data from Track-2 will be suppressed before logging acquirer transactions to the ILF. Valid values are: Y = Suppress discretionary data before logging N = Do not suppress discretionary data (default)

Parameter	Description
SWI-ISS-DISCR-DATA-SUPPRS	A value indicating whether discretionary data from Track-2 will be suppressed before logging issuer transactions to the ILF. Valid values are: Y = Suppress discretionary data before logging N = Do not suppress discretionary data (default)
UPDT-ILF-ON-LATE-RESP	A flag indicating whether the ILF should be updated on a late response from the LINK Switch. Valid values are: Y = Yes N = No (default).

For parameters required for Hardware Security Module (HSM) use, refer to the ***BASE24 Transaction Security Manual***.

3.3 INTERCHANGE CONFIGURATION FILE (ICF)

The LINK (LIS5) Interchange Interface process accesses data maintained in the Interchange Configuration File (ICF) to determine the variable parameters relevant to the LINK network. These parameters define the processing of transactions between the LINK (LIS5) Interchange Interface and the LINK network.

For the LINK (LIS5) Interchange Interface ICF record, data must be configured on the following screens:

- Screen 1
- Screen 2
- Screen 3
- Screen 7
- Screen 8
- Screen 12
- Screen 13

The fields on these screens are described in detail in this section.

In addition, to support Quasi Cash transactions, Non Currency Dispense (NCD) data must be configured on the ICF (Screen 6). This screen is not shown in this section.

Operator access to the ICF is permitted in accordance with the security access levels assigned in the BASE24 CRT access system. For more information on the BASE24 CRT access system, refer to the ***BASE24 User Security Manual***.

For more details on the ICF, refer to the ***BASE24 Base Files Maintenance Manual***.

3.3.1 ICF Screen 1

The ICF screen 1 (Figure 3-1) identifies the Interchange and the Interchange Interface process names and information, such as the Interchange identifiers, station names, default values for certain fields and the report title names to be printed on the RP01SWI reports.

```

BASE24-BASE  INTERCHANGE CONFIG      LLLL      YY/MM/DD  HH:MM  01 OF 13
INTERCHANGE FIID: _____          PROCESS: _____

      SWITCH TYPE:  LINK
INTERCHANGE LOGICAL NET: _____
      REPORTING NAME: _____

      INSTITUTION ID: _____
      SWITCH ID: _____
      STATION 1: _____
      STATION 2: _____

      SIC CODE: _____
      CURRENCY CODE: 826  (UK)
      DEFAULT TERM NUM: _____
      DEFAULT ACQUIRER ID NUM: 00000000000
      CUSTOMER BALANCE DISPLAY: 0  (DON'T DISPLAY OR PRINT)

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
              F12-HELP

```

Figure 3-1 ICF Screen 1

INTERCHANGE FIID

4 (A) M

This field is the institution identifier that uniquely distinguishes the Interchange. It must match the Logical Network ID given to the Interchange. This field and the Process field form the primary key of the file.

PROCESS

16 (A) M

This field specifies the name of the LINK (LIS5) Interchange Interface process defined in the Network Environment File (NEF). The Process and Interchange FIID fields form the primary key of the ICF.

SWITCH TYPE 4 (A) M

This field identifies the Interchange and allows access to the Interchange Interface specific screens. For the LINK (LIS5) Interchange Interface the field must be set to LINK.

INTERCHANGE LOGICAL NET 4 (A) M

This field is the name of the Logical Network to which the LINK (LIS5) Interchange Interface process belongs as defined in the process definition in the Network Environment File (NEF). It is used by the RP01SWI Interchange reports to calculate the net settlement position.

REPORTING NAME 20 (A)

This field contains the name of the LINK network that is to appear as the heading in the LINK (LIS5) Interchange Interface reports.

INSTITUTION ID 15 (A)

The LINK (LIS5) Interchange Interface process does not use this field.

SWITCH ID 15 (A)

The LINK (LIS5) Interchange Interface process does not use this field.

STATION 1 16 (A)

The LINK (LIS5) Interchange Interface process does not use this field.

STATION 2 16 (A)

The LINK (LIS5) Interchange Interface process does not use this field.

SIC CODE 4 (A)

The LINK (LIS5) Interchange Interface process does not use this field.

CURRENCY CODE 3 (N) M

This field identifies the currency code for transactions that the Interchange Interface receives. Valid values are listed in the ***Codes for the Representation of Currencies and Funds, ISO 4217-1981***. For the United Kingdom, a value of 826 (Sterling pound) is specified.

Default Value: 826 (UK Sterling)

Related Fields: IDF CURRENCY CODE
ATD CURRENCY CODE
PTD CURRENCY CODE

DEFAULT TERM NUM

16 (A)

The LINK (LIS5) Interchange Interface process does not use this field.

DEFAULT ACQUIRER ID NUM

11 (N)

This field is used to populate the Acquiring Institution ID (field P-32) in request messages sent to the LINK Interchange, if the PROCESSING MODE (see screen 3) is set to a value of "0".

This field is also used to populate the Authorising Agent Institution ID (field S-113) in response messages sent to the LINK Interchange, if the PROCESSING MODE (see screen 3) is set to a value of "0".

CUSTOMER BALANCE DISPLAY

1 (N) M

Indicates the Interchange preference concerning balance presentation for transactions with incoming responses. The valid values are:

0 = Do not display or print

1 = Display only

2 = Print only

3 = Display and print

Default Value: 0

3.3.2 ICF Screen 2

The ICF screen 2 (Figure 3-2) displays the Interchange settlement information as well as holiday dates.

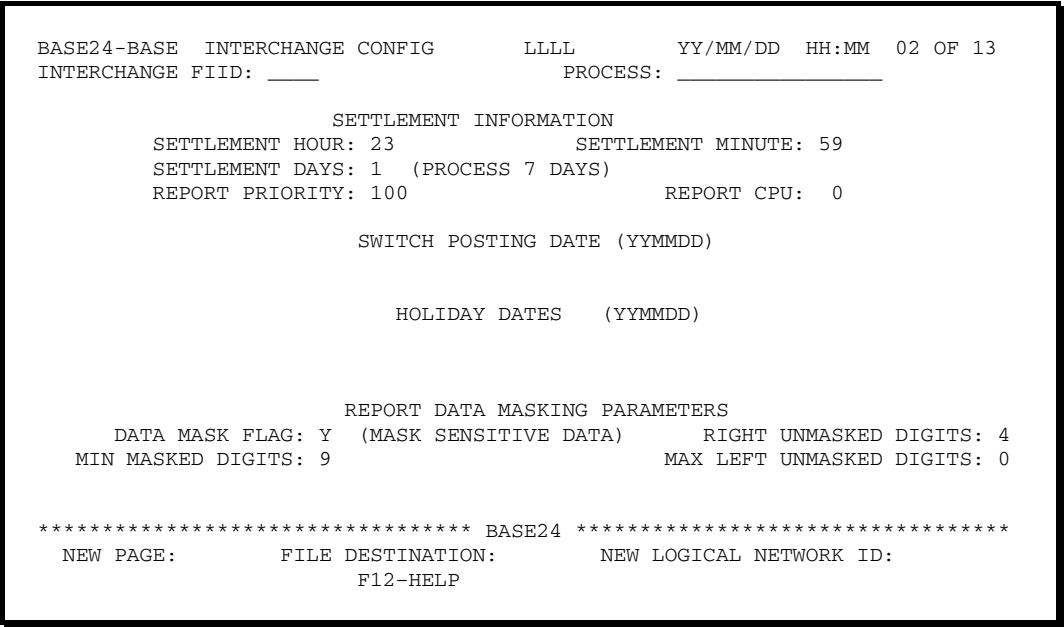


Figure 3-2 ICF Screen 2

SETTLEMENT INFORMATION

The following fields contain Interchange settlement information.

SETTLEMENT HOUR 2 (N) M

This field indicates the hour of settlement (local time) of the Interchange and is used to calculate the Interchange posting date (at warmboot or initialisation).
Valid values are 00–23.
Default Value: 00

SETTLEMENT MINUTE 2 (N) M

This field indicates the minute of settlement (local time) of the Interchange and is used to calculate the Interchange posting date (at warmboot or initialisation).
Valid values are 00–59.
Default Value: 00

SETTLEMENT DAYS

1 (N) M

This field indicates the processing schedule of the Interchange Interface.
The LINK (LIS5) Interchange processes 7 days a week.

Valid values are:

0 = Processes 5 days a week

1 = Processes 7 days a week

Default Value: 1 (7 days a week). However, if this field is blank-filled, the default value is 0 (5 days a week).

REPORT PRIORITY

3 (N) M

This field determines the priority for reports started by the LINK (LIS5) Interchange Interface process.

The value entered in this field must be less than or equal to 255.

Default Value: 100

REPORT CPU

1 (N) M

This field determines the CPU used for reports started by the LINK (LIS5) Interchange Interface.

Default Value: 0

SWITCH POSTING DATE (YYMMDD)

6 (N)

This field displays the current Interchange posting date, in the format YYMMDD.

This date is updated whenever the Interchange cuts over to a new processing date and at initialisation.

The posting date cannot be processed unless six numeric characters representing a valid date are entered.

HOLIDAY DATES (YYMMDD)

16 sets of 6 (N)

The LINK (LIS5) Interchange Interface process does not use this field.

REPORT DATA MASKING PARAMETERS

The values in the following fields specify the report data masking parameters. These parameters support the masking of sensitive information in accordance with PCI data security standards.

DATA MASK FLAG 1 (A) M

A code indicating whether sensitive data should be masked or unmasked.

Valid values are:

Y = Mask sensitive data

N = Do not mask sensitive data.

Default Value: Y

RIGHT UNMASKED DIGITS 1 (N) M

A code defining the number of rightmost digits to be displayed unmasked.

Valid values are:

0–9 = Number of rightmost digits to be displayed unmasked.

Default Value: 4

MIN MASKED DIGITS 1 (N) M

A code defining the minimum number of digits to be masked.

Valid values are:

0–9 = Minimum number of digits to be masked.

Default Value: 9

MAX LEFT UNMASKED DIGITS 1 (N) M

A code defining the maximum number of leftmost digits to be displayed unmasked.

Valid values are:

0–9 = Maximum number of leftmost digits to be displayed unmasked.

Default Value: 0

3.3.3 ICF Screen 3

The ICF screen 3 (Figure 3-3) defines the timers and processing parameters.

BASE24-BASE INTERCHANGE CONFIG
INTERCHANGE FIID: _____

LLLL YY/MM/DD HH:MM 03 OF 13
PROCESS: _____

TIMER LIMITS

NETWORK MANAGEMENT: 30 (SEC)
EXTENDED NETWORK: 30 (SEC)
WAIT FOR TRAFFIC: 600 (SEC)
PERFORMANCE PERIOD: 20 (MIN)

MAXIMUM OUTSTANDING TRANSACTIONS
OUTBOUND: 30
INBOUND: 30

PROCESSING OPTIONS

ACQUIRER: N (Y/N)
ISSUER: N (Y/N)
PROCESSING MODE:
AUTO SIGNON START: N (Y/N)
MAXIMUM TIMEOUTS: 2
MAX SAF RETRY: 5
ACK TO SWITCH: Y (Y/N)
ACK FROM SWITCH: N (Y/N)

NETWORK MANAGEMENT MESSAGE ENABLED: Y (Y = ENABLED, N = DISABLED)
TYPE OF INTERCHANGE REPORTS: 0 (ATM/DETAIL AND SETTLEMENT)
ILF EXTRACT NUMBER: 3

***** BASE24 *****

NEW PAGE:

FILE DESTINATION:
F12-HELP

NEW LOGICAL NETWORK ID:

Figure 3-3 ICF Screen 3

TIMER LIMITS

The following timer fields are used to control specific message activity between the BASE24 network and the Interchange. The values entered in the fields must be checked against the Interchange timer values to ensure the integrity of message traffic.

For more details on the use of these timers in session management processing, refer to the *LINK Switch Service Interchange Standard (LIS5)*.

NETWORK MANAGEMENT 4 (N) M

The maximum number of seconds that the Interchange Interface waits for a response to a network management message; for example, sign on or sign off.

For the LINK (LIS5) Interchange Interface, the recommended value is 30 seconds.

Default Value: 30 (SEC)

EXTENDED NETWORK

4 (N) M

This timer is used to enable the Interchange Interface to sign on again after 30 seconds in certain situations; for example, the Interchange Interface is signed-off or the network management message times out.

For the LINK (LIS5) Interchange Interface, the recommended value is 30 seconds.

Default Value: 60 (SEC)

WAIT FOR TRAFFIC

4 (N) M

The maximum number of seconds that the Interchange Interface waits for traffic on the line from the network before initiating a network management message (echo test).

For the LINK (LIS5) Interchange Interface, the recommended value is 60 seconds.

Default Value: 60 (SEC)

PERFORMANCE PERIOD

4 (N) M

This field specifies the duration (in minutes) of a LINK (LIS5) Interchange performance monitoring period.

Default Value: 20 (MIN)

MAXIMUM OUTSTANDING TRANSACTIONS

The following two fields specify the maximum number of outstanding transactions inbound from and outbound to the Interchange. The combined value determines the number of extended memory pages that will be required for the transactions in process queue.

OUTBOUND

4 (N) M

This field specifies the maximum number of outbound transactions to the LINK network waiting to be processed. This field is used where BASE24 is a transaction acquirer.

Default Value: 15

INBOUND

4 (N) M

This field specifies the maximum number of inbound transactions from the LINK network waiting to be processed. This field is used where BASE24 is a transaction issuer.

Default Value: 15

PROCESSING OPTIONS

The following fields define the processing options for the LINK (LIS5) Interchange Interface.

ACQUIRER **1 (A) M**

This field indicates whether BASE24 will act as an acquirer of LINK transactions. The valid values are:

Y = Yes, send transactions to the Interchange.

N = No, do not send transactions to the Interchange.

Default Value: N

ISSUER **1 (A) M**

This field indicates whether BASE24 will act as issuer for LINK transactions. The valid values are:

Y = Yes, accept transactions from the Interchange.

N = No, do not accept transactions from the Interchange.

Default Value: N

PROCESSING MODE **1 (A)**

This field is Interchange Interface dependant and is used for a variety of purposes.

For the LINK (LIS5) Interchange Interface, this field is used in setting the Acquiring Institution ID (field P-32) in request messages and the Authorising Agent Institution ID (field S-113) in response messages. A value of "0" will cause these fields to be populated with the contents of the DEFAULT ACQUIRER ID NUM (see screen 1). Any other value will cause these fields to be populated with appropriate data from the STM.

AUTO SIGNON START **1 (A) M**

This field specifies whether the Interchange Interface process will automatically sign on to the network at startup. For the LINK (LIS5) Interchange Interface, the recommended value for normal operation is N. Valid values are:

Y = Yes, start up is set to automatically sign on to the Interchange.

N = No, start up is not set to automatically sign on to the Interchange.

Default Value: Y

MAXIMUM TIMEOUTS

4 (N) M

This field specifies the maximum number of consecutive timeouts allowed before network management measures are taken to determine the status of the connection.

Default Value: 2

MAX SAF RETRY

4 (N) M

This field specifies the maximum number of times a transaction from the SAF is transmitted to the Interchange before the record is written to the hard-copy log and deleted from the file.

For the LINK (LIS5) Interchange Interface, the recommended value is 5.

Default Value: 0 (Transactions are transmitted indefinitely until the transmission is successful.)

ACK TO SWITCH

1 (A) M

This field specifies whether the Interchange Interface process acknowledges advice messages from the Interchange.

For the LINK (LIS5) Interchange Interface, this field must be set to Y.
Valid values are:

Y = Yes, acknowledges advices.

N = No, does not acknowledge advices.

Default Value: Y

ACK FROM SWITCH

This field specifies whether the Interchange acknowledges advice messages sent from the Interchange.

For the LINK (LIS5) Interchange Interface, this field must be set to Y.
Valid values are:

Y = Yes, acknowledges advices.

N = No, does not acknowledge advices.

Default Value: N

NETWORK MANAGEMENT MESSAGE ENABLED

1 (A) M

This field specifies whether network management messages between the Interchange and the Interface are enabled or disabled.

For the LINK (LIS5) Interchange Interface, this field must be set to Y.

Valid values are:

Y = Enabled

N = Disabled

Default Value: N

TYPE OF INTERCHANGE REPORTS

1 (N)

This field specifies the types of reports the LINK (LIS5) Interchange requires.

For the LINK (LIS5) Interchange Interface, this field must be set to 0. Valid values are:

0 = ATM (Detail and settlement reports)

1 = POS (Detail and settlement reports)

2 = POS (Detail report)

3 = POS (Debit side detail & settlement reports, including credit side detail report)

Default Value: 0

ILF EXTRACT NUMBER

1 (N)

This field defines the number of ILFs extracted for host reporting and is used by the Super Extract process to indicate the number of ILFs to extract.

Default Value: 3

3.3.4 ICF Screen 7

The ICF screen 7 (Figure 3-4) defines the BASE24-atm allowable transactions.

BASE24-ATM INTERCHANGE CONFIG LLLL YY/MM/DD HH:MM 07 OF 13
INTERCHANGE FIID: _____ PROCESS: _____

ATM ICF DATA

NOT ON US

WITHDRAWAL 0
WITHDRAWAL CC 0
DEPOSIT 0
INQUIRY 0
TRANSFER 0
ELECTRONIC PAYMENT 0
PAYMENT ENCLOSED 0
CASH CHECK 0
MSG TO INSTITUTION 0
PIN CHANGE 0
SPLIT DEPOSIT 0
LOAD VALUE 0

TRANSACTIONS ALLOWED TO SWITCH

WITHDRAWAL N
WITHDRAWAL CC N
DEPOSIT N
INQUIRY N
TRANSFER N
ELECTRONIC PAYMENT N
PAYMENT ENCLOSED N
CASH CHECK N
MSG TO INSTITUTION N
PIN CHANGE N
SPLIT DEPOSIT N
LOAD VALUE N

VALUES FOR NOT ON US ARE '0' THRU '4', TRANS ALLOWED TO SWITCH ARE 'Y' OR 'N'
0 = NOT ALLOWED 1 = INTRACOUNTY 2 = INTRASITE 3 = INTERSATE 4 = INTNAT'L

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

Figure 3-4 ICF Screen 7

ATM ICF DATA

The following fields define the BASE24-atm allowable transactions.

NOT ON US 1 (N) M

Codes defining the circumstances in which incoming transactions from the Interchange are allowed (that is, where BASE24-atm cardholders use ATMs connected to and controlled by institutions serviced by an interchange). For BASE24-atm users with the LINK (LIS5) Interchange Interface, only Inquiry, Withdrawal and Withdrawal CC transactions are currently allowed. A value of 4 is entered against these transactions; all others are set to 0. Valid values are:
0 = Not allowed
1 = Allowed within the county
2 = Allowed within the state
3 = Allowed nationally
4 = Allowed internationally
Default Value: 0

TRANSACTIONS ALLOWED TO SWITCH

1 (A) M

The fields represent the transaction types that are supported by BASE24-atm and can be sent to the Interchange.

For BASE24-atm users with the LINK (LIS5) Interchange Interface, only Inquiry, Withdrawals and Withdrawal CC transactions are allowed. A value of Y is entered against these transactions; all others are set to N. Valid values are:

Y = Allowed

N = Not allowed

Default Value: N

3.3.5 ICF Screen 8

The ICF screen 8 (Figure 3-5) defines the BASE24-atm sharing groups and the business traffic timers.

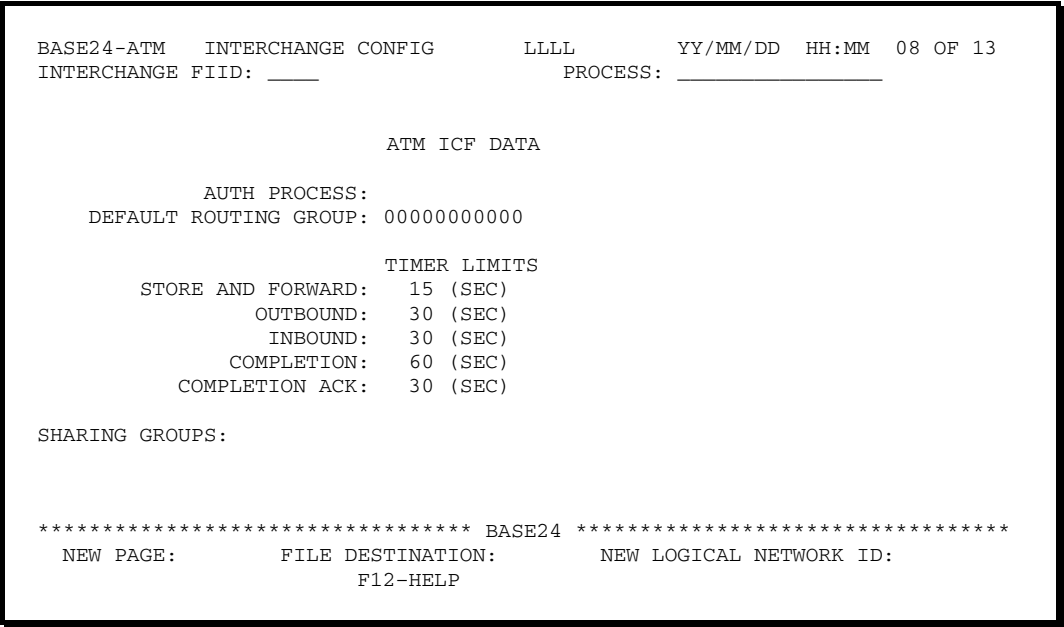


Figure 3-5 ICF Screen 8

ATM ICF DATA

The following fields contain information relevant to BASE24-atm sharing groups and the business traffic timers.

AUTH PROCESS 16 (A)

This field defines the Authorisation process or service to which the LINK (LIS5) Interchange Interface process routes ATM transactions.

DEFAULT ROUTING GROUP 11 (A)

This field contains the ATM routing group identification.
Default Value: 0

TIMER LIMITS

The following fields set the time limits for BASE24-atm transactions.

STORE AND FORWARD 4 (N) M

This field specifies the maximum number of seconds the Interchange Interface will wait for a response to a store-and-forward advice transaction. The value specified in the field is used in the first attempt to send the store-and-forward advice.

For the LINK (LIS5) Interchange Interface, the recommended value is 30 seconds.

Default Value: 30 (SEC)

OUTBOUND 4 (N) M

This timer is used if the institution is an ATM transaction acquirer. It specifies the time limit associated with transaction responses from the Interchange. The Interchange Interface process must receive responses before the timer expires.

For the LINK (LIS5) Interchange Interface, the recommended value is 30 seconds.

Default Value: 15 (SEC)

INBOUND 4 (N) M

This timer is used if the institution is an ATM transaction issuer. It specifies the time limit associated with transaction responses from a BASE24-atm Authorisation process (or a back-end host). The Interchange Interface process must receive responses before the timer expires.

For the LINK (LIS5) Interchange Interface, the recommended value is 10 seconds.

Default Value: 15 (SEC)

COMPLETION 4 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

COMPLETION ACK 4 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

SHARING GROUPS

1 (A)

A maximum of 24 fields is allowed. Each field specifies a one-character code to identify Interchange sharing groups for both incoming and outgoing transactions. Zero is not a valid entry. Codes are not separated by blank spaces; however, unused portions to the right remain blank.

For outgoing transactions to LINK from BASE24, if the acquiring terminal belongs to a sharing group, as indicated in the ATD/TDF record, one or more of the ATD/TDF entries must match the sharing group codes entered in the ICF; otherwise the transaction will be denied.

For incoming transaction requests from LINK, the sharing groups entered in the ICF are passed to the Authorisation process to check that one of the codes matches the sharing groups specified in the IDF record associated with the cardholder's institution. Again, the transaction will be declined if there is not at least one matching sharing group.

Related Fields: IDF SHARING GROUPS
 ATD/TDF SHARING GROUPS

3.3.6 ICF Screen 12

The ICF screen 12 (Figure 3-6) displays the LINK (LIS5) Interchange Interface specific processing parameters.

```

BASE24-BASE  INTERCHANGE CONFIG      LLLL      YY/MM/DD  HH:MM  12 OF 13
INTERCHANGE FIID: _____          PROCESS: _____

                                BASE24 INTERFACE OPTIONS

ACQ STAND-IN AUTH: _ _____      ISS STAND-IN AUTH: _ _____
CUTOVER STATUS:   _ _____      VERSION:   _ _____
FIXED LENGTH FORMAT: _ _____

ITF UPDATE INTERVAL: ____ (MINUTES)  ITF UPDATE MAX TRANSACTIONS: ____
SETTLEMENT TIMER:   ____ (MINUTES)  PROTOCOL TYPE:   _ _____
SELL RATE:          _____      BUY RATE:          _____

PRODUCT INDICATOR (Y/N): _ (ATM)    _ (POS)

                                DATA PREFIX CHARACTERS
1. ____ 2. ____ 3. ____ 4. ____ 5. ____ 6. ____ 7. ____ 8. ____ 9. ____
VALID ENTRIES FOR DATA PREFIX CHARACTERS ARE (0 THRU 9) AND (A THRU F)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3-6 ICF Screen 12

BASE24 INTERFACE OPTIONS

The following fields contain information relevant to the functionality for the LINK (LIS5) Interchange Interface.

ACQ STAND-IN AUTH

1 (A) M

A flag indicating whether BASE24, when acting as the acquirer, is allowed to stand in on behalf of LINK to authorize transactions.

Valid values are as follows:

Y = Stand-in is allowed

N = Stand-in is not allowed

Default Value: N

A description of the code entered is displayed to the right of the field.

ISS STAND-IN AUTH

1 (A) M

A flag indicating whether LINK, when acting as the acquirer, is allowed to stand in for BASE24 to authorize transactions when BASE24 is the issuer.

Valid values are as follows:

Y = Stand-in is allowed

N = Stand-in is not allowed

Default Value: N

A description of the code entered is displayed to the right of the field.

CUTOVER STATUS

1 (A) M

A flag specifying BASE24's relationship to LINK for cutover and reconciliation processing.

For the LINK (LIS5) Interchange Interface, this field must be set to 2. Valid values are as follows:

0 = Equal

1 = Master

2 = Slave

Default Value: 0

A description of the code entered is displayed to the right of the field.

VERSION

2 (N) M

A flag specifying the release level of the BASE24 system.

For the LINK (LIS5) Interchange Interface, this field must be set to 01, unless the interface is running in a BASE24-atm Release 4.0 system. Valid values are as follows:

00 = Release 4.0

01 = Release 5.0 and above

Default Value: 00

FIXED LENGTH FORMAT

1 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

ITF UPDATE INTERVAL

3 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

ITF UPDATE MAX TRANSACTIONS

3 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

SETTLEMENT TIMER

3 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

PROTOCOL TYPE

2 (N) M

A code indicating the type of protocol message header processing to be performed.

Valid values are as follows:

00 = None

01 = SNA/CICS

02 = Bisync Multipoint

Default Value: 00

A description of the code entered is displayed to the right of the field.

SELL RATE

8 (N)

The LINK (LIS5) Interchange Interface process does not use this field.

BUY RATE

8 (N)

The LINK (LIS5) Interchange Interface process does not use this field.

PRODUCT INDICATOR

The LINK (LIS5) Interchange Interface process does not use these fields.

ATM

1 (A) M

POS

1 (A) M

DATA PREFIX CHARACTERS

9 sets of 2 (N)

The LINK (LIS5) Interchange Interface process does not use these fields.

3.3.7 ICF Screen 13

The ICF screen 13 (Figure 3-7) displays the LINK (LIS5) Interchange Interface specific processing parameters.

```

BASE24-BASE INTERCHANGE CONFIG          LLLL          YY/MM/DD  HH:MM  13 OF 13
INTERCHANGE FIID: _____          PROCESS: _____

                                BASE24 INTERFACE STATIONS

      STATION      TYPE      DESCRIPTION      STATION      TYPE      DESCRIPTION
      _____      -      _____

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3-7 ICF Screen 13

BASE24 INTERFACE STATIONS

The following fields contain the names of up to 20 stations and the type of each station with which the LINK (LIS5) Interchange Interface is associated. The stations defined in the ICF must also be defined in the Network Environment File (NEF).

STATION

20 sets of 16 (A)

The symbolic name of the stations to be used for transaction requests and responses to and from the LINK network. At least one station must be defined.

TYPE

20 sets of 1 (N)

A flag indicating the type of station the interface can use. The type of station must be either send/receive or receive only.

Valid values are as follows:

0 = Send/receive

1 = Receive only

Default Value: 0

A description of the code entered is displayed to the right of the field.

3.4 KEY FILE (KEYF)

The KEYF contains the information and parameters required by the BASE24 Interchange Interface processes for PIN encryption, PIN translation, message authentication and dynamic key management.

The ISO Host Interface process and Interchange Interface processes use the KEYF instead of the KEY6 when the LCONF param DES-TRIPLE-SINGLE is either set to a value of N or not present. Otherwise these processes use the KEY6.

The KEY File (KEYF) contains one record for each:

- DPC/Host Interface process
- Interchange Interface process

The key to records in the KEYF is a combination of the values in the DPC/FIID and INTERFACE PROCESS fields.

The following screens are used to access data in the KEYF:

- **KEYF screen 1** contains PIN and MAC processing parameters, key locators to intermediate keys, and key locators to PIN and MAC key exchange keys.
- **KEYF screen 2** contains key locators to inbound and outbound keys for PINs and MACs.
- **KEYF screen 3** contains dynamic key management parameters.
- **KEYF screen 4** contains message encryption parameters.

Screen 4 is not covered here, as the LINK (LIS5) Interchange Interface does not use message encryption.

For more information on the usage of the KEYF, refer to the following:

- *BASE24 Base Files Maintenance Manual*
- *BASE24 Transaction Security Manual*

For more information on configuring the associated TSS data sources, refer to the *BASE24 Transaction Security Services Processing Guide*.

3.4.1 KEYF Screen 1

The KEYF screen 1 (Figure 3–8) contains PIN and MAC processing parameters, key locators to intermediate keys, and key locators to PIN and MAC key exchange keys.

```

BASE24-BASE KEY FILE          EME3          99/08/18  13:40  01 OF 04
DPC/FIID: _____  INTERFACE PROCESS: _____

    ENCRYPT TYPE: 0 (NO ENCRYPTION)  BASE24 ENCRYPT TYPE: 0 (NO ENCRYPTION)
PIN BLOCK FORMAT: 0 (CLEAR)          ANSI PAN FORMAT: 0 (12 RIGHT/NO CHK)
MAC ENCRYPT TYPE: 0 (NO PROCESSING)  PIN PAD CHARACTER: F
    MAC DATA TYPE: 0 (ASCII)          NUMBER OF KEYS: 1 (COMBINED)
    KEY LENGTH: 1 (SINGLE)          FULL MESSAGE MAC: N (SELECTED FIELDS)

                                INTERMEDIATE KEYS

    CLEAR: 0000000000000000          CHECK DIGITS: 0000
    ENCRYPTED: 0000000000000000

                                EXCHANGE KEYS

    PIN KEY: 0000000000000000  0000000000000000  CHECK DIGITS: 0000
    MAC KEY: 0000000000000000  0000000000000000  CHECK DIGITS: 0000

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3–8 KEYF Screen 1

DPC/FIID

4 (AN) M

The FIID of the interchange using this KEYF record. The value in this field matches the value in the INTERCHANGE FIID field on ICF screen 1.

INTERFACE PROCESS

16 (A) M

The name of the interface process associated with the interchange identified in the DPC/FIID field. The value in this field matches the value in the PROCESS field on ICF screen 1.

ENCRYPT TYPE

1 (N) M

Specifies the type of PIN management used by the interchange.

For the LINK (LIS5) Interchange Interface, this field must be set to 1. Valid values are:

- 0 = Clear PINs
- 1 = Security module PIN management
- 2 = Software DES PIN management

Default Value: 0

B24 ENCRYPT TYPE

1 (N) M

Specifies the type of internal PIN encryption between the Interchange and BASE24.

For transactions outbound to LINK, the type of internal PIN encryption is specified in the Standard Internal Message (STM) in the PIN format field.

For the LINK (LIS5) Interchange Interface, this field must be set to 1. Valid values are:

- 0 = Clear PINs
- 1 = Security module PIN management

Default Value: 0

PIN BLOCK FORMAT

1 (N) M

Specifies the PIN block format of the PIN in the inbound and outbound external messages.

For the LINK (LIS5) Interchange Interface, this field must be set to 1. Valid values are:

- 0 = Clear PINs
- 1 = ANSI (PIN/PAN) PIN block
- 3 = PIN/PAD PIN block

While option 1 is called the ANSI PIN block, it is also known as the PIN/PAN PIN block because it includes three PAN formats, only one of which is part of the ANSI standard. The PAN format being used is specified in the ANSI PAN FORMAT field on this screen.

Default Value: 0

ANSI PAN FORMAT**1 (N) M**

If the value in the PIN BLOCK FORMAT field is 1 (ANSI), the value in this field specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message.

For the LINK (LIS5) Interchange Interface, this field must be set to 0. Valid values are:

- 0 = 12 right-most digits of the PAN, excluding the check digit (ANSI standard)
- 1 = 12 right-most digits of the PAN, including the check digit
- 2 = 12 left-most digits of the PAN

The ANSI standard is one of three PAN formats available with PIN/PAN PIN blocks. While this field is called the ANSI PAN FORMAT, it also includes two other PAN formats that are not part of the ANSI standard. The ANSI standard is also known as ISO Format 0.

Default Value: 0

MAC ENCRYPT TYPE**1 (N)**

The LINK (LIS5) Interchange Interface process does not use this field.

PIN PAD CHAR**1 (A) M**

The LINK (LIS5) Interchange Interface process does not use this field.

MAC DATA TYPE**1 (N)**

The LINK (LIS5) Interchange Interface process does not use this field.

NUMBER OF KEYS**1 (N)**

The LINK (LIS5) Interchange Interface process does not use this field.

KEY LENGTH**1 (N)**

Specifies whether single-length or double-length Key Exchange Keys (KEKs) are used with this Interchange Interface.

For the LINK (LIS5) Interchange Interface, this field must be set to 1. Valid values are:

- 1 = Single-length KEKs
- 2 = Double-length KEKs

Default Value: 1.

FULL MESSAGE MAC**1 (A)**

The LINK (LIS5) Interchange Interface process does not use this field.

INTERMEDIATE KEYS

Intermediate keys are used whenever a PIN that is encrypted under a secure key (security module PIN management) must be decrypted into the clear.

CLEAR **16 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

CHECK DIGITS **4 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

ENCRYPTED **16 hexadecimal characters M**

The LINK (LIS5) Interchange Interface process does not use this field.

EXCHANGE KEYS

When PIN or MAC keys are automatically exchanged between the Interchange Interface and the interchange, they are to be encrypted under a key exchange key. These fields contain locators to the key exchange keys used to exchange PIN Block encryption keys between the LINK (LIS5) Interchange Interface and the LINK network. The exchange key must be manually exchanged between the two entities and configured on the TSS database.

PIN KEY **2 sets of 16 hexadecimal characters M**

A locator to the key used to encrypt the Issuer Working Key (IWK) sent to LINK.

Valid values for each character in this field is 0–9 and A–F. This field must contain 32 valid characters.

CHECK DIGITS **4 hexadecimal characters M**

The LINK (LIS5) Interchange Interface process does not use this field.

MAC KEY **2 sets of 16 hexadecimal characters**

A locator to the key used to encrypt the Acquirer Working Key (AWK) received from LINK.

Valid values for each character in this field is 0–9 and A–F. This field must contain 32 valid characters.

CHECK DIGITS **4 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

3.4.2 KEYF Screen 2

The KEYF screen 2 (Figure 3–9) contains key locators to inbound and outbound keys for PINs and MACs.

BASE24-BASE KEY FILE	EME3	99/08/18 13:40 02 OF 04
DPC/FIID: _____	INTERFACE PROCESS: _____	
PIN KEY INFORMATION		
OUTBOUND KEYS:		
PIN KEY 1: 0000000000000000	CHECK DIGITS1: 0000	CURRENT INDEX: 1
PIN KEY 2: 0000000000000000	CHECK DIGITS2: 0000	KEY COUNTER: 000000
INBOUND KEYS:		
PIN KEY 1: 0000000000000000	CHECK DIGITS1: 0000	CURRENT INDEX: 1
PIN KEY 2: 0000000000000000	CHECK DIGITS2: 0000	KEY COUNTER: 000000
MAC KEY INFORMATION		
OUTBOUND KEYS:		
PIN KEY 1: 0000000000000000	CHECK DIGITS1: 0000	CURRENT INDEX: 1
PIN KEY 2: 0000000000000000	CHECK DIGITS2: 0000	KEY COUNTER: 000000
INBOUND KEYS:		
PIN KEY 1: 0000000000000000	CHECK DIGITS1: 0000	CURRENT INDEX: 1
PIN KEY 2: 0000000000000000	CHECK DIGITS2: 0000	KEY COUNTER: 000000
***** BASE24 *****		
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:
	F12-HELP	

Figure 3–9 KEYF Screen 2

PIN KEY INFORMATION

The following fields contain working keys for PINs.

OUTBOUND KEYS

The following fields contain the locators to PIN keys utilised for transactions sent from the interface to the interchange.

PIN KEY1 **16 hexadecimal characters M**

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key locator in this field when the CURRENT INDEX field contains a value of 1.

The associated key value is known as the Acquirer Working Key (AWK) and is dynamically generated by LINK.

The valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS1 **4 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

CURRENT INDEX **1 (N) M**

Indicates which outbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2 **16 hexadecimal characters M**

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key locator in this field when the CURRENT INDEX field contains a value of 2.

The associated key value is known as the Acquirer Working Key (AWK) and is dynamically generated by LINK.

The valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS2 **4 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

KEY COUNTER **6 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

INBOUND KEYS

The following fields contain the locators to PIN keys used in the transactions sent by the interchange to the interface.

PIN KEY1 **16 hexadecimal characters M**

Used for the decryption or translation of PINs in transactions sent from the interchange. The interface process uses the key locator in this field when the CURRENT INDEX field contains a value of 1.

This value is known as the Issuer Working Key (IWK) and is dynamically generated by LINK.

The valid values for each character in this field are 0-9 and A-F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS1 **4 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

CURRENT INDEX **1 (N) M**

Indicates which inbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2 **16 hexadecimal characters M**

Used for the decryption or translation of PINs in transactions sent from the interchange. The interface process uses the key locator in this field when the CURRENT INDEX field contains a value of 2.

This value is known as the Issuer Working Key (IWK) and is dynamically generated by LINK.

The valid values for each character in this field are 0-9 and A-F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS2 **4 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

KEY COUNTER **6 hexadecimal characters**

The LINK (LIS5) Interchange Interface process does not use this field.

3.4.3 KEYF Screen 3

The KEYF screen 3 (Figure 3-10) consists of dynamic key management parameters.

BASE24-BASE KEY FILE

EME399/08/1813:4003 OF 04

DPC/FIID: _____INTERFACE PROCESS: _____

PIN KEY VARIANT: 0MAC KEY VARIANT: 0

LIMITS

PIN KEY TIMER VALUE: 0MAC KEY TIMER VALUE: 0

PIN KEY TIMER INTERVAL: M (MINUTES)MAC KEY TIMER INTERVAL: M (MINUTES)

PIN KEY TRAN: 0MAC KEY TRAN: 0

PIN KEY ERROR: 0MAC KEY ERROR: 0

CONSECUTIVE PIN KEY ERROR: 0CONSECUTIVE MAC KEY ERROR: 0

KMAC SYNCHRONIZATION ERROR: 0

CLEAR OLD KEY TIMER VALUE: 0 (SECONDS)

ORIGINATING ID:

RECEIVING ID:

KEY PROCESSING TYPE: N (NONE)

NOTARIZATION SUPPORTED: 0 (NOT SUPPORTED)

*****BASE24*****

NEW PAGE:FILE DESTINATION:NEW LOGICAL NETWORK ID:

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Figure 3-10 KEYF Screen 3

PIN KEY VARIANT1 (N) M

A code that specifies the key variant to be applied to the PIN key exchange key (KEK) when translating PIN keys.

- For users of Thales e-security (formerly Racal) HSMs this field should be set to 0.
- For users of Atalla HSMs this field should be set to 1.

Valid values are follows:

- 0 = Variant 0. Translate the PIN key directly from the PIN KEK parts.
- 1 = Variant 1. Include the variant 1 constant with the PIN KEK parts when translating the PIN key. The Atalla variant 1 constant is hexadecimal 0800 0000 0000 0000.

Default Value: 0

MAC KEY VARIANT1 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

LIMITS

The following fields are used to set the parameters for dynamic key management for this interface. One set of parameters specifies when the key used for PIN encryption and translation should be changed and a second set specifies when the key used for message authentication should be changed. Both types of keys can be changed based on time intervals, transaction activity, the number of errors since the last key change, or the number of consecutive errors. The MAC key can also be changed based on the number of key synchronization errors since the last key change. These limits can be used individually or in any combination. The counters and timers for the key are reset when the key is changed. If multiple limits are used, the interface initiates a key exchange message and resets all of the counters and timers whenever any limit is exceeded. The value in the KEY PROCESSING TYPE field specifies the type of key management message the interface initiates. The PIN key parameters on this screen must be set to the value 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to the value 1 (hardware PIN encryption). The MAC key parameters on this screen must be set to the value 0 unless the MAC ENCRYPT TYPE field on KEYF screen 1 is set to the value 1 (hardware message authentication).

PIN KEY TIMER VALUE

4 (N) M

The maximum length of time that a PIN key should be used.

The time period can be expressed in minutes, hours, or days, depending on the value in the PIN KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid time periods are as follows:

- 5 through 1500 - if the PIN KEY TIMER INTERVAL field is set to the value M (minutes)
- 1 through 1000 - if the PIN KEY TIMER INTERVAL field is set to the value H (hours)
- 1 through 180 - if the PIN KEY TIMER INTERVAL field is set to the value D (days)

Default Value: 0

The interface does not use this parameter if the value in this field is set to 0. This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to the value 1 (hardware PIN encryption).

BASE24 products check the values in the PIN KEY TIMER VALUE and PIN KEY TIMER INTERVAL fields when the KEYF record is added or updated. If the time is evenly divisible by a longer interval, BASE24 products change the value in the PIN KEY TIMER INTERVAL field to the longer interval and redisplay the value in the PIN KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60.

MAC KEY TIMER VALUE

4 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

PIN KEY TIMER INTERVAL

1 (A) M

A code specifying the unit of time used with the value in the PIN KEY TIMER VALUE field to set the maximum length of time that a PIN key should be used.

BASE24 products change the value in this field to the largest possible unit of time when the KEYF record is added or updated. Refer to the PIN KEY TIMER VALUE field for additional information on how this field is used. Valid values are as follows:

D = Days
H = Hours
M = Minutes (default)

MAC KEY TIMER INTERVAL **1 (A) M**

The LINK (LIS5) Interchange Interface process does not use this field.

PIN KEY TRAN **6 (N) M**

The maximum number of transactions that can be performed with the current PIN key before changing the key.

Valid values are as follows:

0 (default) = The parameter is not used.

50–100000 = The maximum number of transactions.

This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to the value 1 (hardware PIN encryption).

MAC KEY TRAN **6 (N) M**

The LINK (LIS5) Interchange Interface process does not use this field.

PIN KEY ERROR **4 (N) M**

The maximum number of PIN translation errors that can occur with the present PIN key before changing the key.

Valid values are as follows:

0 (default) = This parameter is not used.

5–9999 = The maximum number of errors.

This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to the value 1 (hardware PIN encryption).

MAC KEY ERROR **4 (N) M**

The LINK (LIS5) Interchange Interface process does not use this field.

CONSECUTIVE PIN KEY ERROR **4 (N) M**

The maximum number of consecutive PIN translation errors that can occur with the present PIN key before changing the key.

A successful PIN translation, as well as a change in the PIN key, resets the counter associated with this limit. For the LINK (LIS5) Interchange Interface, this field must be set to 5. Valid values are as follows:

0 (default) = This parameter is not used.

5–9999 = The maximum number of errors.

This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to the value 1 (hardware PIN encryption).

CONSECUTIVE MAC KEY ERROR 4 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

KMAC SYNCHRONIZATION ERROR 4 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

CLEAR OLD KEY TIMER VALUE 4 (N) M

The maximum length of time, in seconds, that an old key can be used before it is automatically cleared by the interface process.

The INBOUND PIN KEY and OUTBOUND PIN KEY fields on KEYF screen 2 and the INBOUND MAC KEY, and OUTBOUND MAC KEY fields on KEYF screen 2 can contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key (1 or 2) is currently in use. At the time a new key is placed in service, the other key becomes the old key. For example, if Inbound PIN Key number 1 is currently in use when the BASE24 product receives a new key, the new key is stored as Inbound PIN Key number 2. Inbound PIN Key number 2 becomes the current key and Inbound PIN Key number 1 becomes the old key. (If Inbound PIN Key number 2 is the current key, it becomes the old key and the new key is stored as Inbound PIN Key number 1.)

When processing a transaction, BASE24 products use the new key for PIN processing or message authentication. If the security device used by BASE24 products for PIN processing or message authentication detects an error because the DPC or interchange is still using the old key, it can repeat the PIN processing or message authentication using the old key before the BASE24 product declines the transaction. The value in this field controls how long the old key is available. If the old key is not available, a second PIN processing or message authentication attempt is not made. Valid values are as follows:

0 (default) = The old key is not cleared.

1–9999 = The number of seconds before the old key is cleared.

This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to the value 1 (hardware PIN encryption) or the MAC ENCRYPT TYPE field on KEYF screen 1 is set to the value 1 (hardware message authentication).

ORIGINATING ID**16 (AN) M**

The LINK (LIS5) Interchange Interface process does not use this field.

RECEIVING ID**16 (AN) M**

Identifies the receiver of dynamic key management messages.

The values in the RECEIVING ID field and this field need to be agreed upon by the sender and receiver. The sender's originating ID is the receiver's receiving ID and the sender's receiving ID is the receiver's originating ID. These values do not need to match any specific values in the BASE24 database.

For the LINK (LIS5) Interchange Interface, this field is used in setting the Message Security Code (field S-96) in network management request messages sent to LINK.

KEY PROCESSING TYPE**1 (A) M**

A code identifying the type of dynamic key management processing this interface can perform.

For the LINK (LIS5) Interchange Interface, this field must be set to C. Valid values are as follows:

C = Co-network. Each interface is responsible for generating its outbound key and can request its inbound key from LINK.

M = Main. This interface is responsible for all key generation.

N = None. This interface does not generate or receive keys.

S = Secondary. This interface can request the main process to generate keys, but does not generate keys itself.

Default Value: N

A description of the code entered is displayed to the right of the KEY PROCESSING TYPE field.

NOTARIZATION SUPPORTED**1 (N) M**

The LINK (LIS5) Interchange Interface process does not use this field.

3.5 KEY6 FILE (KEY6)

The Key File (KEY6) contains the information and parameters required by release 6.0 BASE24 Host Interface and Interchange Interface processes running in release 5.x or 6.0 networks for PIN encryption, PIN translation, message authentication and dynamic key management.

The KEY6 allows for double length encryption keys which are required when translating PINS from encryption under a double length key using the Triple DES algorithm to encryption under a single length key using the Single DES algorithm and vice versa.

The KEY File (KEY6) contains one record for each:

- DPC/Host Interface process
- Interchange Interface process

The key to records in the KEY6 is a combination of the values in the DPC/FIID and INTERFACE PROCESS fields.

The following screens are used to access records in the KEY6:

- **Screen 1** allows PIN encryption, PIN translation, and message authentication parameters, intermediate keys, PIN key exchange keys, and message authentication code (MAC) key exchange keys.
- **Screen 2** details inbound and outbound keys for PINs.
- **Screen 3** details inbound and outbound keys for MACs.
- **Screen 4** contains dynamic key management parameters.

Screen 3 is not covered here, as the LINK (LIS5) Interchange Interface does not use message authentication.

For more information on the usage of the KEY6, refer to the following:

- *BASE24 BASE Files Maintenance Manual*
- *BASE24 Transaction Security Manual*

3.5.1 KEY6 Screen 1

The KEY6 screen 1 (Figure 3-11) contains PIN encryption, PIN translation, and message authentication parameters, intermediate keys, PIN key exchange keys, and message authentication code (MAC) key exchange keys.

```

BASE24-BASE KEY6 FILE          LLLL      YY/MM/DD  HH:MM  01 OF 04
DPC/FIID: _____ INTERFACE PROCESS: _____

  ENCRYPT TYPE: 0 (NO ENCRYPTION)  BASE24 ENCRYPT TYPE: 0 (NO ENCRYPTION)
PIN BLOCK FORMAT: 0 (CLEAR)        ANSI PAN FORMAT: 0 (12 RIGHT/NO CHK)
MAC ENCRYPT TYPE: 0 (NO PROCESSING)  PIN PAD CHARACTER: F
  MAC DATA TYPE: 0 (ASCII)        NUMBER OF KEYS: 1 (COMBINED)
    KEY LENGTH: 1 (SINGLE)          FULL MESSAGE MAC: N (SELECTED FIELDS)
  MAC KEY LENGTH: 1 (SINGLE)

                                INTERMEDIATE KEYS

      CLEAR: 000000000000000000    CHECK DIGITS: 0000
    ENCRYPTED: 000000000000000000

                                EXCHANGE KEYS

      PIN KEY: 000000000000000000  000000000000000000  CHECK DIGITS: 0000
      MAC KEY: 000000000000000000  000000000000000000  CHECK DIGITS: 0000

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3-11 KEY6 Screen 1

DPC/FIID

4 (AN) M

The FIID of the interchange using this KEY6 record. The value in this field matches the value in the INTERCHANGE FIID field on ICF screen 1.

INTERFACE PROCESS

16 (A) M

The name of the interface process associated with the interchange identified in the DPC/FIID field. The value in this field matches the value in the PROCESS field on ICF screen 1.

ENCRYPT TYPE

1 (N) M

Specifies the type of PIN management used by the interchange.

For the LINK (LIS5) Interchange Interface, this field must be set to 1. Valid values are:

0 = Clear PINs

1 = Security module PIN management

2 = Software DES PIN management

Default Value: 0

BASE24 ENCRYPT TYPE

1 (N) M

Specifies the type of internal PIN encryption between the Interchange and BASE24.

For transactions outbound to LINK, the type of internal PIN encryption is specified in the Standard Internal Message (STM) in the PIN format field.

For the LINK (LIS5) Interchange Interface, this field must be set to 1. Valid values are:

0 = Clear PINs

1 = Security module PIN management

Default Value: 0

PIN BLOCK FORMAT

1 (N) M

Specifies the PIN block format of the PIN in the inbound and outbound external messages.

For the LINK (LIS5) Interchange Interface, this field must be set to 1. Valid values are:

0 = Clear PINs

1 = ANSI (PIN/PAN) PIN block

3 = PIN/PAD PIN block

While option 1 is called the ANSI PIN block, it is also known as the PIN/PAN PIN block because it includes three PAN formats, only one of which is part of the ANSI Standard. The PAN format being used is specified in the ANSI PAN FORMAT field on this screen.

Default Value: 0

ANSI PAN FORMAT

1 (N) M

If the value in the PIN BLOCK FORMAT field is 1 (ANSI), the value in this field specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message.

For the LINK (LIS5) Interchange Interface, this field must be set to 0. Valid values are:

0 = 12 right-most digits of the PAN, excluding the check digit (ANSI Standard)

1 = 12 right-most digits of the PAN, including the check digit

2 = 12 left-most digits of the PAN

The ANSI Standard is one of three PAN formats available with PIN/PAN PIN blocks. While this field is called the ANSI PAN FORMAT, it also includes two other PAN formats that are not part of the ANSI Standard.

Default Value: 0

MAC ENCRYPT TYPE

1 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

PIN PAD CHARACTER

1 (A) M

The LINK (LIS5) Interchange Interface process does not use this field.

MAC DATA TYPE

1 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

NUMBER OF KEYS

1 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

KEY LENGTH

1 (N) M

Specifies whether single-length or double-length Key Exchange Keys (KEKs) are used with this Interchange Interface.

For the LINK (LIS5) Interchange Interface, this field must be set to 2. Valid values are:

1 = Single-length KEKs

2 = Double-length KEKs

Default Value: 1.

FULL MESSAGE MAC

1 (A) M

The LINK (LIS5) Interchange Interface process does not use this field.

MAC KEY LENGTH

1 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

INTERMEDIATE KEYS

Intermediate keys are used whenever a PIN that is encrypted under a secure key (security module PIN management) must be decrypted into the clear or when a PIN is in the clear and needs to be encrypted under a secure key.

CLEAR

16 hexadecimal characters

The LINK (LIS5) Interchange Interface process does not use this field.

CHECK DIGITS

4 hexadecimal characters

The LINK (LIS5) Interchange Interface process does not use this field.

ENCRYPTED

16 hexadecimal characters

Security module encrypted intermediate key. This key is used for the interim Triple DES solution when a PIN block encrypted using a double-length key (Triple DES) is received from an interchange. The PIN block is translated to encryption under the intermediate key (Single DES) by the security module.

- For users of Thales e-security (formerly Racal) HSMs the keys stored in this field need to be encrypted under LMK pair 06/07 with no variant applied.
- For users of Atalla HSMs the keys stored in this field need to be encrypted under MFK variant 1.

Valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

Default Value: 0000000000000000

This field is mandatory.

EXCHANGE KEYS

When PIN or MAC keys are automatically exchanged between the Interchange Interface and the interchange, they are to be encrypted under the Key Exchange Key (KEK) for the interchange. The key exchange key, also known as the Zone Master Key (ZMK), must be manually exchanged between LINK and the LINK member.

PIN KEY **2 sets of 16 hexadecimal characters M**

A double-length key used to encrypt the Issuer Working Key (IWK) sent to LINK.

- For users of Thales e-security (formerly Racal) HSMs the key stored in this field need to be encrypted under LMK pair 04/05 variant A6/5A.
- For users of Atalla HSMs the keys stored in this field need to be encrypted under MFK variant 0.

Valid values for each character in this field is 0–9 and A–F. This field must contain 32 valid characters.

Default Value: 0000000000000000 0000000000000000

CHECK DIGITS **4 hexadecimal characters M**

The LINK (LIS5) Interchange Interface process does not use this field.

MAC KEY **2 sets of 16 hexadecimal characters M**

A double-length key used to decrypt the Acquirer Working Key (AWK) received from LINK.

- For users of Thales e-security (formerly Racal) HSMs the key stored in this field need to be encrypted under LMK pair 04/05 variant A6/5A.
- For users of Atalla HSMs the keys stored in this field need to be encrypted under MFK variant 0.

Valid values for each character in this fields is 0–9 and A–F. This field must contain 16 valid characters.

Default Value: 0000000000000000 0000000000000000

CHECK DIGITS **4 hexadecimal characters M**

The LINK (LIS5) Interchange Interface process does not use this field.

3.5.2 KEY6 Screen 2

The KEY6 screen 2 (Figure 3-12) consists of inbound and outbound working keys for PINs and MACs.

BASE24-BASEKEY6 FILELLLLYY/MM/DDHH:MM02 OF 04
DPC/FIID: _____INTERFACE PROCESS: _____

PIN KEY INFORMATION

OUTBOUND KEYS:

PIN KEY 1: 0000000000000000
0000000000000000

CHECK DIGITS1: 0000CURRENT INDEX: 1

PIN KEY 2: 0000000000000000
0000000000000000

CHECK DIGITS2: 0000KEY COUNTER: 000000

INBOUND KEYS:

PIN KEY 1: 0000000000000000
0000000000000000

CHECK DIGITS1: 0000CURRENT INDEX: 1

PIN KEY 2: 0000000000000000
0000000000000000

CHECK DIGITS2: 0000KEY COUNTER: 000000

***** BASE24 *****

NEW PAGE:FILE DESTINATION:NEW LOGICAL NETWORK ID:
F12-HELP

Figure 3-12 KEY6 Screen 2

PIN KEY INFORMATION

The following fields contain working keys for PINs.

OUTBOUND KEYS

The following fields contain the PIN keys utilised for transactions sent from the interface to the interchange.
If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1 **2 sets of 16 hexadecimal characters M**

A double-length key used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1.

This value is known as the Acquirer Working Key (AWK) and is dynamically generated by LINK.

- For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07 variant A6/5A.
- For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Valid values for each character in this field is 0–9 and A–F. This field must contain 32 valid characters.

Default Value: 0000000000000000 0000000000000000

CHECK DIGITS1 **4 hexadecimal characters M**

The check digits corresponding to the value in the PIN KEY1 field. The valid values for each character in these fields are 0–9 and A–F. The LINK (LIS5) Interchange Interface uses check digits to verify a new AWK received from LINK.

Default Value: 0000.

CURRENT INDEX **1 (N) M**

Indicates which outbound PIN encryption key is currently being used by the interface.

Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2 **2 sets of 16 hexadecimal characters M**

A double-length key used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2.

This value is known as the Acquirer Working Key (AWK) and is dynamically generated by LINK.

- For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07 variant A6/5A.
- For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Valid values for each character in this field is 0–9 and A–F. This field must contain 32 valid characters.

Default Value: 0000000000000000 0000000000000000

CHECK DIGITS2 **4 hexadecimal characters M**

The check digits corresponding to the value in the PIN KEY2 field. The valid values for each character in these fields are 0–9 and A–F. The LINK (LIS5) Interchange Interface uses check digits to verify a new AWK received from LINK.

Default Value: 0000

KEY COUNTER **6 hexadecimal characters M**

The LINK (LIS5) Interchange Interface process does not use this field.

INBOUND KEYS

The following fields contain the PIN keys used in the transactions sent by the interchange to the interface.

If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1 **2 sets of 16 hexadecimal characters M**

A double-length key used for the decryption or translation of PINs in transactions sent from the interchange to the BASE24 transaction processing system. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1.

This value is known as the Issuer Working Key (IWK) and is dynamically generated by LINK.

- For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07 variant A6/5A.
- For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Valid values for each character in this field is 0–9 and A–F. This field must contain 32 valid characters.

Default Value: 0000000000000000 0000000000000000

CHECK DIGITS1 **4 hexadecimal characters M**

The check digits corresponding to the value in the PIN KEY1 field. The valid values for each character in these fields are 0–9 and A–F. LINK uses check digits to verify a new IWK received from the LINK (LIS5) Interchange Interface.

Default Value: 0000

CURRENT INDEX **1 (N) M**

Indicates which inbound PIN encryption key is currently being used by the interface.

Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2 **2 sets of 16 hexadecimal characters M**

A double-length key used for the decryption or translation of PINs in transactions sent from the interchange to the BASE24 transaction processing system. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2.

This value is known as the Issuer Working Key (IWK) and is dynamically generated by LINK.

- For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07 variant A6/5A.
- For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Valid values for each character in this field is 0–9 and A–F. This field must contain 32 valid characters.

Default Value: 0000000000000000 0000000000000000

CHECK DIGITS2 **4 hexadecimal characters M**

The check digits corresponding to the value in the PIN KEY1 field. The valid values for each character in these fields are 0–9 and A–F. LINK uses check digits to verify a new IWK received from the LINK (LIS5) Interchange Interface.

Default Value: 0000

KEY COUNTER **6 hexadecimal characters M**

The LINK (LIS5) Interchange Interface process does not use this field.

3.5.3 KEY6 Screen 3

Not used by the LINK (LIS5) Interchange Interface.

3.5.4 KEY6 Screen 4

The KEY6 screen 4 (Figure 3-13) consists of dynamic key management parameters.

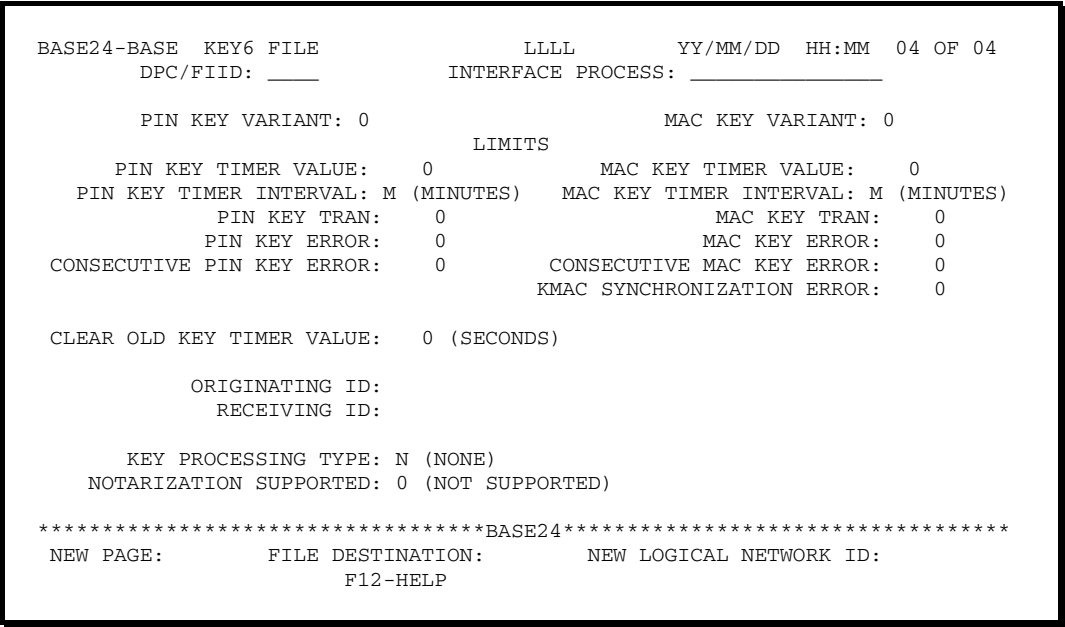


Figure 3-13 KEY6 Screen 4

PIN KEY VARIANT 1 (N) M

A code that specifies the key variant to be applied to the PIN key exchange key (KEK) when translating PIN keys.

- For users of Thales e-security (formerly Racal) HSMs this field should be set to 0.
- For users of Atalla HSMs this field should be set to 1.

Valid values are follows:

0 = Variant 0. Translate the PIN key directly from the PIN KEK parts.

1 = Variant 1. Include the variant 1 constant with the PIN KEK parts when translating the PIN key. The Atalla variant 1 constant is hexadecimal 0800 0000 0000 0000.

Default Value: 0

MAC KEY VARIANT 1 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

LIMITS

The following fields are used to set the parameters for dynamic key management for this interface. One set of parameters specifies when the key used for PIN encryption and translation should be changed and a second set specifies when the key used for message authentication should be changed. Both types of keys can be changed based on time intervals, transaction activity, the number of errors since the last key change, or the number of consecutive errors. The MAC key can also be changed based on the number of key synchronization errors since the last key change. These limits can be used individually or in any combination. The counters and timers for the key are reset when the key is changed. If multiple limits are used, the interface initiates a key exchange message and resets all of the counters and timers whenever any limit is exceeded. The value in the KEY PROCESSING TYPE field specifies the type of key management message the interface initiates. The PIN key parameters on this screen must be set to the value 0 unless the ENCRYPT TYPE field on KEY6 screen 1 is set to the value 1 (hardware PIN encryption). The MAC key parameters on this screen must be set to the value 0 unless the MAC ENCRYPT TYPE field on KEY6 screen 1 is set to the value 1 (hardware message authentication).

PIN KEY TIMER VALUE

4 (N) M

The maximum length of time that a PIN key should be used.

The time period can be expressed in minutes, hours, or days, depending on the value in the PIN KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid time periods are as follows:

- 5 through 1500 - if the PIN KEY TIMER INTERVAL field is set to the value M (minutes)
- 1 through 1000 - if the PIN KEY TIMER INTERVAL field is set to the value H (hours)
- 1 through 180 - if the PIN KEY TIMER INTERVAL field is set to the value D (days)

Default Value: 0

The interface does not use this parameter if the value in this field is set to 0. This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEY6 screen 1 is set to the value 1 (hardware PIN encryption).

BASE24 products check the values in the PIN KEY TIMER VALUE and PIN KEY TIMER INTERVAL fields when the KEY6 record is added or updated. If the time is evenly divisible by a longer interval, BASE24 products change the value in the PIN KEY TIMER INTERVAL field to the longer interval and redisplay the value in the PIN KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60.

MAC KEY TIMER VALUE

4 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

PIN KEY TIMER INTERVAL

1 (A) M

A code specifying the unit of time used with the value in the PIN KEY TIMER VALUE field to set the maximum length of time that a PIN key should be used.

BASE24 products change the value in this field to the largest possible unit of time when the KEY6 record is added or updated. Refer to the PIN KEY TIMER VALUE field for additional information on how this field is used. Valid values are as follows:

D = Days
H = Hours
M = Minutes (default)

MAC KEY TIMER INTERVAL **1 (A) M**

The LINK (LIS5) Interchange Interface process does not use this field.

PIN KEY TRAN **6 (N) M**

The maximum number of transactions that can be performed with the current PIN key before changing the key.

Valid values are as follows:

0 (default) = The parameter is not used.

50–100000 = The maximum number of transactions.

This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEY6 screen 1 is set to the value 1 (hardware PIN encryption).

MAC KEY TRAN **6 (N) M**

The LINK (LIS5) Interchange Interface process does not use this field.

PIN KEY ERROR **4 (N) M**

The maximum number of PIN translation errors that can occur with the present PIN key before changing the key.

Valid values are as follows:

0 (default) = This parameter is not used.

5–9999 = The maximum number of errors.

This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEY6 screen 1 is set to the value 1 (hardware PIN encryption).

MAC KEY ERROR **4 (N) M**

The LINK (LIS5) Interchange Interface process does not use this field.

CONSECUTIVE PIN KEY ERROR **4 (N) M**

The maximum number of consecutive PIN translation errors that can occur with the present PIN key before changing the key.

A successful PIN translation, as well as a change in the PIN key, resets the counter associated with this limit. For the LINK (LIS5) Interchange Interface, this field must be set to 5. Valid values are as follows:

0 (default) = This parameter is not used.

5–9999 = The maximum number of errors.

This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEY6 screen 1 is set to the value 1 (hardware PIN encryption).

CONSECUTIVE MAC KEY ERROR 4 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

KMAC SYNCHRONIZATION ERROR 4 (N) M

The LINK (LIS5) Interchange Interface process does not use this field.

CLEAR OLD KEY TIMER VALUE 4 (N) M

The maximum length of time, in seconds, that an old key can be used before it is automatically cleared by the interface process.

The INBOUND PIN KEY and OUTBOUND PIN KEY fields on KEY6 screen 2 and the INBOUND MAC KEY, and OUTBOUND MAC KEY fields on KEY6 screen 3 can contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key (1 or 2) is currently in use. At the time a new key is placed in service, the other key becomes the old key. For example, if Inbound PIN Key number 1 is currently in use when the BASE24 product receives a new key, the new key is stored as Inbound PIN Key number 2. Inbound PIN Key number 2 becomes the current key and Inbound PIN Key number 1 becomes the old key. (If Inbound PIN Key number 2 is the current key, it becomes the old key and the new key is stored as Inbound PIN Key number 1.)

When processing a transaction, BASE24 products use the new key for PIN processing or message authentication. If the security device used by BASE24 products for PIN processing or message authentication detects an error because the DPC or interchange is still using the old key, it can repeat the PIN processing or message authentication using the old key before the BASE24 product declines the transaction. The value in this field controls how long the old key is available. If the old key is not available, a second PIN processing or message authentication attempt is not made. Valid values are as follows:

0 (default) = The old key is not cleared.

1–9999 = The number of seconds before the old key is cleared.

This parameter must be set to the value 0 unless the ENCRYPT TYPE field on KEY6 screen 1 is set to the value 1 (hardware PIN encryption) or the MAC ENCRYPT TYPE field on KEY6 screen 1 is set to the value 1 (hardware message authentication).

ORIGINATING ID**16 (AN) M**

The LINK (LIS5) Interchange Interface process does not use this field.

RECEIVING ID**16 (AN) M**

Identifies the receiver of dynamic key management messages.

The values in the RECEIVING ID field and this field need to be agreed upon by the sender and receiver. The sender's originating ID is the receiver's receiving ID and the sender's receiving ID is the receiver's originating ID. These values do not need to match any specific values in the BASE24 database.

For the LINK (LIS5) Interchange Interface, this field is used in setting the Message Security Code (field S-96) in network management request messages sent to LINK.

KEY PROCESSING TYPE**1 (A) M**

A code identifying the type of dynamic key management processing this interface can perform.

For the LINK (LIS5) Interchange Interface, this field must be set to C. Valid values are as follows:

C = Co-network. Each interface is responsible for generating its outbound key and can request its inbound key from LINK.

M = Main. This interface is responsible for all key generation.

N = None. This interface does not generate or receive keys.

S = Secondary. This interface can request the main process to generate keys, but does not generate keys itself.

Default Value: N

A description of the code entered is displayed to the right of the KEY PROCESSING TYPE field.

NOTARIZATION SUPPORTED**1 (N) M**

The LINK (LIS5) Interchange Interface process does not use this field.

3.6 SWITCH TERMINAL FILE (STF)

The LINK (LIS5) Interchange Interface will use the STF to identify a terminal as located in “Financial Inclusion” or as “Unknown/Not Financial Inclusion”.

If an STF already exists for another one of the customer's interfaces, then the customer just needs to add their LINK records to that file.

On outbound messages from the LINK (LIS5) Interchange Interface, the STF will be searched. If a record is found, the first position of the Switch Terminal ID will be moved into the POS Data Entry (P-61) sub-field 13 (Financial Inclusion).

One screen is used to access records in the STF. The fields on this screen used by the LINK (LIS5) Interchange Interface are described in the section below.

Operator access to the STF is permitted in accordance with the security access levels assigned in the BASE24 CRT access system. For more information on the BASE24 CRT access system, refer to the ***BASE24 User Security Manual***.

For more details on the STF, refer to the ***BASE24 Base Files Maintenance Manual***.

3.6.1 STF Screen 1

The STF screen 1 (

Figure 3–14) contains switch terminal and merchant information.

```

BASE24-SWI  SWITCH  TERMINAL FILE      TES9      07/05/05  05:06  01 OF 01
  SWITCH FIID: _____  MERCHANT ID: _____
  TERMINAL FIID: _____  TERMINAL ID: _____  PROD ID: ____

SWITCH MERCHANT INFORMATION                SWITCH TERMINAL INFORMATION

  ID: _____                          ID: _____
  OFFSET: _____                      OFFSET: _____
  LENGTH: _____                      LENGTH: _____

                                ON PREMISE FLAG: _ (Y/N)
                                EMV CERTIFIED FLAG: _ (Y/N)
                                CARDHOLDER ACTIVATED TERMINAL: _ (NOT USED)
                                TERMINAL INPUT CAPABILITIES: _ (NOT USED)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3–14 STF Screen 1

SWITCH FIID

4 (A) M

This field is the institution identifier that uniquely distinguishes the Interchange. The value entered in this field must be identical to the value entered in the INTERCHANGE FIID field on ICF screen 1.

Default Value: None

MERCHANT ID

19 (A)

This field identifies the merchant. This field can be filled with asterisks.

Default Value: None

TERMINAL FIID

1 (A)

The FIID associated with the terminal.

Default Value: None

TERMINAL ID 16 (A)

This field identifies the BASE24 terminal identifier. The value entered in this field must be left-justified and blank-filled.

Default Value: None

PROD ID 2 (A) M

This field identifies the product with which the terminal is associated. This field must be set to '01' (ATM).

Default Value: None

SWITCH MERCHANT INFORMATION

The LINK (LIS5) Interchange Interface process does not use these fields.

SWITCH TERMINAL INFORMATION

The following fields contain information about the interchange's terminal.

ID 19 (A)

The LINK (LIS5) Interchange Interface will use the first position of the Switch Terminal ID field to indicate whether the corresponding terminal identified as Financial Inclusion or not. The rest of the field should be space filled. Valid values are as follows:

0 = Unknown/Not Financial Inclusion

1 = Financial Inclusion

Default Value: 0

OFFSET 2 (N)

The LINK (LIS5) Interchange Interface process does not use this field.

LENGTH 2 (N)

The LINK (LIS5) Interchange Interface process does not use this field.

The LINK (LIS5) Interchange Interface process does not use the remaining fields on STF Screen 1.

3.7 LOG MESSAGE CONFIGURATION FILE (LMCF)

The LMCF contains a record for each modified log message in the system. The LINK (LIS5) Interchange Interface process opens the LMCF, retrieves its modified log messages, and closes the LMCF. Whenever the LINK (LIS5) Interchange Interface process generates a log message, it checks the LMCF records in memory to determine whether the message requires modification.

3.8 TOKEN FILE (TKN)

To log or extract tokens, the Token File (TKN) must be configured appropriately. The token parameters can be set on the TKN file maintenance screen.

The following tokens used by the LINK (LIS5) Interchange Interface can be logged or extracted:

- 03 - ATM 5.0 token
- 06 - PIN Change token
- 24 - ATM Flag1 token
- 25 - Surcharge token
- 30 - Issuer Fee/Rebate token
- A5 - Non-Currency Dispense token
- AG - ATM Data1 token
- B0 - Acquirer Switch token
- B1 - Issuer Switch token
- BE - Original Currency token
- BF - Pre-pay Receipt token
- BI - Pre-pay Top-up token

For more details, refer to the ***BASE24 Tokens Manual***.

3.9 OTHER BASE24 FILES

Certain fields in other BASE24 files must be set to appropriate values to ensure that the required data is sent in messages to the LINK Interchange.

3.9.1 Institution Definition File (IDF)

The FIID field on screen 1 of the IDF is moved into the Card Acceptor ID Code (P-42) field sent in 0200/0420 ATM messages.

The INSTITUTION ID NUMBER field on screen 1 of the IDF is moved into the Authorising Agent ID Code (S-113) field sent in 0110/0210/0420 ATM messages.

3.9.2 Cardholder Authorisation File (CAF)

The appropriate ACCOUNT NUMBER field on screens 3 and 4 of the CAF is moved into the Account Identification 1 (S-102) field sent in 0110/0210 ATM messages.

3.9.3 Terminal Data File (ATD/TDF)

The following ATD/TDF fields are moved into the specified ISO fields in 0100/0200/0420 ATM messages:

Screen	Screen Field Name	ISO Field Number	ISO Field Name
1	TERM ID	P-41	Card Acceptor Terminal Id
1	INST ID NUM	P-32	Acquiring Institution Identification Code
6	AMT CURRENCY CODE	P-49	Currency Code, Transaction

3.9.3.1 Terminal Location

The following ATD/TDF fields are moved into the Card Acceptor Name/Location (P-43) field in 0100/0200/0420 ATM messages:

Screen	Screen Field Name	ISO Field Number	ISO Field Location
1	LOCATION	P-43	Positions 01-23
1	CITY	P-43	Positions 24-36
1	STATE	P-43	Positions 37-38
1	COUNTRY	P-43	Positions 39-40

LINK defines positions 24-38 as the ‘locality’, the component of the address that is the geographical area. In effect, acquirers may chose to use the two characters of state information on the screen as a continuation of the city, provided that no issuers rely on values of the state field. For example, ‘CHELSEA, LONDON’ could be entered on the screen as ‘CHELSEA, LOND’ in the city field and ‘ON’ in the state field. Acquirers must enter data on the screen in at least ‘LOCATION’ and ‘CITY’.

3.9.3.2 Remote Terminals

Records in the ATD/TDF must be configured with an appropriate Branch ID value to indicate either a branch ATM or a remote ATM. To identify the location of an ATM, the LINK (LIS5) Interchange Interface uses the Branch ID from the TDF/ATD in conjunction with the LCONF parameter REMOTE-TERMINAL-BRANCH-ID.

The LCONF parameter (retrieved on initialisation) contains a four-character identifier for remote ATMs. The parameter may contain any alphanumeric characters and asterisks (asterisks are used as “wild-card” characters). If the parameter is not present, or is invalid, then it is assumed that remote ATMs are not supported by the system.

For request and reversal messages sent to the LINK Interchange, the LCONF parameter is compared to the contents of the BRCH-ID field in the STM (retrieved from the ATD/TDF for ATMs directly-connected to BASE24). An asterisk in the LCONF parameter indicates a match against any value in the corresponding character position in the BRCH-ID field.

- If the values match, then the LINK (LIS5) Interchange Interface assumes that the transaction originates from a remote ATM.
- If the values do not match, then the LINK (LIS5) Interchange Interface assumes that the transaction originates from a branch ATM.

The LINK (LIS5) Interchange Interface uses the result of the comparison to populate field P-61 (POS Data) sub-field 4 (Operating Environment) with an appropriate value in the message sent to the LINK network.

3.9.3.3 Co-Owned Terminals

Records in the ATD/TDF must be configured with an appropriate Branch ID value to indicate whether a remote ATM is co-owned by the LINK acquirer and the site operator. To identify whether an ATM is co-owned, the LINK (LIS5) Interchange Interface uses the Branch ID from the TDF/ATD in conjunction with the LCONF parameter CO-OWNED-TERMINAL-BRANCH-ID.

The LCONF parameter (retrieved on initialisation) contains a four-character identifier for remote ATMs. The parameter may contain any alphanumeric characters and asterisks (asterisks are used as “wild-card” characters). If the parameter is not present, or is invalid, then it is assumed that remote ATMs are not co-owned.

For request and reversal messages sent to the LINK Interchange, the LCONF parameter is compared to the contents of the BRCH-ID field in the STM (retrieved from the ATD/TDF for ATMs directly-connected to BASE24). An asterisk in the LCONF parameter indicates a match against any value in the corresponding character position in the BRCH-ID field.

- If the values match, then the LINK (LIS5) Interchange Interface assumes that the transaction originates from a remote ATM that is co-owned.
- If the values do not match, then the LINK (LIS5) Interchange Interface assumes that the transaction originates from a remote ATM that is not co-owned.

The LINK (LIS5) Interchange Interface uses the result of the comparison to populate field P-61 (POS Data) sub-field 4 (Operating Environment) with an appropriate value in the message sent to the LINK network.

3.9.4 Surcharge File (SURF)

The Surcharge File (SURF) contains one record for each card group targeted by a terminal group for transaction acquirer fees in a particular currency.

Card groups are separated according to transaction routing method, e.g., either CPF or SPROUTE. Basic configuration of surcharges is driven by the relationship between terminal groups (i.e., TDF Surcharge Profile IDs) and card groups (i.e., Issuers). Within this relationship, specific card types, transaction types, account types and surcharge calculation methods are used to determine the surcharge amount.

3.9.5 Card Prefix File (CPF)

For acquired Mobile Phone Top-Up (MPTU) transactions, customers are unable to authorise the “funds” portion of the transaction for on-us cards. This is because the LINK (LIS5) Interchange Interface does not support the sending of MPTU transactions to the LINK network where the funds have already been authorised. Therefore, on-us MPTU transactions must be routed differently to other on-us transactions (i.e., sent to the LINK network instead of being authorised by BASE24-atm or send to a back-end Host system).

BASE24-atm Release 6.0 includes an enhancement to provide enhanced routing capabilities for MPTU transactions. This enhancement provides a standard mechanism to allow on-us MPTU transactions to be routed differently to other on-us transactions. Acquirers should set the MTU PREFIX ROUTING field on CPF Screen 4 to a different value to that in the PREFIX ROUTING field on CPF Screen 1. This ensures that on-us MPTU transactions access a different entry in the IDF Routing Table to other on-us transactions, thereby allowing different authorisation parameters to be configured. Acquirers should utilize this enhancement to allow on-us MPTU transactions acquired by BASE24-atm to be routed to the LINK (LIS5) Interchange process.

4. STARTUP AND CONTROL

This section provides information on the configuration of the LINK (LIS5) Interchange Interface process. It also describes the processing involved for initialisation, network control commands, timers and the network management message flow between the LINK (LIS5) Interchange Interface process and the LINK network.

4.1 PROCESS CONFIGURATION

The following paragraphs contain an explanation on how the LINK (LIS5) Interchange Interface process is configured. They provide information about the startup logic of the LINK (LIS5) Interchange Interface process and the method in which the LINK (LIS5) Interchange Interface process logs messages.

4.1.1 Startup

In the Network Environment File (NEF), the STARTUP attribute for the PROCESS object type specifies the startup logic for an application process. Because of the length of time required for initialisation, the LINK (LIS5) Interchange Interface process should be assigned a startup logic of AUTOMATIC in its process definition. AUTOMATIC means the LINK (LIS5) Interchange Interface process is started automatically when the XPNET logical network is initiated. This ensures that the LINK (LIS5) Interchange Interface process is ready when it receives its first transaction.

4.1.2 Log Messages

Log messages are routed to certain destinations, based on the routing code that appears in the log messages. The LINK (LIS5) Interchange Interface process routes all of its log messages with a standard routing code of zero.

The LINK (LIS5) Interchange Interface process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

For a complete description of log messages output by the LINK (LIS5) Interchange Interface, refer to Appendix C.

4.1.3 Transactional Interface Monitoring (T.I.M.)

The LINK (LIS5) Interchange Interface supports the Transactional Interface Monitoring (T.I.M.) enhancement. T.I.M. offers the ability to monitor, in real-time, the actual transaction health between BASE24 and the LINK network. This is achieved by keeping track of a number of transaction conditions in the interchange, for both acquirer and issuer processing. The T.I.M. monitoring facility can be started, stopped and have its counters reset using NCPCOM commands.

This enhancement works in conjunction with products such as NET24-OM, ENGWARD and PROGNOSIS. The critical information is held in appropriate Measure counters, enabling products that interface to the Measure utility available with HP NonStop (formally Tandem) hardware to provide display and reporting data.

4.1.4 Tokens

Additional disk space may be required on the HP NonStop (formerly Tandem) for the Release 5.0 or above BASE24 database depending on the number of tokens being logged to the files.

The Release 5.0 or above BASE24 user can indicate what tokens should be logged by the LINK (LIS5) Interchange Interface to its Interface Log File (ILF). The user can also indicate to the Release 5.0 or above BASE24-atm Authorisation process what tokens should be logged to the Transaction Log File (TLF).

When deciding whether or not to log a token to any Release 5.0 or above log file, the user should consider the resulting length of the log record with the new token added. If logging a new token to a log file increases the length of a log file record so that fewer records can fit in the maximum HP NonStop (formerly Tandem) data block, more data blocks (i.e., more disk space) will be required to hold the same number of records. This will increase the user's disk space requirements.

4.2 INITIALISATION

During initialisation processing, the LINK (LIS5) Interchange Interface performs the steps listed below. If a problem occurs during initialisation, the interface process will log a message and abort.

- Reads the name of the BASE24 Logical Network Configuration File (LCONF) into memory.
- Opens the LCONF and reads the assigns and parameters specified in section 3.
- Opens the Log Message Configuration File (LMCF), reads modified log messages and closes the LMCF.
- Opens the logging facility for writing log messages.
- Opens the Interchange Configuration File (ICF) and reads the appropriate record for configuration details (keyed by process name).
- Opens the Token File (TKN) if a TKN assign is found in the LCONF record.
- Initialises internal memory for the amount of space required to perform processing.
- Initialises its timer table.
- Locates either the KEYF or KEY6, dependent on the contents of the DES-TRIPLE-SINGLE parameter, using the LCONF assign KEYF or KEY6.
- Opens the Key File (KEYF or KEY6), and reads the appropriate record for security details (keyed by process name).
- Builds the Process Control Table (PCT) using information from the ICF and KEYF or KEY6.
- Determines the BASE24-atm posting date by comparing the current time with the ATM-LN-CUTOVER parameter value from the LCONF.
- Opens the Interchange Log File (ILF) for the previous, current and next day.
- Opens the Store-and-Forward File (SAF).
- Opens the Switch Terminal File (STF) if a STF assign is found in the LCONF record. Reads the appropriate STF records into memory. The MAX-STF-RECORDS parameter is used to allocate the extended memory segment.
- Updates the ICF record with the current posting date then closes the ICF.
- Initialises a security device.
- Reads the appropriate TKN records into memory.

- Starts a reports timer.
- Starts a performance timer.
- Closes the LCONF and the TKN file.
- Generates the log message ‘APPLICATION INITIALISATION SUCCESSFUL’.

The values of the LCONF parameters LINK-START-TRACE-NUM and INCR-LINK-TRACE-NUM-BY determine the initial value of the Systems Trace Audit Number (STAN) on startup. For each parameter, the LINK (LIS5) Interchange Interface defaults the value to 1, if the parameter is not present or is incorrect. Refer to section 5.3.2 for further information on this processing.

4.3 NETWORK CONTROL COMMANDS

This section contains the description and entry syntax of all network control text commands that the LINK (LIS5) Interchange Interface process supports. The LINK (LIS5) Interchange Interface process supports the following text commands:

- DEBUG
- DUMP
- LOGON
- LOGOFF
- PERFORMANCE
- PIN KEY CHANGE
- PMONON
- PMONOFF
- PMONRESET
- STATUS
- SWL
- TRACE ON
- TRACE OFF
- WARMBOOT

The above commands are issued to the LINK (LIS5) Interchange Interface by means of the DELIVER PROCESS command.

For more details, refer to the *NET24-XPNET Network Control Operations Guide* and the *NET24-XPNET Network Control Command Reference Manual*.

More details of the above commands are given on the following pages.

4.3.1 DEBUG Command

A restricted use operator command that places the LINK (LIS5) Interchange Interface process into the Inspect programming tool provided with HP NonStop (formerly Tandem) hardware, and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process).

Syntax

DELIVER PROCESS *object-name*, "DEBUG"

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

Processing

When the BASE24 process receives the DEBUG command, it performs the following processing steps:

1. Reads the message.
2. Places the LINK (LIS5) Interchange Interface process under control of the Inspect programming tool and returns an Inspect prompt to the home terminal.

For more information on Inspect, refer to the appropriate HP NonStop (formerly Tandem) documentation.

4.3.2 DUMP Command

The DUMP command directs the LINK (LIS5) Interchange Interface process to request a programmatic dump.

Syntax

DELIVER PROCESS *object-name*, "DUMP"

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

Processing

A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The LINK (LIS5) Interchange Interface process can continue running after the dump is generated. The information resulting from the dump can be read using RDEBUG. RDEBUG allows an image of the process environment (known as a dump) to be saved in an unstructured disk file called the dump file. The RDEBUG program allows interactive analysis of the dump file at a later time.

The LINK (LIS5) Interchange Interface process performs the following processing steps when it receives a DUMP command from an XPNET network control facility:

1. Creates a dump file containing an image of the process' environment.
2. Logs messages to indicate the status of the programmatic dump.

At this point, an operator can read and analyze the dump file. The dump file contains results of the programmatic dump. If necessary, analysis can be performed interactively through a terminal using the RDEBUG program at a later time. Information from the dump file can also be printed for further analysis.

4.3.3 LOGON Command

A general use command that initiates a logon sequence for one or more stations. It causes the LINK (LIS5) Interchange Interface process to send a logon message to each station indicated in the message.

Syntax

DELIVER PROCESS *object-name*, "LOGON *station-name*"

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

station-name — The symbolic station name defined for the LINK (LIS5) Interchange Interface process in the Interchange Configuration File (ICF).

Processing

If ALL is specified in this variable instead of a specific symbolic station name, a logon sequence is initiated for all stations defined to the process in the ICF. If an individual station is specified, a logon sequence is initiated for that station only.

4.3.4 LOGOFF Command

A general use command that initiates a logoff sequence for one or more stations. It causes the LINK (LIS5) Interchange Interface process to send a logoff message to each station indicated in the message.

Syntax

DELIVER PROCESS *object-name*, "LOGOFF *station-name*"

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

station-name — The symbolic station name defined for the LINK (LIS5) Interchange Interface process in the Interchange Configuration File (ICF).

Processing

If ALL is specified in this variable instead of a specific symbolic station name, a logoff sequence is initiated for all stations defined to the process in the ICF. If an individual station is specified, the logoff sequence is initiated for that station only.

4.3.5 PERFORMANCE Command

A general use command that causes the performance statistics kept in memory by the LINK (LIS5) Interchange Interface process be sent to the EMS collector process. The LINK (LIS5) Interchange Interface process tracks performance statistics for each station assigned to it. These statistics are gathered on an interval basis - the interval length being established in the Interchange Configuration File (ICF) - and statistics are available for the last five intervals.

Syntax

```
DELIVER PROCESS object-name, "PERFORMANCE"
```

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

Processing

This command requests the performance statistics kept in memory by the LINK (LIS5) Interchange Interface process. The performance period is defined in the ICF record (screen 3) and is specified in minutes. The following is an example of a performance display consisting of a series of log messages:

```
*** ATM Performance Statistics Summary ***
Mins in period: <#> <Outbd/Inbd> rqst limit: <#>

      REQUESTS - TIMEOUTS - PERCENT - AVERAGE
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>

*** POS Performance Statistics Summary ***
Mins in period: <#> <Outbd/Inbd> rqst limit: <#>

      REQUESTS - TIMEOUTS - PERCENT - AVERAGE
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
```

These log messages are described in Appendix C of this manual.

4.3.6 PMONON Command

A general use command that will mark the Performance Monitoring System as active.

Syntax

```
DELIVER PROCESS object-name, "PMONON"
```

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

Processing

This command will cause the LINK (LIS5) Interchange Interface process to start Performance Monitoring and initialise the Performance Monitoring counters. When successfully completed, a message "Performance Monitoring is now on." is logged.

4.3.7 PMONOFF Command

A general use command that will mark the Performance Monitoring System as inactive.

Syntax

```
DELIVER PROCESS object-name, "PMONOFF"
```

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

Processing

This command will cause the LINK (LIS5) Interchange Interface process to stop Performance Monitoring. When successfully completed, a message "Performance Monitoring is now off." is logged.

4.3.8 PMONRESET Command

A general use command that will initialise all the Performance Monitoring variables to zero.

Syntax

```
DELIVER PROCESS object-name, "PMONRESET"
```

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

Processing

This command will cause the LINK (LIS5) Interchange Interface process to reset Performance Monitoring counters. When successfully completed, a message “Performance Monitoring counters have been reset” is logged.

4.3.9 PIN KEY CHANGE Command

A general use command that will either send a new Issuer Working Key (IWK) to LINK or request a new Acquirer Working Key (AWK) from LINK.

Syntax

DELIVER PROCESS *object-name*, “PIN KEY CHANGE *parameter*”

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

parameter — The type of key to be changed. Valid values are:

- O (outbound) - Acquirer Working Key
- I (inbound) - Issuer Working Key
- B (both) - Both AWK and IWK

Processing

This command will cause the LINK (LIS5) Interchange Interface process to initiate dynamic key exchange:

- If the parameter in the command contains “O”, the LINK (LIS5) Interchange Interface process will send a “key change request” 0800 network management message to LINK for the AWK.
- If the parameter in the command contains “I”, the LINK (LIS5) Interchange Interface process will generate a new IWK and send a “key change” 0800 network management request message to LINK.
- If the parameter in the command contains “B”, the LINK (LIS5) Interchange Interface process will perform both of the functions described above.

4.3.10 STATUS Command

A general use operator command that causes the LINK (LIS5) Interchange Interface process to send miscellaneous information on its current processing parameters to the EMS collector process.

Syntax

```
DELIVER PROCESS object-name, "STATUS"
```

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

Processing

This command requests the status of the LINK (LIS5) Interchange Interface process. The LINK (LIS5) Interchange Interface can log the following information:

```
BASE24-atm DATE:  yymmdd
LINK Date:  yymmdd
# Outbound Requests Pending of # Maximum
# Inbound Requests Pending of # Maximum
# Timers Active of # Maximum
SAF Cycle in progress
File:  SAF-name, # Percent Full
File:  ILF-name, # Records, # Percent Full
xxx Station is up/down and Logged on/off with # timeouts of # max
whether or not Performance is running
```

These log messages are described in Appendix C of this manual.

4.3.11 SWL Command

A general use operator command that changes the event message logging facility from the current facility to the indicated facility.

Syntax

```
DELIVER PROCESS object-name, "SWL logging-facility"
```

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

logging-facility — The facility to which logging is being changed.

Processing

The LINK (LIS5) Interchange Interface begins sending event messages to the logging facility specified in the SWL command.

Note: This command is not the same as the command used to switch log files for the XPNET process. For more on the command used to switch log files for the XPNET process, refer to the *NET24-XPNET Network Control Command Reference Manual*.

4.3.12 TRACE ON Command

A restricted use operator command that activates a trace function within an LINK (LIS5) Interchange Interface process. The LINK (LIS5) Interchange Interface process contains trace log messages that are output to the BASE24 log. Trace log messages are requested only in a test environment. A trace log message is output when specified points are reached in the LINK (LIS5) Interchange Interface software.

Syntax

```
DELIVER PROCESS object-name, "TRACE ON"
```

object-name — The symbolic name assigned to the process in the XPNET node.

Processing

The LINK (LIS5) Interchange Interface process performs the following processing steps when it receives a TRACE ON command:

1. Reads the command.
2. Activates the trace function within the process.

4.3.13 TRACE OFF Command

A restricted use operator command that turns off the trace function within an LINK (LIS5) Interchange Interface process. The LINK (LIS5) Interchange Interface process contains trace log messages that are output to the BASE24 log. Trace log messages are requested only in a test environment. A trace log message is output when specified points are reached in the LINK (LIS5) Interchange Interface software.

Syntax

```
DELIVER PROCESS object-name, "TRACE OFF"
```

object-name — The symbolic name assigned to the process in the XPNET node.

Processing

The LINK (LIS5) Interchange Interface process performs the following processing steps when it receives a TRACE OFF command:

1. Reads the command.
2. Deactivates the trace function within the process.

4.3.14 WARMBOOT Command

A general use operator command that causes the LINK (LIS5) Interchange Interface process to reinitialize.

Syntax

```
DELIVER PROCESS object-name, "9501"
```

```
DELIVER PROCESS object-name, "WARMBOOT"
```

object-name — The symbolic name assigned to the LINK (LIS5) Interchange Interface process in the XPNET node.

Processing

This command warm boots the LINK (LIS5) Interchange Interface process. Warmboot processing is similar to process initialization, except that a warmboot only includes those steps required to update the file information carried by the LINK (LIS5) Interchange Interface process in memory. A message is logged to the BASE24 log indicating that the warmboot is completed and the new status implemented, or the warmboot was aborted and previous status retained.

4.4 NETWORK MANAGEMENT MESSAGE FLOWS

Network management messages (NMMs) are used to control the message processing within the network.

This section describes the message flows for the following:

- Administrative Advice
- Echo Test
- Logon
- Logoff
- Business Day Cutover
- Dynamic Key Exchange.

Two options are available for each type of member connection (specific issuer/acquirer connection or combined issuer/acquirer connection) in respect of the use of network management messages.

The BASE24 LINK (LIS5) Interchange Interface supports option one (combined zone message) only. Under option one, the LINK Switch and the member will always use a single network management message to perform an action in both acquirer and issuer zones (with the exception of “key change” and “key change request” messages).

Further information about network management message options is given in the ***LINK Switch Service Interchange Standard – LIS5 Security Standard***.

4.4.1 Administrative Advice

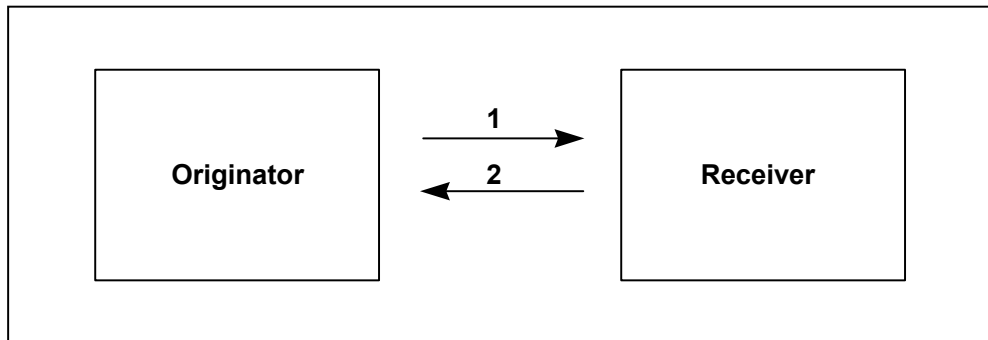


Figure 4-1 Administrative Advice

1. If either the BASE24 LINK (LIS5) Interchange Interface module or the LINK Switch receives an invalid message (e.g., the message cannot be identified or a mandatory data item is absent), the recipient of the invalid message formats and sends a 0620 Administrative Advice message to the sender. The Administrative Advice is identified by the value 900 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
2. If the LINK (LIS5) Interchange Interface module is the recipient of the 0620 message, the message is logged to the BASE24 event log to indicate that it has been received. No further processing of the message will be performed.

4.4.2 Echo Test

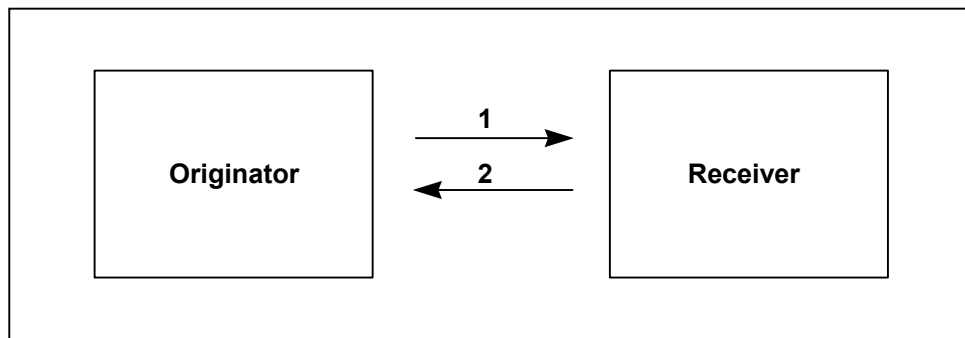


Figure 4-2 Echo Test

1. When either the BASE24 LINK (LIS5) Interchange Interface module or the LINK Switch sends an echo test message it is identified by the value 301 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
2. The receiver of the echo test message sends an echo test response. The echo test response message is identified by the value 301 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.

If the LINK (LIS5) Interchange Interface is the recipient of the echo test message, then the process will set a Network Management Message timer for each of the interface's send/receive stations that is down.

4.4.3 Logon Messages

Stations to the LINK network are defined in the ICF. These stations must be defined as being either issuer-only, acquirer-only, or both. The network control LOGON request is issued either for a specific station or for all stations defined in the ICF. For more information on the syntax of the network control command, refer to section 4.3.3.

4.4.3.1 BASE24 Initiated Logon

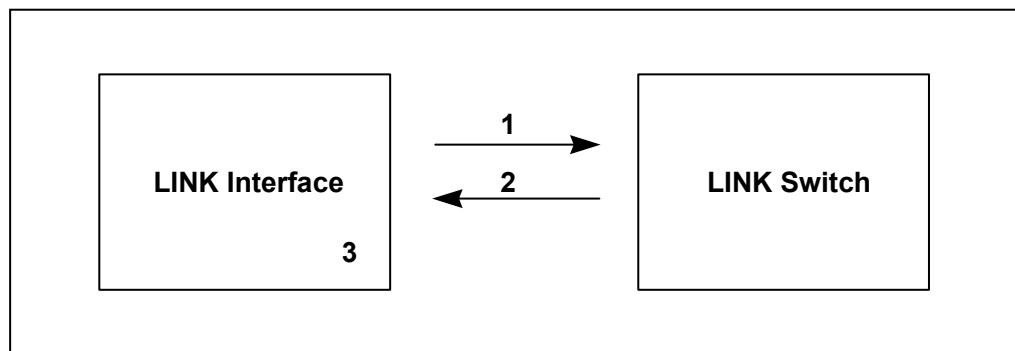


Figure 4-3 BASE24 Initiated Logon

1. If the LINK (LIS5) Interchange Interface module can send network management messages to the LINK Switch (the NETWORK MANAGEMENT MESSAGE ENABLED field in the ICF is set to Y), the LINK (LIS5) Interchange Interface module sends a logon message to each of the send/receive stations. The logon message is identified by the value 001 in the Network Management Information Code (S-70) field of the LINK (LIS5) message. In addition, the LINK (LIS5) Interchange Interface module sets a Network Management timer for each station, and marks each receive-only station as up.
2. LINK returns a logon response from each station. The logon response is identified by the value 001 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
3. The LINK (LIS5) Interchange Interface module receives each logon response and checks the Response Code (P-39) field in the message:
 - If the Response Code field contains zeros, indicating an approval, the LINK (LIS5) Interchange Interface module marks the LINK Switch as up and logged on. The LINK (LIS5) Interchange Interface module also marks the corresponding station up and sets a Wait-for-Traffic timer for that station.

- If the Response Code field does not contain zeros, the LINK (LIS5) Interchange Interface module logs a message, marks the LINK Switch as logged off and ceases all attempts to communicate with the station.

4.4.3.2 LINK Initiated Logon

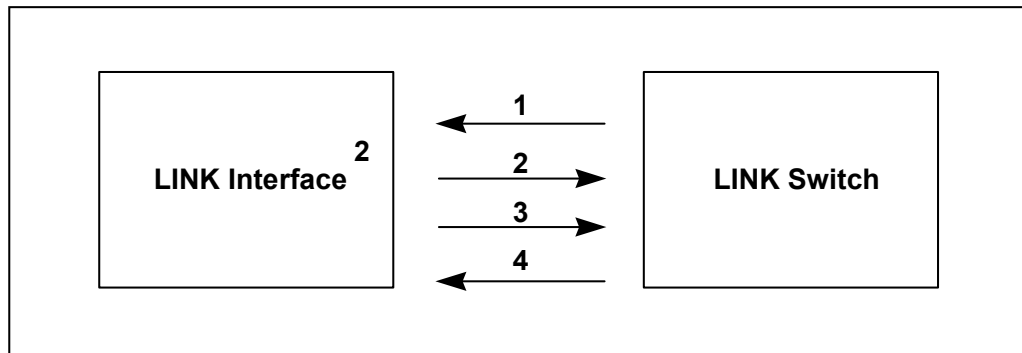


Figure 4-4 LINK Initiated Logon

1. LINK sends a logon message to re-establish communications. The logon message is identified by the value 001 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
2. The LINK (LIS5) Interchange Interface module receives the message and performs the following steps:
 - Marks the LINK Switch as logged on.
 - Sets a Wait-for Traffic timer for each station defined for the LINK Switch.
 - Sends a logon response to the LINK Switch. The logon response is identified by the value 001 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
3. The LINK (LIS5) Interchange Interface sends an echo test message (section 4.4.2) and sets a Network Management Message timer for each of the LINK Switch's send/receive stations that is down. The echo test message is identified by the value 301 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
4. The LINK Switch sends an echo test response. The echo test response message is identified by the value 301 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.

4.4.4 Logoff Messages

The network control LOGOFF request is issued either for a specific station or for all stations defined in the ICF. For more information on the syntax of the network control command, refer to section 4.3.4.

4.4.4.1 BASE24 Initiated Logoff

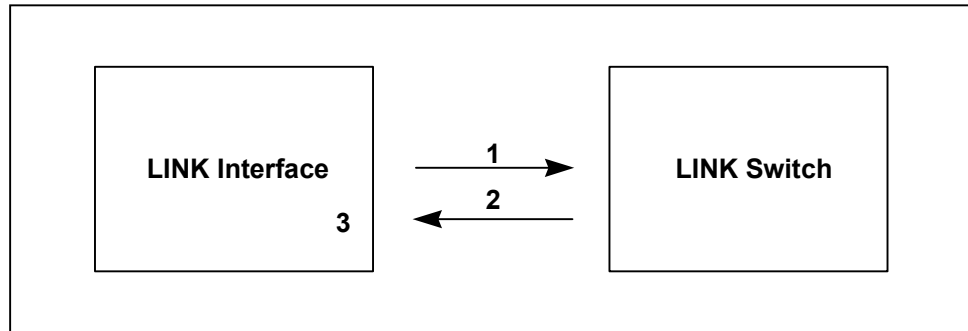


Figure 4-5 BASE24 Initiated Logoff

1. The LINK (LIS5) Interchange Interface module receives a logoff command from the Network Control Supervisor (NCS) screen. It sends a logoff message to each of LINK's send/receive stations and starts a Network Management Message timer for each. The logoff message is identified by the value 002 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
2. LINK returns a logoff response from one or each station. The logoff response is identified by the value 002 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
3. The LINK (LIS5) Interchange Interface module marks LINK as logged off and clears the Network Management Message timers, Extended Network Management Message timers and Wait-for Traffic timers for the LINK Switch and its stations. All other timers are left running to ensure that transactions in progress complete.

4.4.4.2 LINK Initiated Logoff

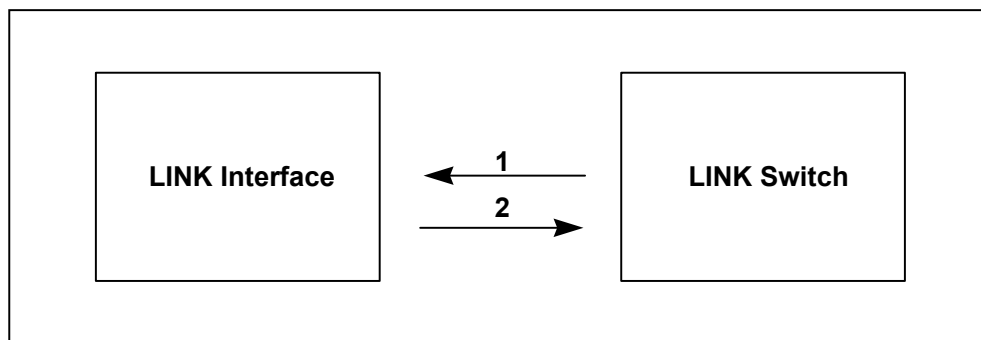


Figure 4-6 LINK Initiated Logoff

1. LINK sends a logoff message to BASE24. The logoff message is identified by the value 002 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
2. The LINK (LIS5) Interchange Interface module receives the message and marks the LINK Switch as logged off. The Network Management Message timers, Extended Network Management Message timers and Wait-for Traffic timers are cleared. All other timers are left running to ensure that transactions in progress complete.
3. The LINK (LIS5) Interchange Interface module sends a logoff response to LINK on the same station. The logoff response is identified by the value 002 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.

4.4.5 LINK Initiated Business Day Cutover

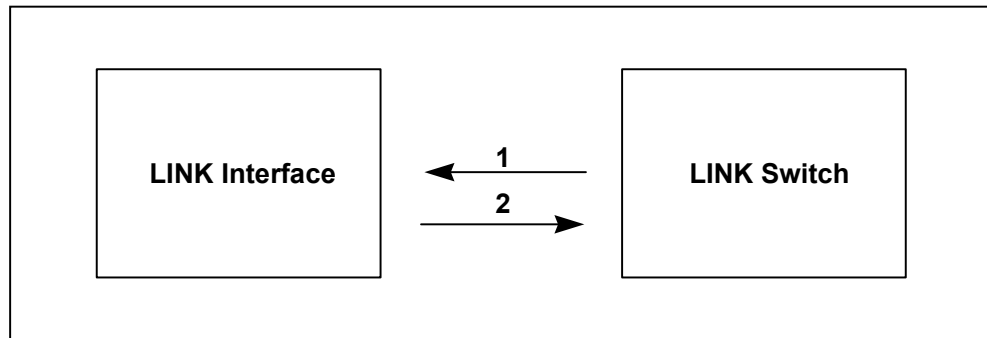


Figure 4-7 LINK Initiated Business Day Cutover

1. LINK sends a Cutover message to BASE24. The cutover message is identified by the value 201 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
 - The LINK (LIS5) Interchange Interface module assumes that a cutover message has not been processed for the LINK Switch processing date as there is no date in the external message.
 - The LINK (LIS5) Interchange Interface module updates the current business date for the LINK Switch.
 - The LINK (LIS5) Interchange Interface module opens and begins logging transaction records to a new ILF.
 - The LINK (LIS5) Interchange Interface may purge the oldest ILF from the system, based on the value of the LCONF parameter SW-ILF-FILE-RETENTION.
2. The LINK (LIS5) Interchange Interface module returns a cutover response. The cutover response is identified by the value 201 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.

Note: Settlement totals messages are not supported.

4.4.6 Dynamic Key Exchange

The LINK Switch is responsible for the generation and issue of new Acquirer Working Keys (AWKs) to Institutions, via a Key Change message. LINK may also inform an Institution, via a Key Change Request message, that a new Issuer Working Key (IWK) is to be generated for the Institution.

The LINK member (Institution) is responsible for the generation and issue of new Issuer Working Keys (IWKs) to Institutions, via a Key Change message. The LINK member may also inform the LINK Switch, via a Key Change Request message, that a new Acquirer Working Key (AWK) is to be generated for the Institution.

4.4.6.1 BASE24 Initiated Key Change Request

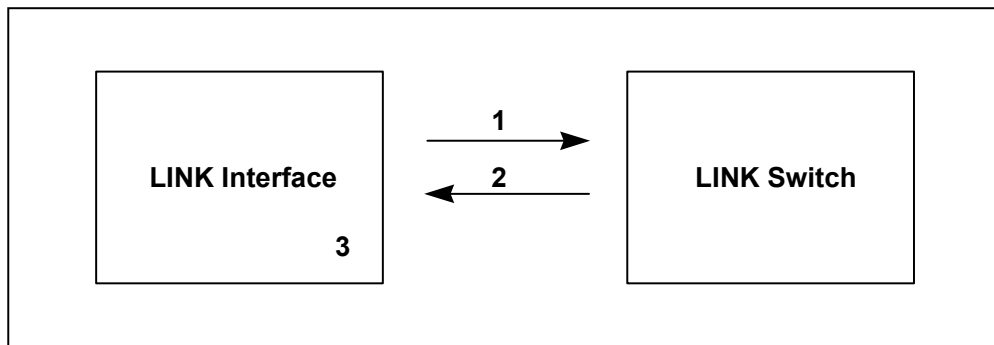


Figure 4-8 BASE24 Initiated Key Change Request

1. The LINK (LIS5) Interchange Interface module performs the following processing:
 - Formats a Key Change Request 0800 message.
 - Sends the Key Change Request 0800 message to LINK. The Key Change Request message is identified by the value 181 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.

- Starts a Network Management Message timer for the Key Change Request 0800 message. If this timer expires before a 0810 response is received from LINK, the LINK (LIS5) Interchange Interface module increments the timeout counter for the station and starts an Extended Network Management Message timer for the Key Change Request 0800 message. If the maximum number of timeouts for the station has been exceeded, defined by the field MAXIMUM TIMEOUTS on the ICF, it marks the station down. If the Extended Network Management Message timer expires, the LINK (LIS5) Interchange Interface module sends the Key Change Request 0800 message on the next available station.
2. The LINK Switch receives the Key Change Request 0800 message, processes it and returns a Key Change Request 0810 message. The Key Change Request message is identified by the value 181 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
 3. The LINK (LIS5) Interchange Interface module receives the 0810 response:
 - If the response code is '00', the Key Change Request has been approved and the LINK (LIS5) Interchange Interface will delete the associated timer.
 - If the response code <> '00', the Key Change Request has been denied. In this case, the timer will not be deleted and the timer will expire. The LINK (LIS5) Interchange Interface module will resend the Key Change Request 0800 message as part of timeout processing.

4.4.6.2 LINK Switch Initiated Key Change

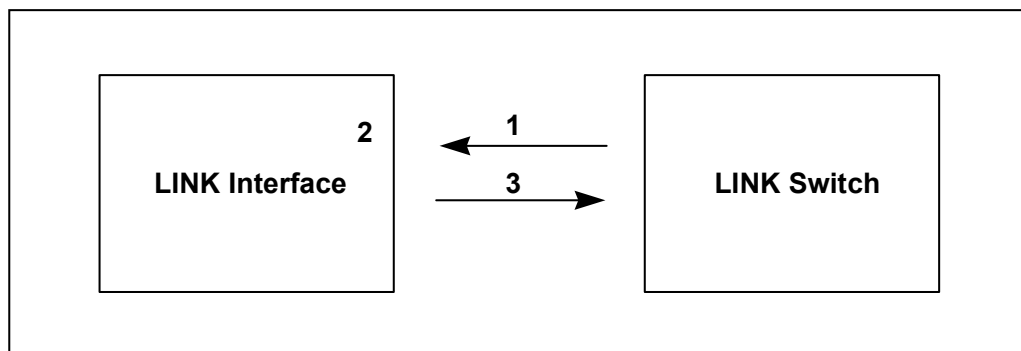


Figure 4-9 LINK Initiated Key Change

1. The LINK (LIS5) Interchange Interface module receives a Key Change 0800 message. The Key Change message is identified by the value 161 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
2. The LINK (LIS5) Interchange Interface module validates the check digits received in the Key Change 0800 message.
3. The LINK (LIS5) Interchange Interface module continues processing depending on the response code generated.
 - If the check digits generated by the LINK (LIS5) Interchange Interface module do not match the check digits sent in the Key Change 0800 message, return a Key Change 0810 message with the response code field (P-39) set to '76'. The Key Change message is identified by the value 161 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.

- If the check digits are valid, update the database with the new key and check digits. Continue processing depending on the response code generated.

If an error occurs, return a Key Change 0810 message with the response code field (P-39) set to '76'.

If the database was updated successfully the LINK (LIS5) Interchange Interface module returns an approved Key Change 0810 message to LINK and starts a clear old key timer. During key exchange if the security device cannot verify the PIN because of a key error, the transaction is re-tried using the old key, if one exists. This timer controls how long the old key exists and is set by the LINK (LIS5) Interchange Interface module when the CLEAR OLD KEY TIMER VALUE on the KEYF/KEY6 contains a non-zero value. On expiry of the clear old key timer the LINK (LIS5) Interchange Interface module clears the old Acquirer and Issuer Working Key and check digits.

4.4.6.3 LINK Initiated Key Change Request

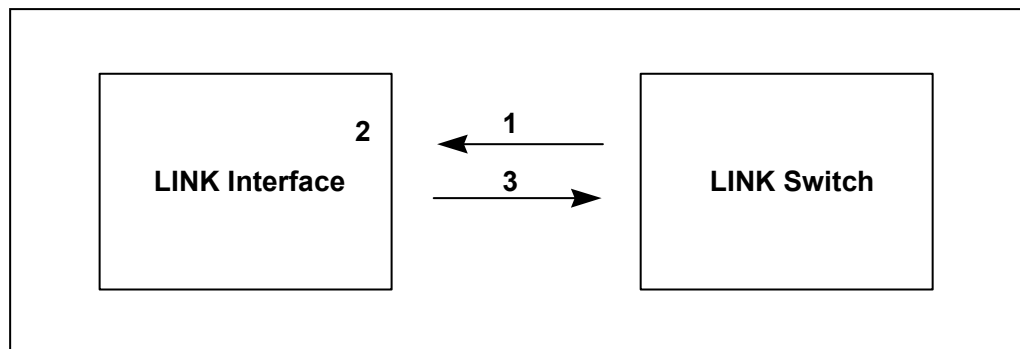


Figure 4-10 LINK Initiated Key Change Request

1. The LINK (LIS5) Interchange Interface module receives a Key Change Request 0800 message. The Key Change Request message is identified by the value 181 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
2. The LINK (LIS5) Interchange Interface module validates the Key Change Request 0800 message.
3. The LINK (LIS5) Interchange Interface module returns a Key Change Request 0810 message, identified by the value 181 in the Network Management Information Code (S-70) field of the LINK (LIS5) message. The response code field (P-39) will be set to '00'. Should an error be encountered, the response code field (P-39) will be set to '30'.

4.4.6.4 BASE24 Initiated Key Change

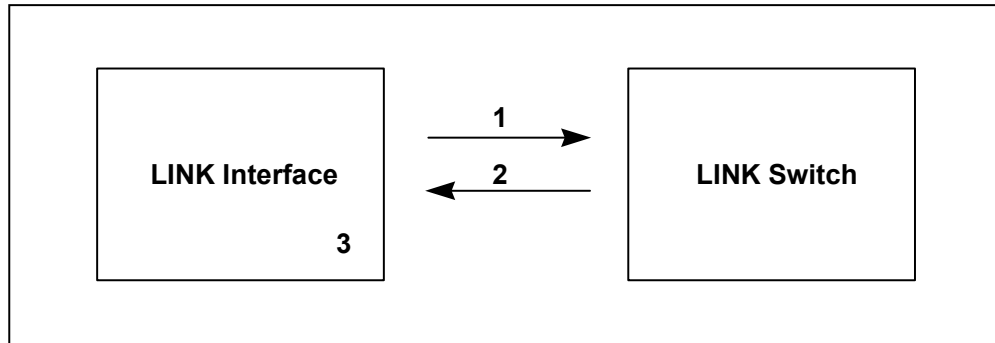


Figure 4-11 BASE24 Initiated Key Change

1. The LINK (LIS5) Interchange Interface module sends a Key Change 0800 message. The Key Change message is identified by the value 161 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
2. The LINK Switch validates the check digits received in the Key Change 0800 Message, and returns a Key Change 0810 Message. The Key Change Message is identified by the value 161 in the Network Management Information Code (S-70) field of the LINK (LIS5) message.
3. The LINK (LIS5) Interchange Interface module continues processing depending on the response code generated.
 - If the response code field (P-39) is set to '76', then the LINK Switch failed to validate the new key.
 - If the check digits are valid, update the database with the new key and check digits.

4.4.6.5 Key Exchange at Sign-On

Two options are available to members in respect of how Acquirer Working Keys (AWKs) and Issuer Working Keys (IWKs) are exchanged at sign-on. Keys can be generated on sign-on or keys can be generated on request.

The BASE24 LINK (LIS5) Interchange Interface supports option one (keys generated on request) only. Under option one, the party that initiates the sign-on manages the key exchange process. A new AWK or IWK will be returned to the party initiating the sign-on following receipt of a network management message requesting a new key.

Where LINK initiates the sign-on, the following message flow occurs:

- Logon messages are exchanged (see section 4.4.3.2).
- If BASE24 is the acquirer for LINK transactions, key change messages are exchanged (see section 4.4.6.2).
- If BASE24 is the issuer for LINK transactions, key change request messages are exchanged (see section 4.4.6.3).
- If key change request messages have been exchanged, further key change messages are exchanged (see section 4.4.6.4).

Where the LINK (LIS5) Interchange Interface initiates the sign-on, the following message flow occurs:

- Logon messages are exchanged (see section 4.4.3.1).
- If BASE24 is the issuer for LINK transactions, key change messages are exchanged (see section 4.4.6.4).
- If BASE24 is the acquirer for LINK transactions, key change request messages are exchanged (see section 4.4.6.1).
- If key change request messages have been exchanged, further key change messages are exchanged (see section 4.4.6.2).

Further information about key exchange options at sign-on is given in the ***LINK Switch Service Interchange Standard – LIS5 Security Standard*** and in the ***Network Management Message Flows Document***.

4.5 TIMERS

This section describes each timer used by the LINK (LIS5) Interchange Interface. Each timer description includes the timer's function, and how and when the timer is set.

4.5.1 Outbound Requests Timer

An outbound request timer is set each time a 0100 message is sent to LINK. The number of seconds used for this timer is taken from the OUTBOUND field on ICF screen 8 for BASE24-atm transactions. The outbound request timer is used to indicate whether a transaction sent to LINK has timed out.

If the timer expires before a response is received from LINK (LIS5) for the 0100 message, the request is denied and the appropriate performance and timeout counters are incremented. If the expiration of this timer causes the maximum allowed timeouts to be reached, the connection to LINK is marked down.

4.5.2 Inbound Requests Timer

An inbound request timer is set each time a 0200 message is sent to a BASE24 process for authorisation. The number of seconds used for this timer is taken from the INBOUND field on ICF screen 8. The inbound request timer is used to determine that an external transaction has timed out.

If the timer expires before a response is received from the authorizing process, the request is denied. The appropriate performance and timeout counters are incremented.

4.5.3 Store-And-Forward Timer

A store-and-forward timer is set each time a store-and-forward message is sent to LINK. The number of seconds used for this timer is taken from the STORE AND FORWARD field on ICF screen 8 for BASE24-atm transactions. The store and forward timer is used to determine that a Store-and-Forward File (SAF) record sent to LINK has timed out.

If the timer expires before an acknowledgment is received from LINK, the SAF timeout counter is incremented. If the SAF timeout counter is equal to the maximum number of retries set in the MAX SAF RETRY field on ICF screen 3, the record is written to the log and deleted from the SAF. If the MAX SAF RETRY field is set to 0, the SAF message will be sent indefinitely until it is successfully processed.

4.5.4 Network Management Timer

A network management timer is set each time a 0800 message is sent to a station. The number of seconds used for this timer is taken from the **TIMER LIMITS NETWORK MANAGEMENT** field on ICF screen 3. The network management timer is used to monitor the communications between BASE24 and LINK.

If the network management timer expires before a response is received from LINK, an extended network management timer is set.

4.5.5 Extended Network Management Timer

An extended network management timer is set each time a network management timer expires without a response being received from LINK. The number of seconds used for this timer is taken from the **TIMER LIMITS EXTENDED NETWORK** field on ICF screen 3.

The network management message is repeated at the interval specified by the extended network management timer until an acknowledgment is received from LINK or the communications line is marked up.

4.5.6 Wait For Traffic Timer

A wait for traffic timer is set each time a station is marked up and each time a previously set wait for traffic timer expires. The number of seconds used for this timer is taken from the **TIMER LIMITS WAIT FOR TRAFFIC** field on ICF screen 3.

If the timer expires and message traffic has occurred on the line, a new wait for traffic timer is started.

If the timer expires and no message traffic has occurred on the line, a network management logon message is sent. The line is marked up or down, depending on the results of the logon.

4.5.7 Performance Timer

A performance timer is set at the start of each new performance interval. The number of minutes for this timer is taken from the **TIMER LIMITS PERFORMANCE PERIOD** field on ICF screen 3. The performance timer triggers the end of one performance period and the beginning of the next.

When the performance timer expires, the **LINK (LIS5) Interchange Interface** process:

- Moves the interval four performance statistics to interval five.
- Moves the interval three performance statistics to interval four.
- Moves the interval two performance statistics to interval three.
- Moves the interval one performance statistics to interval two.
- Initialises the interval one performance statistics to zero.
- Starts a new performance timer.

4.5.8 Report Startup Timer

The report startup timer is used to start the **SETL01** and **STAT09** report programs. It is set at initialisation and at interchange cutover. The value for this timer is taken from the **SW-RPT-STARTUP** param in the **LCONF**.

When the report startup timer expires, the obey file containing the commands to run the report programs are executed.

4.5.9 Key Exchange Timer

The key exchange timer is used to specify the maximum length of time that a PIN block encryption key (**AWK** or **IWK**) should be used. The value for this time is taken from the **PIN KEY TIMER** and **PIN KEY TIMER INTERVAL** fields on **KEYF** screen 3 or **KEY6** screen 4.

When the key exchange timer expires, a key change request message (for the **AWK**) or a key change message (for the **IWK**) is sent to **LINK**.

4.5.10 Clear Old Key Timer

The clear old key timer is used to specify the maximum length of time that an old PIN block encryption key (AWK or IWK) can be used before it is automatically cleared by the LINK (LIS5) Interchange Interface process. The value for this timer is taken from the CLEAR OLD KEY TIMER VALUE field on KEYF screen 3 or KEY6 screen 4.

When the clear old key timer expires, the “old” AWK or IWK is removed from the KEYF or KEY6 record.

4.5.11 Key Management Request Timer

A key management request timer is set each time a 0800 change key or change key request message is sent to a station. The number of seconds used for this timer is taken from the TIMER LIMITS NETWORK MANAGEMENT field on ICF screen 3.

If the network management timer expires before a response is received from LINK, an extended key management request timer is set.

4.5.12 Extended Key Management Request Timer

An extended key management request timer is set each time a key management request timer expires without a response being received from LINK. The number of seconds used for this timer is taken from the TIMER LIMITS EXTENDED NETWORK field on ICF screen 3.

The network management message is repeated at the interval specified by the extended key management timer until an acknowledgment is received from LINK.

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5. TRANSACTION HANDLING

The following functions related to transaction processing are described in this section.

- BASE24 Acquirer Message Flows
- BASE24 Issuer Message Flows
- Transaction Matching
- Transaction Security
- Transaction Types
- Acquirer Fees & Issuer Pre-Notification
- Interchange Log File (ILF) Processing
- Store-and-Forward (SAF) Processing
- Token Usage
- Response Code Translation

5.1 BASE24 ACQUIRER MESSAGE FLOWS

This section describes transaction message flows where BASE24 is acting as an acquirer system.

The following message flows outbound from BASE24-atm to LINK are shown:

- Account Balance Request
- PIN Management Request/Reversal
- Financial Transaction Request/Reversal
- Timeout of PIN Management Request Message
- Timeout of Financial Transaction Request Message

5.1.1 Account Balance Request

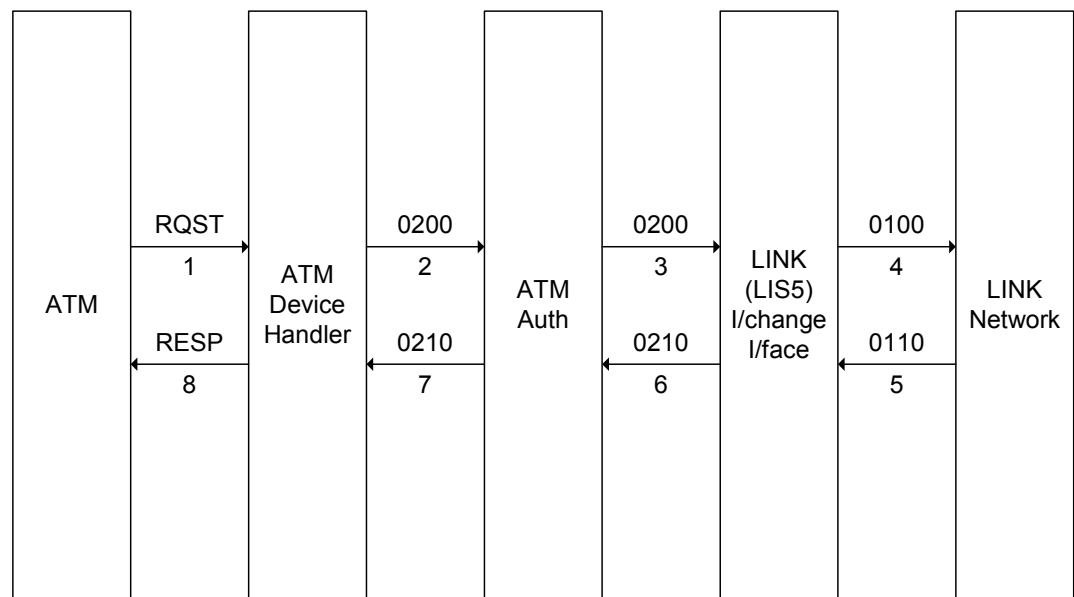


Figure 5-1 Account Balance Request Message to LINK

1. The customer enters the transaction. The ATM sends a native-mode request message to the Device Handler module.
2. The Device Handler module reformats the native-mode request message into an STM type 0200 account balance request. The Device Handler sets an authorisation timer and sends the 0200 request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the request message and routes it to the BASE24 LINK (LIS5) Interchange Interface module.
4. The LINK (LIS5) Interchange Interface module reformats the internal message into an external message, sets an outbound request timer, and sends the 0100 request message to LINK.
5. The LINK (LIS5) Interchange Interface module receives the 0110 response message from LINK.
6. The LINK (LIS5) Interchange Interface module reforms the external message from LINK into an STM, deletes its outbound request timer, logs the transaction to the ILF and forwards the 0210 response message to the BASE24-atm Authorisation process.
7. The BASE24-atm Authorisation process receives the 0210 response message, forwards it to the Device Handler module and logs it to the TLF.

8. The Device Handler module deletes its authorisation timer, formats the 0210 response message into a native-mode response and sends it to the ATM.

5.1.2 PIN Management Request/Reversal

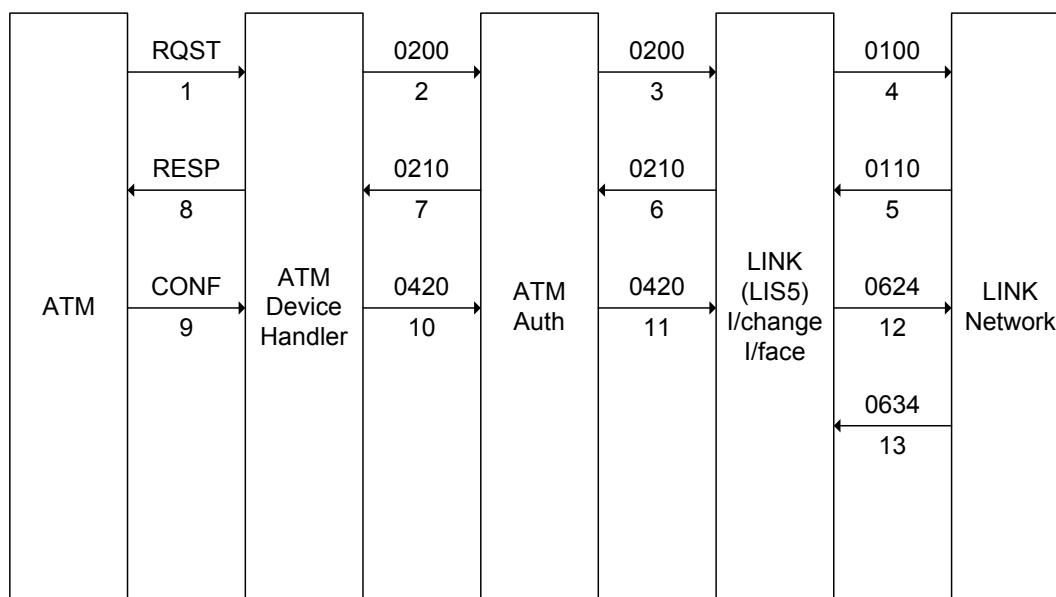


Figure 5-2 PIN Management Request Message to LINK

1. The customer enters the transaction. The ATM sends a native-mode request message to the Device Handler module.
2. The Device Handler module reformats the native-mode request message into a Standard Internal Message (STM) type 0200 financial request. The Device Handler sets an authorisation timer and sends the 0200 STM request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the request message. The BASE24-atm Authorisation process routes the STM to the BASE24 LINK (LIS5) Interchange Interface module.
4. The LINK (LIS5) Interchange Interface module reformats the internal message into an external message, sets an outbound request timer, and sends the 0100 PIN management message to LINK.
5. The LINK (LIS5) Interchange Interface module receives the 0110 response message from LINK. The LINK (LIS5) Interchange Interface module reforms the external message from LINK into an STM, deletes its outbound request timer, logs the transaction to the Interchange Log File (ILF) and forwards the 0210 response message to the BASE24-atm Authorisation process.

6. The Authorisation module receives the 0210 response message and logs it to the Transaction Log File (TLF). The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN).
7. The Authorisation module forwards the 0210 response message to the Device Handler module
8. The Device Handler module deletes its authorisation timer, reformats the 0210 response message into a native-mode response and sends it to the ATM.
9. The ATM sends a native-mode confirmation message to the Device Handler module indicating the success or failure of the PIN Management action. For instance, if an EMV Issuer Script could not be processed, the ATM would indicate a failure.
10. If a PIN Management failure occurs, the Device Handler module will format and send a 0420 reversal STM to the BASE24-atm Authorisation process.
11. The BASE24-atm Authorisation process forwards the 0420 reversal message to the LINK (LIS5) Interchange Interface module, if applicable.
12. The LINK (LIS5) Interchange Interface module locates the ILF record for the original transaction and updates the record with the reversal information. It reformats the internal message into an external message and logs this message to the Store and Forward File (SAF) to be sent to LINK during store-and-forward processing, if applicable.
13. LINK sends a 0634 acknowledgement message to the BASE24 LINK (LIS5) Interchange Interface module. Upon receipt of the 0634 message, the LINK (LIS5) Interchange Interface module deletes the record from the SAF.

5.1.3 Financial Transaction Request/Reversal

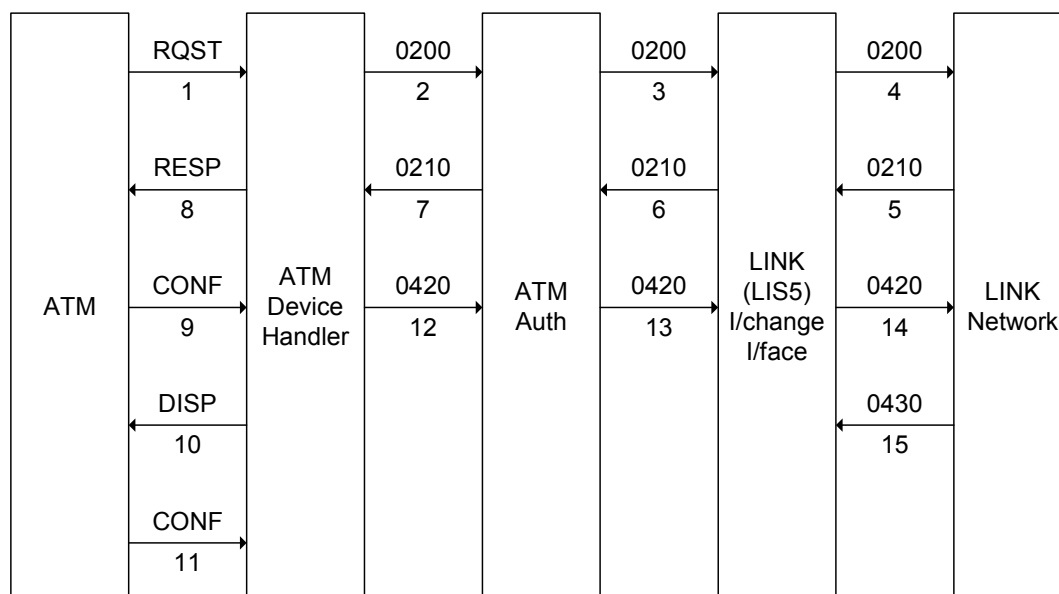


Figure 5-3 Financial Transaction Request Message to LINK

1. The customer enters the transaction. The ATM sends a native-mode request message to the Device Handler module.
2. The Device Handler module reformats the native-mode request message into a Standard Internal Message (STM) type 0200 financial request. This message contains the Surcharge Data Token with the Terminal Profile ID from the Terminal Data File (TDF). The Device Handler sets an authorisation timer and sends the 0200 STM request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the request message. It uses the Terminal Profile ID from the Surcharge Data Token and the Card Prefix to determine whether a surcharge is to be applied. The BASE24-atm Authorisation process routes the STM to the BASE24 LINK (LIS5) Interchange Interface module.
4. The LINK (LIS5) Interchange Interface module reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to LINK. The message includes the Surcharge Fee, if required.

5. The LINK (LIS5) Interchange Interface module receives the 0210 response message from LINK, which may include an issuer fee or rebate. The LINK (LIS5) Interchange Interface module reforms the external message from LINK into an STM, deletes its outbound request timer, logs the transaction to the Interchange Log File (ILF) and forwards the 0210 response message to the BASE24-atm Authorisation process. If an issuer fee is present, the Issuer Fee/Rebate Token is appended to the STM.
6. The Authorisation module receives the 0210 response message, forwards it to the Device Handler module and logs it to the Transaction Log File (TLF). The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN).
7. The Device Handler module deletes its authorisation timer formats the appropriate native-mode surcharge and/or issuer fee notification screen and sends it to the ATM to be displayed (if applicable). Otherwise, the transaction will complete as in standard processing by reformatting the 0210 response message into a native-mode response and sending it to the ATM, but may need to include a new screen to notify the customer of a net charge of zero.
8. The ATM displays the corresponding surcharge and/or issuer fee screen if applicable. Once the customer selects to proceed or to cancel, the ATM sends a native-mode confirmation message to the Device Handler module.
9. The Device Handler module receives the ATM native-mode confirmation message, indicating whether the customer wishes to proceed or to cancel.
10. If the customer chose to proceed, the Device Handler module will send the corresponding dispense/print information to the ATM in the native-mode ATM response message. If the customer chose to cancel the transaction, the Device Handler module will send the corresponding information to cancel the transaction to the ATM in the native-mode ATM response message (i.e., to display the correct screen as well as any print information if a receipt is configured for a denied transaction).
11. The ATM sends a native-mode message to the Device Handler module indicating that the transaction has been completed.
12. If the customer chose to cancel the transaction, the Device Handler module will format and send a 0420 reversal STM to the BASE24-atm Authorisation process.
13. The BASE24-atm Authorisation process forwards the 0420 reversal message to the LINK (LIS5) Interchange Interface module, if applicable.
14. The LINK (LIS5) Interchange Interface module locates the ILF record for the original transaction and updates the record with the reversal information. It reformats the internal message into an external message and logs this message to the Store and Forward File (SAF) to be sent to LINK during store-and-forward processing, if applicable.

15. LINK sends a 0430 acknowledgement message to the BASE24 LINK (LIS5) Interchange Interface module. Upon receipt of the 0430 message, the LINK (LIS5) Interchange Interface module deletes the record from the SAF.

5.1.4 Timeout of PIN Management Request Message

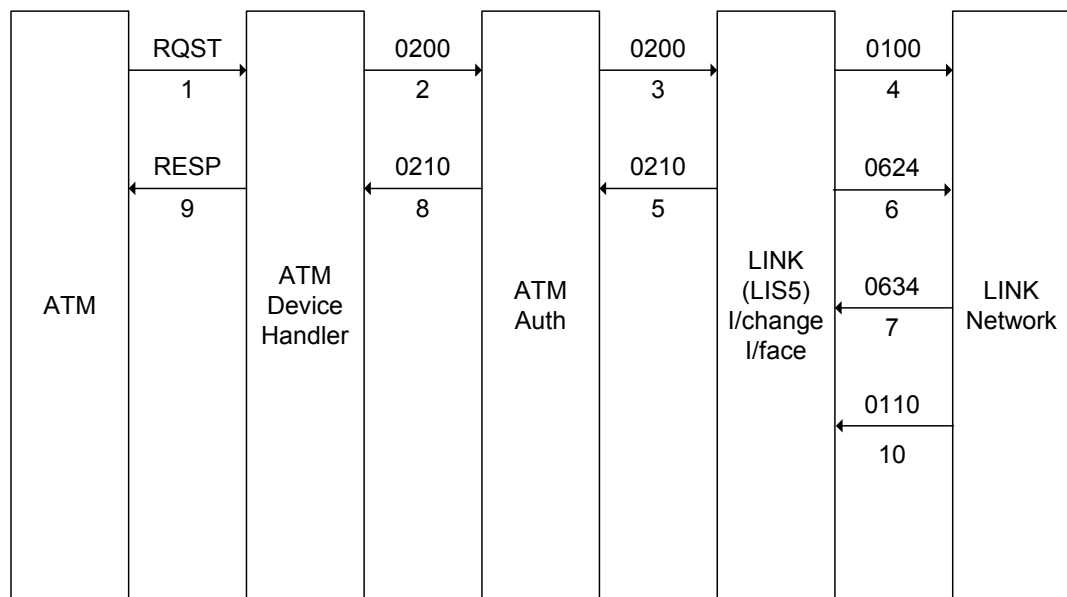


Figure 5-4 Timeout of PIN Management Request Message to LINK

1. The customer enters the transaction. The ATM sends a native-mode request message to the Device Handler module.
2. The Device Handler module reformats the native-mode request message into an STM type 0200 financial request. The Device Handler sets an authorisation timer and sends the 0200 request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the request message and routes it to the BASE24 LINK (LIS5) Interchange Interface module.
4. The LINK (LIS5) Interchange Interface module reformats the internal message into an external message, sets an outbound request timer, and sends the 0100 request message to LINK.
5. When the inbound request timer expires, the LINK (LIS5) Interchange Interface module attempts to route the transaction to an alternate destination. If an alternate destination is not configured, the LINK (LIS5) Interchange Interface module formats a 0210 internal response message denying the transaction, returns the response to the BASE24-atm Authorisation process and logs a copy of the transaction to the ILF. The LINK (LIS5) Interchange Interface module formats a 0624 PIN Management Failure Notification advice and adds it to the SAF to be sent to LINK during store and forward processing.

6. The LINK (LIS5) Interchange Interface retrieves the 0624 PIN Management Failure Notification advice from the SAF and sends it to LINK.
7. LINK sends a 0634 acknowledgement message to the BASE24 LINK (LIS5) Interchange Interface module.
8. The BASE24-atm Authorisation process receives the 0210 response message, forwards it to the Device Handler module and logs it to the TLF.
9. The Device Handler module deletes its authorisation timer, formats the 0210 response message into a native-mode response and sends it to the ATM.
10. LINK may send a late 0110 response to the LINK (LIS5) Interchange Interface module. The LINK (LIS5) Interchange Interface module then drops the late response.

5.1.5 Timeout of Financial Transaction Request Message

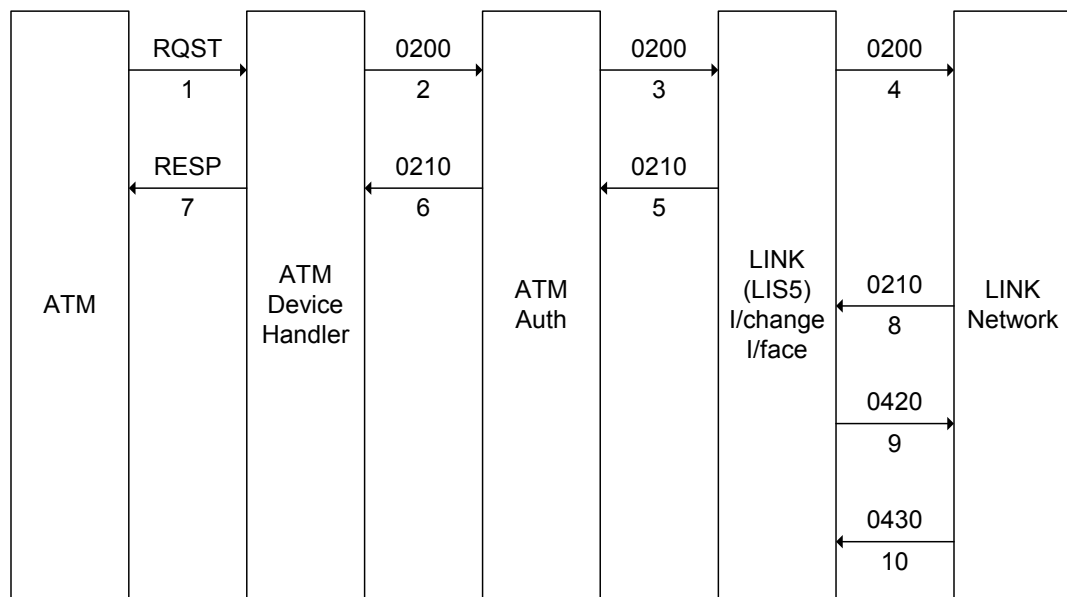


Figure 5-5 Timeout of Financial Transaction Request Message to LINK

1. The customer enters the transaction. The ATM sends a native-mode request message to the Device Handler module.
2. The Device Handler module reformats the native-mode request message into an STM type 0200 financial request. The Device Handler sets an authorisation timer and sends the 0200 request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the request message and routes it to the BASE24 LINK (LIS5) Interchange Interface module.
4. The LINK (LIS5) Interchange Interface module reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to LINK.
5. When the inbound request timer expires, the LINK (LIS5) Interchange Interface module attempts to route the transaction to an alternate destination. If an alternate destination is not configured, the LINK (LIS5) Interchange Interface module formats a 0210 internal response message denying the transaction, returns the response to the BASE24-atm Authorisation process and logs a copy of the transaction to the ILF.
6. The BASE24-atm Authorisation process receives the 0210 response message, forwards it to the Device Handler module and logs it to the TLF.

7. The Device Handler module deletes its authorisation timer, formats the 0210 response message into a native-mode response and sends it to the ATM.
8. LINK sends a late 0210 response to the LINK (LIS5) Interchange Interface module. If the LCONF parameter UPDT-ILF-ON-LATE-RESP is set to 'Y', the LINK (LIS5) Interchange Interface module updates the ILF record with information from the 0210 response message. If the transaction was denied, the LINK (LIS5) Interchange Interface module then drops the late response.
9. If the transaction was a financial transaction request (message type 0200) which was approved by LINK, the LINK (LIS5) Interchange Interface module formats a 0420 external reversal advice message and adds it to the SAF to be sent to LINK during store-and-forward processing.
10. LINK sends a 0430 acknowledgement message to the BASE24 LINK (LIS5) Interchange Interface module.

5.2 BASE24 ISSUER MESSAGE FLOWS

This section describes transaction message flows where BASE24 is acting as an issuer system.

The following message flows inbound from LINK to BASE24-atm are shown:

- Account Balance Request
- PIN Management Request/Reversal
- Financial Transaction Request/Reversal
- Timeout of PIN Management Request Message
- Timeout of Financial Transaction Request Message.

5.2.1 Account Balance Request Message

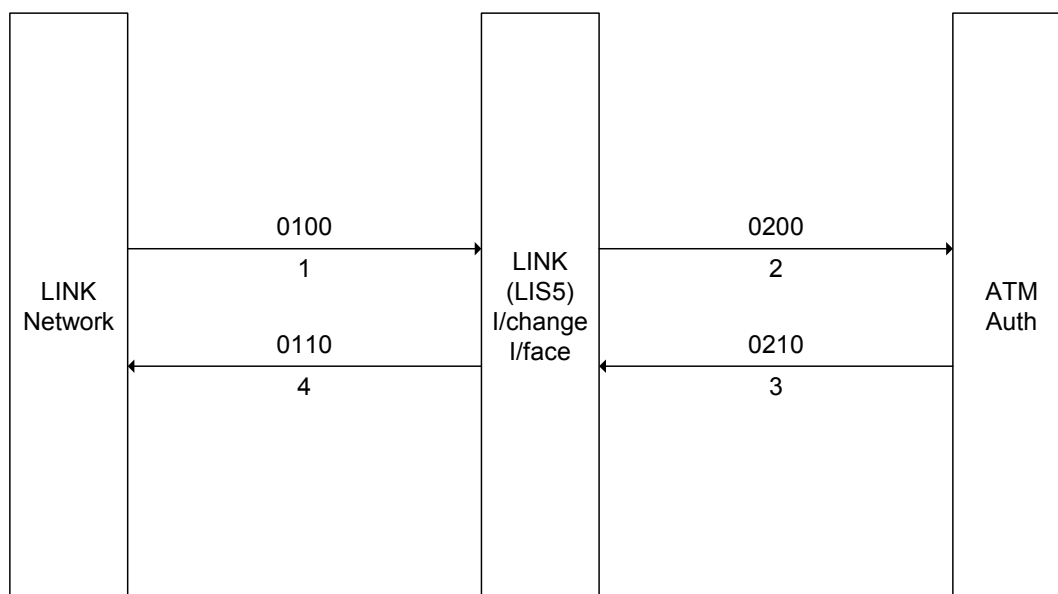


Figure 5-6 Account Balance Request Message from LINK

1. The LINK (LIS5) Interchange Interface module receives the external request message from LINK.
2. The LINK (LIS5) Interchange Interface module formats the external message from LINK into an STM type 0200, sets an inbound request timer, and forwards the request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the 0200 request message and either authorises the transaction or routes it to the issuer interface for authorisation. When authorisation is complete the BASE24-atm Authorisation process logs the transaction to the TLF. The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN). The BASE24-atm Authorisation process sends the 0210 response to the LINK (LIS5) Interchange Interface module.
4. The LINK (LIS5) Interchange Interface module deletes the inbound request timer, formats a 0110 external response message, returns the response to LINK to be completed and logs a copy of the transaction to the ILF.

5.2.2 PIN Management Request/Reversal

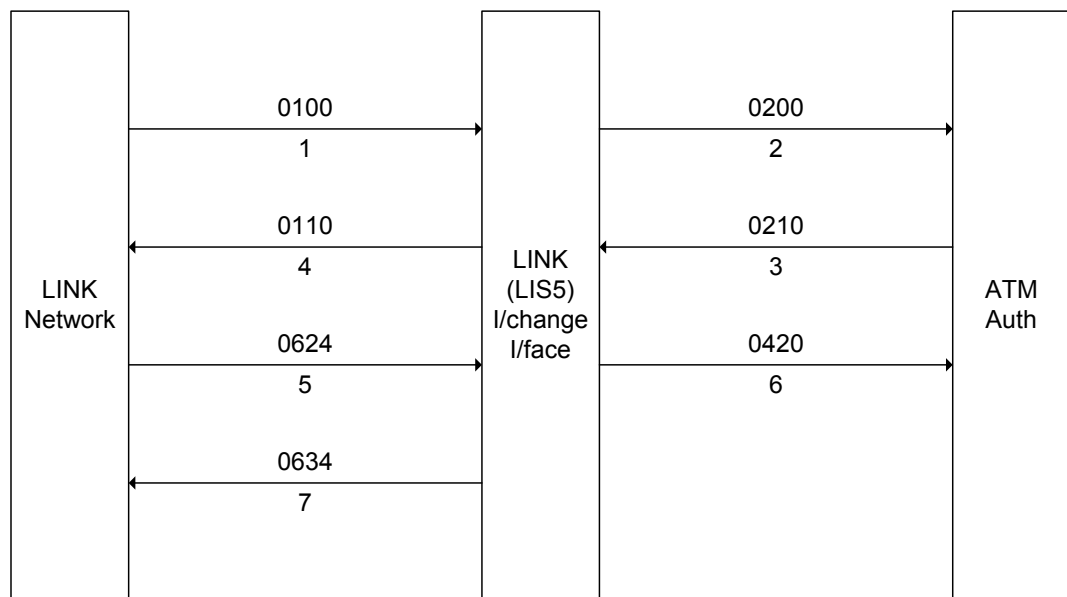


Figure 5-7 PIN Management Request Message from LINK

1. The LINK (LIS5) Interchange Interface module receives the external request message from LINK.
2. The LINK (LIS5) Interchange Interface module formats the external message from LINK into an STM type 0200, sets an inbound request timer, and forwards the request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the 0200 request message and either authorises the transaction or routes it to the issuer interface for authorisation. When authorisation is complete the BASE24-atm Authorisation process logs the transaction to the TLF. The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN). The BASE24-atm Authorisation process sends the 0210 response to the LINK (LIS5) Interchange Interface module.
4. The LINK (LIS5) Interchange Interface module deletes the inbound request timer, formats a 0110 external response message, returns the response to LINK and logs a copy of the transaction to the ILF.
5. If the transaction failed to complete as authorised, LINK returns a 0624 external message to the LINK (LIS5) Interchange Interface module.

6. The LINK (LIS5) Interchange Interface module updates the ILF record for the original transaction, formats a 0420 internal message and sends the internal message to the BASE24-atm Authorisation process.
7. The LINK (LIS5) Interchange Interface module sends a 0634 message to LINK.

5.2.3 Financial Transaction Request/Reversal

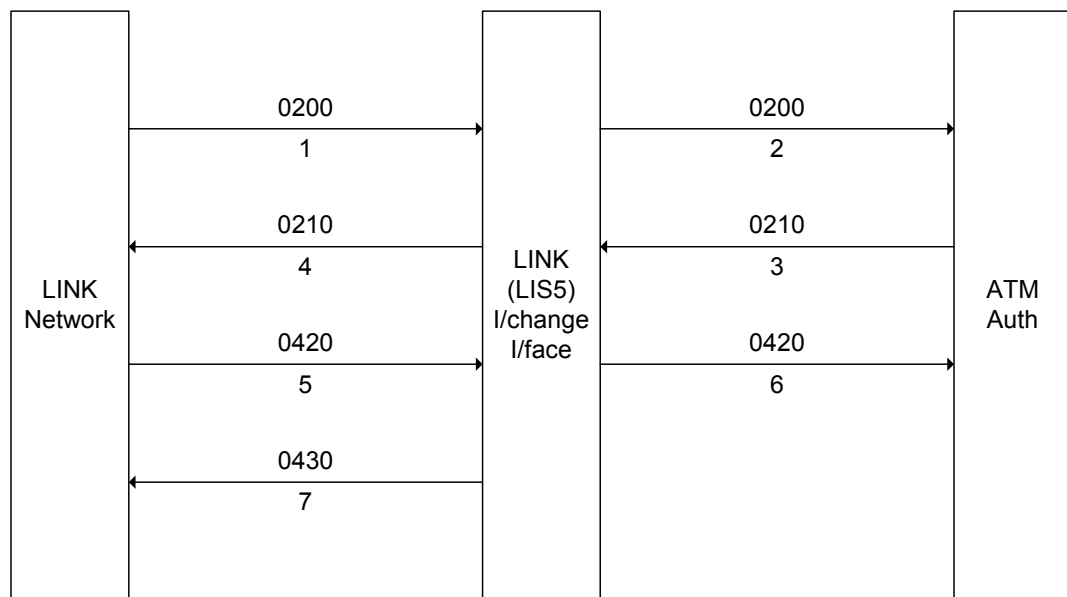


Figure 5-8 Financial Transaction Request Message from LINK

8. The LINK (LIS5) Interchange Interface module receives the external request message from LINK.
9. The LINK (LIS5) Interchange Interface module formats the external message from LINK into an STM type 0200, sets an inbound request timer, and forwards the request message to the BASE24-atm Authorisation process.
10. The BASE24-atm Authorisation process receives the 0200 request message and either authorises the transaction or routes it to the issuer interface for authorisation. When authorisation is complete the BASE24-atm Authorisation process logs the transaction to the TLF. The token information may or may not be logged to the TLF, depending on the configuration of the Token File (TKN). The BASE24-atm Authorisation process sends the 0210 response to the LINK (LIS5) Interchange Interface module.

11. The LINK (LIS5) Interchange Interface module deletes the inbound request timer, formats a 0210 external response message, returns the response to LINK and logs a copy of the transaction to the ILF. If the incoming 0200 financial request message contained a surcharge (i.e., the message contains field P-28 with a positive value), it is not permitted to return a 0210 financial response containing an issuer charge. Otherwise, if the Issuer Fee/Rebate Token is present, then the LINK (LIS5) Interchange Interface will use the value to add an issuer fee to the external response message. Alternatively, an issuer fee may be added using the DEFAULT-ISSUER-FEE value specified in the LCONF.
12. If the transaction failed to complete as authorised, LINK returns a 0420 external message to the LINK (LIS5) Interchange Interface module.
13. The LINK (LIS5) Interchange Interface module updates the ILF record for the original transaction, formats a 0420 internal message and sends the internal message to the BASE24-atm Authorisation process.
14. The LINK (LIS5) Interchange Interface module sends a 0430 message to LINK.

5.2.4 Timeout of PIN Management Request Message

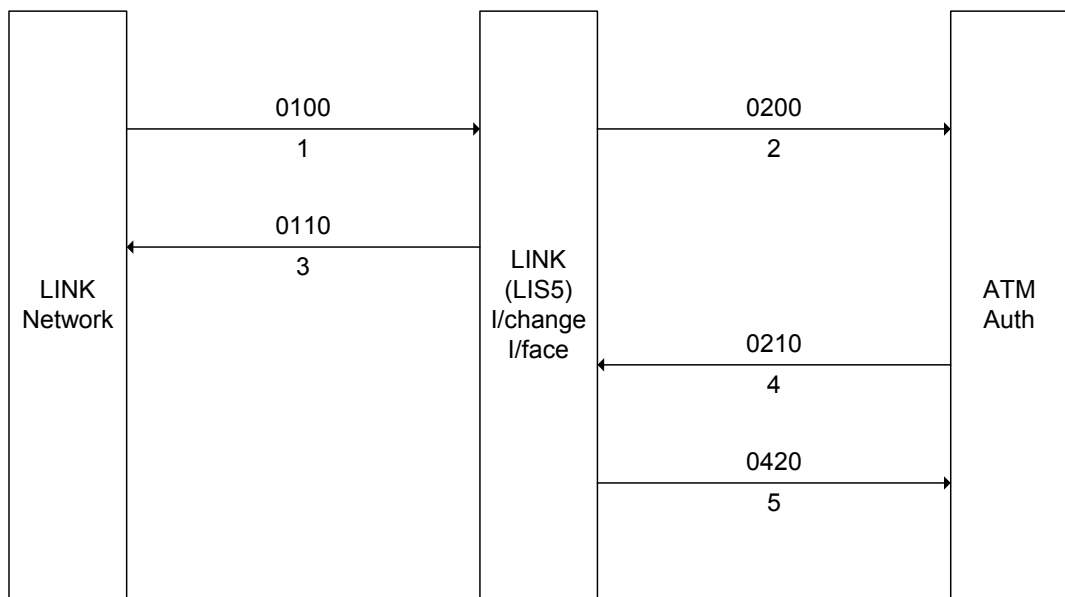


Figure 5-9 Timeout of PIN Management Request Message from LINK

1. The LINK (LIS5) Interchange Interface module receives the external request message from LINK.
2. The LINK (LIS5) Interchange Interface module formats the external message from LINK into an STM type 0200, sets an inbound request timer, and forwards the request message to the BASE24-atm Authorisation process.
3. When the inbound request timer expires, the LINK (LIS5) Interchange Interface module formats a 0110 external response message denying the transaction, sends the 0110 message to LINK and logs a copy of the transaction to the ILF.
4. The BASE24-atm Authorisation process sends a late 0210 response to the LINK (LIS5) Interchange Interface module. If the transaction was denied, the LINK (LIS5) Interchange Interface module drops the late response.
5. If the transaction was a PIN management request (message type 0200) which was approved, the LINK (LIS5) Interchange Interface module formats a 0420 internal reversal message and sends the reversal to a BASE24-atm Authorisation process.

5.2.5 Timeout of Financial Transaction Request Message

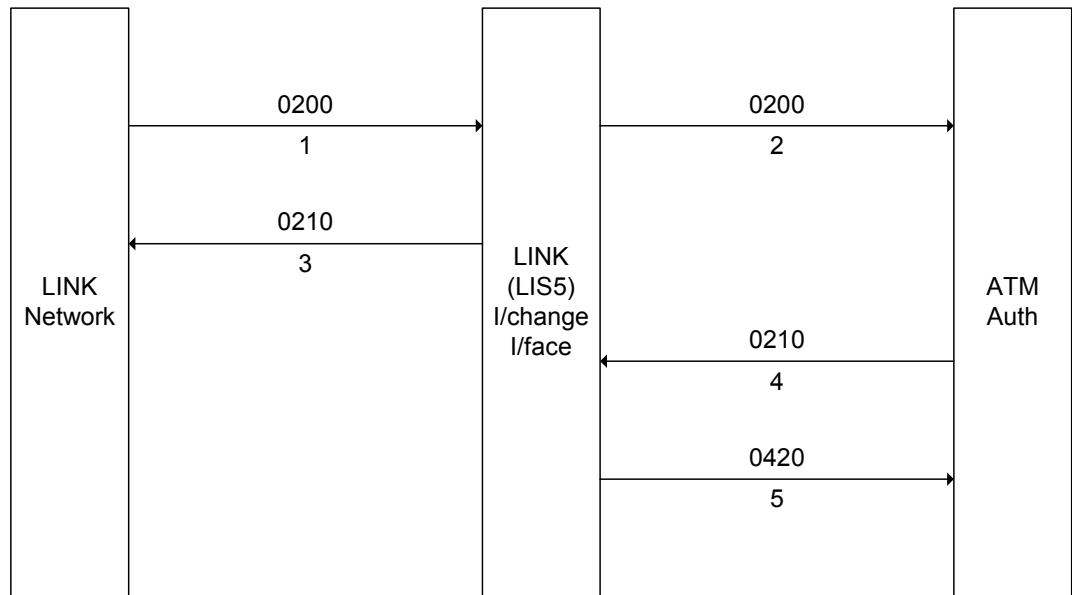


Figure 5-10 Timeout of Financial Transaction Request Message from LINK

1. The LINK (LIS5) Interchange Interface module receives the external request message from LINK.
2. The LINK (LIS5) Interchange Interface module formats the external message from LINK into an STM type 0200, sets an inbound request timer, and forwards the request message to the BASE24-atm Authorisation process.
3. When the inbound request timer expires, the LINK (LIS5) Interchange Interface module formats a 0210 external response message denying the transaction, sends the 0210 message to LINK and logs a copy of the transaction to the ILF.
4. The BASE24-atm Authorisation process sends a late 0210 response to the LINK (LIS5) Interchange Interface module. If the transaction was denied, the LINK (LIS5) Interchange Interface module drops the late response.
5. If the transaction was a financial transaction request (message type 0200) which was approved, the LINK (LIS5) Interchange Interface module formats a 0420 internal reversal message and sends the reversal to a BASE24-atm Authorisation process.

5.3 TRANSACTION MATCHING

In order to trace transactions, e.g., matching a reversal message to the corresponding response message, the LINK (LIS5) Interchange Interface uses a combination of fields from the external message. The data elements from the external message that are used to trace transactions are listed below.

- Field P-2 is the PAN
- Field P-42 is the Terminal Id
- Field P-37 is the Retrieval Reference Number
- Field P-12 is the Local Transaction Time
- Field P-13 is the Local Transaction Time

In order to match a response message to the corresponding request message, the LINK (LIS5) Interchange Interface also uses the Systems Trace Audit Number (field P-11).

5.3.1 Mobile Phone Top-Up (MPTU) Transactions

For acquired Mobile Phone Top-Up (MPTU) transactions, a flag indicating that the transaction has been acquired on BASE24 is also used to trace transactions. Customers who acquire MPTU transactions are unable to authorise the “funds” portion of the transaction for on-us cards. This is because the LINK (LIS5) Interchange Interface does not support the sending of MPTU transactions to the LINK network where the funds have already been authorised. Therefore, MPTU transactions sent to the LINK network indicate that the funds have not been authorised. Because of this, on-us transactions are subsequently routed back to the acquirer by the LINK network, in order to obtain funds authorisation. If the member uses the same LINK (LIS5) Interchange Interface process to send/receive both messages, then this flag ensures that message matching (in memory and on the ILF) is performed correctly, i.e., the flag allows the LINK (LIS5) Interchange Interface process to differentiate between the original MPTU transaction request (sent to the LINK network), and the subsequent funds authorisation request (received from the LINK network).

5.3.2 Definition of a Unique STAN

To ensure a unique Systems Trace Audit Number (STAN) is used when multiple LINK (LIS5) Interchange Interface processes exist in a single BASE24 system, two LCONF parameters need to be configured. These are LINK-START-TRACE-NUM and INCR-LINK-TRACE NUM-BY. These parameters are optional, as they have no value for a BASE24 system that operates with a single LINK (LIS5) Interchange Interface process. If these parameters are not present, then the LINK (LIS5) Interchange Interface process will assume that they have a value of 1.

The LINK-START-TRACE-NUM needs to be defined per LINK (LIS5) Interchange Interface process, and should be set to a different value for each LINK (LIS5) Interchange Interface process operating within a BASE24 system. The INCR-LINK-TRACE NUM-BY needs to be generic to all LINK (LIS5) Interchange Interface processes, and should be set to the number of LINK (LIS5) Interchange Interface processes that are configured.

The Trace Number maintained by a LINK (LIS5) Interchange Interface process is a 6-digit value sent in all messages to LINK. The first one or two digits always contain the same value for a single LINK (LIS5) Interchange Interface process, and reflect the value in the LINK-START-TRACE-NUM parameter, as follows:

- If there are less than 10 LINK (LIS5) Interchange Interface processes configured in the system, then the LINK-START-TRACE-NUM value is used to set the first digit of the Trace Number.
- If there are 10 or more LINK (LIS5) Interchange Interface processes configured in the system, then the LINK-START-TRACE-NUM value is used to set the first and second digits of the Trace Number.

The number of LINK (LIS5) Interchange Interface processes configured in the system is specified by the INCR-LINK-TRACE NUM-BY parameter. This allows the process to determine whether the LINK-START-TRACE-NUM parameter should be used to set the first 1 or 2 digits of the Trace Number.

The other 4 or 5 digits of the Trace Number comprise the value that changes, and this value is always incremented by 1. If 5 digits are used, and the value exceeds 65535, then the number is reset to the value in the LINK-START-TRACE-NUM parameter, followed by 00001. If 4 digits are used, and the value exceeds 9999, then the number is reset to the value in the LINK-START-TRACE-NUM parameter, followed by 0001.

5.4 TRANSACTION SECURITY

PIN blocks are encrypted using Triple DES in the interface with the LINK network.

The processing of PIN blocks depends on whether the BASE24 system is supporting the “full” Triple DES solution (Transaction Security Services supported) or the “interim” Triple DES solution (Transaction Security Services not supported).

5.4.1 Triple DES “Full” Solution

This processing is applicable to BASE24 Release 6.0 systems only.

The double-length Acquirer Working Key (AWK) and Issuer Working Key (IWK) are dynamically exchanged between the LINK (LIS5) Interchange Interface and the interchange. These values are stored in the TSS database, and locators to these keys are stored in the KEYF (see section 3.4).

A PIN-based transaction acquired by BASE24 and forwarded to the BASE24 LINK (LIS5) Interchange Interface contains a PIN block encrypted under a key identified by a key locator in the internal STM/PSTM message. At the LINK (LIS5) Interchange Interface, the PIN is translated from being encrypted under this key to being encrypted under the double-length AWK. The transaction is then passed to the LINK network.

For a PIN-based transaction request acquired by another LINK member and received at the BASE24 LINK (LIS5) Interchange Interface, the PIN block in the external message is encrypted under the double-length IWK. At the LINK (LIS5) Interchange Interface, the PIN and the key locator to the Inbound ZMK from the KEYF are moved to the BASE24 Standard Internal Message (STM).

Further details of the TSS are given in the ***BASE24 Transaction Security Services Processing Guide***. Appendix E, “TSS Migration”, describes the steps required to configure the BASE24 LINK (LIS5) Interchange Interface for use with TSS.

5.4.2 Triple DES “Interim” Solution

The double-length Acquirer Working Key (AWK) and Issuer Working Key (IWK) are dynamically exchanged between the LINK (LIS5) Interchange Interface and the interchange. These values are stored on screen 1 of the KEY6 (see section 3.5). A single length intermediate PIN block encryption key (KPE) is stored on screen 1 of the KEY6 file.

A PIN-based transaction acquired by BASE24 and forwarded to the BASE24 LINK (LIS5) Interchange Interface contains a PIN block encrypted under a single length key held in the internal STM/PSTM message. At the LINK (LIS5) Interchange Interface, the PIN is translated from being encrypted under this key to being encrypted under the double length Acquirer Working Key (AWK) from screen 2 of the KEY6. The transaction is then passed to the LINK network.

For a PIN-based transaction request acquired by another LINK member and received at the BASE24 LINK (LIS5) Interchange Interface, the PIN block in the external message is encrypted under the double-length Issuer Working Key (IWK) on screen 2 of the KEY6. At the LINK (LIS5) Interchange Interface, the PIN block is translated from being encrypted under this key to being encrypted under the single length intermediate PIN block encryption key (KPE) from screen 1 of the KEY6. The PIN block and the single length Intermediate PIN block encryption key (KPE) are moved to the BASE24 Standard Internal Message (STM).

5.5 TRANSACTION TYPES

This section describes specific processing performed on ATM transactions by the LINK (LIS5) Interchange Interface.

5.5.1 Acquirer Processing

When a transaction message is sent to the LINK network, the BASE24-atm transaction code is mapped to a LINK processing code, as shown below.

BASE24-atm Transaction Code		LINK Processing Code	
10	Cash Withdrawal	01	Cash Withdrawal
10	Cash Withdrawal	11	Quasi Cash (see Note 1)
10	Cash Withdrawal	59	Mobile Phone Top-Up (see Note 2)
30	Balance Enquiry	31	Balance Enquiry
81	PIN Change	90	PIN Management (PIN change)
82	EMV PIN Unblock	91	PIN Management (PIN unblock)

Note 1: A BASE24-atm cash withdrawal transaction will be mapped to a LINK Quasi Cash processing code if the NCD token is present in the STM and the Pre-pay Top-up token is not present in the STM.

Note 2: A BASE24-atm cash withdrawal transaction will be mapped to a LINK Mobile Phone Top-Up processing code if the Pre-pay Top-up token is present in the STM.

BASE24 does not generate an E-Cash Load processing code in request messages to the LINK Interchange.

The “from” and “to” account types in the LINK processing code are both set to “00”.

Other BASE24-atm transaction types may be routed to an alternate destination for authorisation, (e.g., the ATM Authorisation process).

5.5.2 Issuer Processing

When a transaction message is received from LINK (LIS5), the LINK processing code is mapped to a BASE24-atm transaction, as shown below.

LINK Processing Code		BASE24-atm Transaction Code	
00	Goods or Service Purchase (see Note 1)	10	Cash Withdrawal
01	Cash Transaction	10	Cash Withdrawal
11	Quasi Cash (see Note 2)	10	Cash Withdrawal
13	E-Cash Load	10	Cash Withdrawal

Note 1: When a transaction message is received from LINK (LIS5), the LINK "to" account type is mapped to a BASE24-atm "to" account type of '00'.

Note 2: A funds authorisation for a LINK Mobile Phone Top-Up transaction will be received with a processing code of '00', and the Pre-pay Top-up token will be added to the STM.

Note 3: A LINK Quasi Cash transaction will be mapped to a BASE24-atm cash withdrawal, and the NCD token will be added to the STM.

When a transaction message is received from LINK (LIS5), the LINK account type is mapped to a BASE24-atm account type, as shown below.

LINK Account Code	BASE24 Account Code	Description
00	00	No account type
10	11	Saving account
20	01	Cheque account
30	31	Credit account
50	60	Other account
60	00	E-Cash load/default account
61	00	E-Cash load/default account
80	00	Pre-pay/default account

5.6 SURCHARGING AND PRE-NOTIFICATION

LIS5 provides Acquirers with the ability to surcharge transactions. Acquirers may also provide cardholders with an “incentive”, i.e., a negative surcharge, to use their ATMs.

LIS5 also provides Issuers with the ability to inform cardholders of an issuer charge or an issuer rebate. This is known as issuer pre-notification.

Within BASE24, the support provided for LINK’s requirements is known as Surcharge Interactive Response Notification.

5.6.1 Acquirer Surcharging

LINK has defined Field P-28, Amount, Transaction Fee, to be used for the surcharge/incentive amount in financial request, reversal and repeat reversal message types. The field is conditional and is only used when a surcharge is applied.

Field P-28 carries a sign, which can be set to ‘D’ (debit) or ‘C’ (credit). The LINK implementation of surcharging uses ‘D’ for a normal surcharge (value greater than 0) and ‘C’ for an incentive (value less than or equal to 0). The sign is changed to ‘C’ and ‘D’ respectively for a reversal (for value not equal to 0 only).

BASE24 carries surcharges in the Surcharge Data Token (Token ID 25). The surcharge amount is held in TRAN-FEE, which is a signed, numeric field. For a full description of the Surcharge Data Token, refer to the **BASE24 Tokens Manual**.

For BASE24-atm Authorisation to assess a surcharge, it must find a record in the Surcharge File (SURF) containing fee information. The SURF contains one record for each card group targeted by a terminal group for transaction acquirer fees. Acquirer fees (or surcharges) are assessed by an institution for particular transactions initiated at selected ATMs.

Card groups are separated according to transaction routing method, either Card Prefix File (CPF) or SPROUTE. Basic configuration of surcharges is driven by the relationship between terminal groups (i.e., TDF Surcharge Profile IDs) and card groups (i.e., issuers). Within this relationship, specific card types, transaction types, account types and surcharge calculation methods are used to determine the surcharge amount. For a full description of the SURF, refer to the **BASE24 Base Files Maintenance Manual**.

5.6.2 Issuer Pre-Notification

LINK has defined Field P-30, Amount, Transaction Processing Fee, to be used for the pre-notification of issuer charges. This field is transmitted in financial transaction response messages from issuers, and contains the 'issuer charge or issuer rebate' to be displayed on the acquirer's ATM screen. Field P-30 may be absent in financial transaction responses for non-debit card transactions, indicating an "unknown" issuer fee.

Field P-30 carries a sign, which can be set to 'D' (debit) or 'C' (credit). The BASE24 LINK implementation of issuer charging uses 'D' for normal issuer charges (value greater than 0) and 'C' for issuer rebates (value less than or equal to 0). This field is not changed for reversal processing.

BASE24 carries issuer fees/rebates in the Issuer Fee/Rebate Token (Token ID 30). The issuer fee/rebate amount is held in TRAN-FEE, which is a signed, numeric field. For a full description of the Issuer Fee/Rebate Token, refer to the **BASE24 Tokens Manual**.

A process-specific LCONF parameter (DEFAULT-ISSUER-FEE) may be configured to specify a default issuer fee value. If present, the parameter must contain an eight-digit numeric value. A second process-specific LCONF parameter (ISSUER-FEE-ACCT-TYPES) may be configured to allow issuers to specify the account type(s) to which an issuer fee should be applied. The parameter may contain up to 20 account types, each containing two numeric digits, and delimited by a semi-colon (;). The LINK (LIS5) Interchange Interface reads the parameters on initialisation and warmboot. If either parameter is not present, or is invalid, then the Interface will assume a default issuer fee of zero.

For response messages sent to the LINK network, the LINK (LIS5) Interchange Interface checks for the presence of the Issuer Fee/Rebate token.

- If the token is present, then the value in the Issuer Fee/Rebate token will be used to populate field P-30 in the 0210 response message sent to the LINK Interchange as long as either or both of the following conditions are satisfied:

The Issuer Fee/Rebate token contains a negative value;

The Surcharge Data Token is not present, or is present but contains zeros or a negative value.

- If the token is not present, then the default issuer fee value will be used to populate field P-30 in the 0210 response message sent to the LINK Interchange as long as both of the following conditions are satisfied:

The Surcharge Data Token is not present, or is present but contains zeros or a negative value;

The SAVE-ACCT.FROM-ACCT-TYP field in the STM matches one of the account types specified in the new LCONF parameter.

Otherwise, field P-30 in the 0210 response message sent to the LINK Interchange will be set to zeroes.

Note that the value of any Issuer Fee/Rebate will not be displayed on the Transaction Log File (TLF) perusal screen.

Although it is possible to configure the application of an issuer charge by account type, there is no facility for varying the value of the issuer charge. It is not possible for issuers to configure negative issuer fees (rebates) on their BASE24-atm systems.

5.7 INTERCHANGE LOG FILE (ILF) PROCESSING

Transaction messages handled by the LINK (LIS5) Interchange Interface are logged to the ILF.

5.7.1 ILF Record Structure

The transactions logged to the ILF by the LINK (LIS5) Interchange Interface follow a standard structure that contains a fixed length portion (818 bytes), followed in the case of a timeout by the external message, followed by any tokens configured for logging to the ILF.

Each ILF record has the following format:

Bytes 1–55	Primary Key
Bytes 56–98	ICF Record Information
Bytes 99–150	External Message Information
Bytes 151–818	STM Message Data
Bytes 819–4072	External Message and token data (variable length depending on the token data)

The primary key for the LINK (LIS5) ILF contains the following information:

- Field P-12 is the Local Transaction Time
- Field P-2 is the PAN
- Field P-37 is the Retrieval Reference Number
- Field P-13 is the Local Transaction Date
- Field P-42 is the Terminal Id

The LINK (LIS5) Interchange Interface provides users with the capability of identifying when the ILF is becoming full. The LINK (LIS5) Interchange Interface maintains a counter of records written to the ILF. Each time the counter exceeds the value of the LCONF parameter ILF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter ILF-NOTIFY-PERCENTAGE, a warning message is logged.

5.7.2 Data Suppression

The Payment Card Industry Data Security Standards (PCI:DSS) do not allow institutions to store sensitive transaction data on their systems. This includes the discretionary data from the card's magnetic stripe and/or chip (Track-2). This requirement affects LINK members, and therefore impacts the LINK (LIS5) Interchange Interface.

Acquirers can configure the LINK (LIS5) Interchange Interface to ensure that this data is suppressed, i.e., not logged to the ILF, using the LCONF parameter SWI-ACQ-DISCR-DATA-SUPPRS.

Issuers can configure the LINK (LIS5) Interchange Interface to ensure that this data is suppressed, i.e., not logged to the ILF, using the LCONF parameter SWI-ISS-DISCR-DATA-SUPPRS.

Note that issuers should suppress the Track-2 discretionary data only if the LINK (LIS5) Interchange Interface is running in a BASE24 Release 6.0 system. Earlier releases of BASE24-atm use the discretionary data in reversal message processing, and the track data is retrieved from the ILF when the LINK (LIS5) Interchange Interface receives an external reversal advice.

5.8 STORE-AND-FORWARD (SAF) PROCESSING

The following is a description of the Store-and-Forward (SAF) processing performed by the LINK (LIS5) Interchange Interface. Only reversal advices and authorisation advices are placed on the SAF.

1. If a reversal or force post transaction (advice) is received from BASE24, an external message will be formatted and placed on the LINK (LIS5) Interchange Interface SAF for processing. In addition, if a late authorisation response is received from LINK, then an external reversal advice will also be formatted and placed on the LINK (LIS5) SAF for processing.
2. When the LINK (LIS5) Interchange Interface determines that there are transactions on the SAF (the SAF counter is greater than zero), the LINK (LIS5) Interchange Interface will send an external reversal advice or force post (advice) transaction to LINK. The ICF SAF timer is set in the LINK (LIS5) Interchange Interface for an acknowledgment to the transaction.
3. The LINK network will process the reversal advice or force post (advice) transaction and send an acknowledgment message to the BASE24 LINK (LIS5) Interchange Interface. The LINK (LIS5) Interchange Interface process will delete the transaction from the SAF.
4. If an acknowledgment is not received before the SAF acknowledgment timer expires at the BASE24 LINK (LIS5) Interchange Interface, the Interface will attempt to send the transaction again as a repeat advice (message type 0421) until an acknowledgment is received or until the maximum number of SAF retries has been reached (specified in the ICF).

The LINK (LIS5) Interchange Interface provides users with the capability of identifying when the SAF is becoming full. The LINK (LIS5) Interchange Interface maintains a counter of records written to the SAF. Each time the counter exceeds the value of the LCONF parameter SAF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter SAF-NOTIFY-PERCENTAGE, a warning message is logged.

5.9 TOKEN USAGE

The LINK (LIS5) Interchange Interface supports the following BASE24 tokens:

- Generic Switch token (Token Ids = B0 and B1)
- Original Currency token (Token Id = BE)
- Surcharge token (Token Id = 25)
- Issuer Fee/Rebate token (Token Id = 30)
- ATM 5.0 token (Token Id = 03)
- ATM Flag1 token (Token Id = 24)
- ATM Data1 token (Token Id = AG)
- Non-Currency Dispense token (Token Id = A5)
- Pre-pay Receipt token (Token Id = BF)
- Pre-pay Top-up token (Token Id = BI)
- PIN Change token (Token Id = 06)

These tokens are discussed below.

5.9.1 Generic BASE24 Switch Token

The generic switch token is a variable-length token definition that can be used by any BASE24 interchange interface. The LINK (LIS5) Interchange Interface supports the generic switch token.

Two token IDs are available for the switch token:

- B0 (Acquirer Switch token)
- B1 (Issuer Switch token)

For the LINK (LIS5) Interchange Interface, the generic switch token includes the following data:

- Processing Code (Field P-3)
- Systems Trace Audit Number (Field P-11)
- POS Entry Mode (Field P-22)
- POS Condition Code (Field P-25)
- Forwarding Institution ID (Field P-33)
- Response Code (Field P-39)
- Reversal Reason Code (Field P-60)
- Point of Service Data (Field P-61)
- Issuer Trace ID (Field S-126)

For requests, advices and advice repeats received from LINK, the interface adds the Acquirer Switch token (B0).

For reversal advices and reversal advice repeats received from LINK, the interface checks for the presence of the Acquirer Switch token (B0) in the ILF. If the token exists, the interface updates the token; otherwise the interface will add the token.

For responses received from LINK, the interface adds the Issuer Switch token (B1).

For information on the LINK-specific definition of the Generic Switch token, refer to Appendix B.

5.9.2 Original Currency Token

Card management systems such as CMS require certain information that is not available in standard BASE24 transaction log file records. In particular, for transactions where currency conversion occurs, details of the currency in which the transaction was acquired are not written to the Transaction Log File (TLF).

Transactions acquired in currencies other than sterling may be forwarded to issuers via the LINK network. This includes Foreign Currency Dispense (FCD) transactions. The message sent to the issuer contains the amount in both the transaction currency (non-sterling) and the cardholder billing currency (sterling). The LINK (LIS5) Interchange Interface creates the Original Currency token on transaction messages received from the LINK network, where the transaction currency is different to the cardholder billing currency.

5.9.3 Surcharge Token

This token is used to pass an acquirer surcharge or an acquirer incentive between the LINK (LIS5) Interchange Interface and the rest of the BASE24-atm system. This token is also used to pass the acquirer commission in Foreign Currency Dispense (FCD) transactions from the LINK (LIS5) Interchange Interface to the rest of the BASE24-atm system.

5.9.4 Issuer Fee/Rebate Token

This token is used to pass an issuer fee or an issuer rebate between the LINK (LIS5) Interchange Interface and the rest of the BASE24-atm system.

5.9.5 ATM 5.0 Token

This token is used to pass assorted data items specific to ATM processing. The data items used by the LINK (LIS5) Interchange Interface are listed below.

- Error Flag
- Card Verify Flag
- Complete Track Data Flag

5.9.6 ATM Flag1 Token

This token is used to pass assorted data items specific to ATM processing. The data items used by the LINK (LIS5) Interchange Interface are listed below.

- Cash Deposit Flag

The LINK (LIS5) Interchange Interface uses this field to differentiate between a cash-only deposit and a cash/cheque or cheque-only deposit.

5.9.7 ATM Data1 Token

This token is used to pass assorted data items specific to ATM processing. The LINK (LIS5) Interchange Interface uses the Terminal Location Indicator to identify whether the transaction originates from a counter terminal in a bank branch.

5.9.8 Non-Currency Dispense Token

This token is used to pass information on a Quasi Cash transaction between the LINK (LIS5) Interchange Interface and the rest of the BASE24-atm system.

5.9.9 Pre-pay Receipt token

This token is used to pass information to be printed on the customer receipt in a Mobile Phone Top-Up transaction from the LINK (LIS5) Interchange Interface to the rest of the BASE24-atm system.

For the LINK (LIS5) Interchange Interface, the Pre-pay Receipt token includes the following datum:

- Generic Message (field S-121, sub-field 002) in field OPER-MSG of the token

5.9.10 Pre-pay Top-up token

This token is used to pass information on a Mobile Phone Top-Up transaction between the LINK (LIS5) Interchange Interface and the rest of the BASE24-atm system.

For the LINK (LIS5) Interchange Interface, the Pre-pay Top-up token includes the following data:

- Response Code (field P-39) in field RESP-CDE of the token
- Issuer Transaction ID (field S-121, sub-field 003) in field REF of the token
- Issuer Account Number (field S-121, sub-field 004) in field PHN-NUM of the token
- Product ID (field S-123, sub-field 002). Pre-pay product ids are assumed to be of form '300n'. Byte 8 of field TRK2 of the token is used to carry the value of 'n'
- Pre-pay Account Number (field S-123, sub-field 007) in bytes 9 to 19 of TRK2 of the token
- Service Provider ID (field S-123, sub-field 015) in bytes 2 to 7 of TRK2 of the token.

5.9.11 PIN Change token

This token is used to carry information about the new PIN requested by a customer through a BASE24 system.

An acquirer LINK (LIS5) Interchange Interface populates the New PIN (field S-123, sub-field 013) from NEW-PIN-1 of the token.

An issuer LINK (LIS5) Interchange Interface populates the PIN change token with the following data:

- '1' in NEW-PIN-FRMT indicating ANSI X9.8 PIN block format
- '1' in PIN-CNT of the token
- '16' in NEW-PIN-SIZE of the token indicating an encrypted PIN
- New PIN (field S-123, sub-field 013) in NEW-PIN-1 of the token.

5.10 RESPONSE MESSAGE PROCESSING

This section describes the processing performed by the LINK (LIS5) Interchange Interface on response messages.

5.10.1 Response Code Translation

When sending a response message to the LINK network, BASE24 maps the BASE24 internal response code to an ISO response code supported by the LINK network. When receiving a response message from the LINK network, BASE24 maps the ISO response code used by the LINK network to a BASE24 internal response code.

For a complete description of response code mappings performed by the LINK (LIS5) Interchange Interface, refer to Appendix A.

5.10.2 Response Code “85”

A response code of ‘85’ must be used for approved non-financial transactions, i.e., balance enquiries and PIN management transactions. However, a response code of ‘85’ in field 039 may not be returned by issuers until all acquirers have migrated to LIS5 2008-1.

The LCONF parameter LINK-ACQUIRERS-MIGRATED indicates whether all acquirers have migrated to LIS5 2008-1. This allows issuers to configure whether or not to use response code of ‘85’ in approved non-financial response messages sent to LINK.

5.11 REVERSAL MESSAGE PROCESSING

This section describes the processing performed by the LINK (LIS5) Interchange Interface on reversal messages.

5.11.1 Reversal Code Translation

When sending a reversal message to LINK, BASE24 maps the BASE24 reversal code to a reversal reason code supported by the LINK network, and places the value in field P-60. When receiving a reversal message from LINK, BASE24 retrieves the reversal reason code used by the LINK network from field P-60, and maps the value to a BASE24 internal reversal code.

5.11.2 Track-2 Data

Field P-35 (Track-2 Data) must not be included in reversal messages sent to the LINK network. However, some issuers still expect to receive Track-2 data in reversal messages, so acquirers will continue to send field P-35 until all issuers have migrated to LIS5 2008-1.

The LCONF parameter LINK-ISSUERS-MIGRATED indicates whether all issuers have migrated to LIS5 2008-1. This allows acquirers to configure whether or not to include field P-35 in reversal messages sent to LINK.

6. CUTOVER AND REPORTING

Within the BASE24 system, interchange cutover processing is a transaction management and reporting function that produces reports for each business day. These reports can be used in conjunction with BASE24-atm reports to reconcile the differences between the BASE24 processing day and the LINK network processing day.

6.1 SETTLEMENT WITH LINK

Settlement with LINK is performed offline from BASE24 using extracts from transaction files (e.g., the ILF).

6.2 INTERCHANGE REPORTS

The BASE24 LINK (LIS5) Interchange Interface can be configured to produce the standard BASE24 Interchange reports.

The LINK (LIS5) Interchange Interface produces two reports of interchange transaction activity:

- SETL01 Clearings Statement Report
- STAT09 Transaction Detail Report

For details of the standard BASE24 Interchange reports, refer to the ***BASE24-atm Settlement and Reporting Manual***.

6.2.1 SETL01 Report

The Clearings Statement Report for the LINK (LIS5) Interchange Interface reflects the debit, credit, and net position of the LINK network to BASE24.

This report contains the totals of all transactions logged to the ILF. The LINK (LIS5) Interchange Interface calculates totals as an acquirer and as an issuer.

6.2.2 STAT09 Report

The Transaction Detail Report for the LINK (LIS5) Interchange Interface provides a complete listing of all interchange activity. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the LINK (LIS5) Interchange Interface during its processing day. The information provided for each transaction includes the cardholder number, transaction type, transaction time, posting date, and all transaction amounts. This report also includes a summary of the transaction detail information contained in the report.

To comply with the requirements of the PCI data security standards, some or all of the cardholder number in the report may be displayed masked with asterisks (*). The setting of the DATA MASK FLAG on Screen 2 of the ICF record determines whether the data in this field is masked. If masking is enabled for the interface process, then the values of the other three report data masking parameters on ICF Screen 2 determine which digits are masked.

6.2.3 Configuration

The support of Interchange reports for the LINK (LIS5) Interchange Interface process is not mandatory. If it is required to produce the standard BASE24 Interchange reports, then they can be defined in the Logical Network Configuration File (LCONF) and produced for the LINK (LIS5) Interchange Interface process.

The LCONF assigns required to specify the settlement report production are:

- RPT
- RPT-ERROR-FILE
- SW-RPT01-PROG-NAME
- SW-RPT09-PROG-NAME
- SW-RPT-OBEY-FILE

The LCONF parameter required for settlement report production is:

- SW-RPT-STARTUP

The LCONF parameters also used in settlement report production are:

- ACCT-NUM-TYP
- DATE-DISPLAY-TYPE

For a detailed explanation of the above assigns and parameters, refer to section 3.2.

6.2.4 Starting Reports

The LINK (LIS5) Interchange Interface is responsible for starting the interface reports. It does so by creating an obey file to start the report programs and then running the obey file automatically at the appropriate time.

Using the time specified in the LCONF param SW-RPT-STARTUP, the LINK (LIS5) Interchange Interface sets a timer to expire when the reports are to be run. When the timer expires, the LINK (LIS5) Interchange Interface creates and runs an obey file containing the commands necessary to run the report programs using Tandem Advanced Command Language (TACL).

Under normal circumstances, the report programs start and stop without operator intervention. However, if the LINK (LIS5) Interchange Interface does not start the programs, they can be started manually using TACL by running the report obey file created by the LINK (LIS5) Interchange Interface.

If the report programs abend or fail, they can be restarted in the same way.

7. EMV SUPPORT

The BASE24 LINK (LIS5) EMV module is an additional licensable module, built on top of the Switch 6.0 version of the LINK (LIS5) Interchange Interface.

The source files of the standard LINK (LIS5) Interchange Interface include a number of EMV-specific ‘hooks’ or user exits. These provide an interface between the standard LINK (LIS5) Interchange Interface and the LINK EMV modules. The LINK (LIS5) EMV modules are bound together with the standard LINK (LIS5) Interchange Interface code to provide support for LINK (LIS5) EMV transactions, in addition to non-EMV transactions.

For details of the BASE24 LINK (LIS5) EMV modules, refer to the ***BASE24-atm EMV Support Manual***.

The Release 6.0 LINK (LIS5) Interchange Interface is NOT backwards compatible with the Release 5.1.2 ICC product, i.e., it is not possible to bind the Release 6.0 interface with the Release 5.1.2 LINK (LIS5) ICC extension module.

7.1 LIS5 MESSAGE

For EMV support, fields used in magnetic stripe transaction processing may contain different values, and fields not used in magnetic stripe transaction processing (typically using the tertiary data area of the ISO 8583 message) contain EMV-specific data.

7.1.1 Fields Containing New Values for EMV Transactions

Fields used in magnetic stripe transaction processing contain different values to indicate that a transaction is being performed at a chip-capable terminal. However, chip data is included in authorisation messages only if the terminal has successfully read the chip.

For EMV transactions, the following new values in existing fields are supported:

- Value ‘05’ (chip-read) is supported for digits 1 and 2 of the Point of Service Entry Mode (field P-22).
- Values ‘55’ (ICC-capable Branch ATM) and ‘57’ (ICC-capable Remote ATM) are supported for the Point of Service Condition Code (field P-25).
- Values ‘2’ (chip card – “fallback” transaction) and ‘1’ (chip card – not a “fallback” transaction) are supported for the “Status of Last Chip Attempt” sub-field of the Advice/Reversal Reason Code (field P-60.3).

7.1.2 Fields Containing New Values for EMV Transactions

Where a transaction is performed using a chip card and additional chip data is included in authorisation messages, extra fields are used in the LIS5 message.

The third (tertiary) bit map contains the following data elements required for the support of EMV transactions:

Bit	Name
130	Terminal Capability Profile
131	Terminal Verification Results
132	Unpredictable Number
133	Terminal Serial Number
134	Issuer Application Data
136	Cryptogram
137	Application Transaction Counter
138	Application Interchange Profile
139	Issuer Authentication Data
142	Issuer Script
144	Cryptogram Transaction Type
145	Terminal Country Code
146	Terminal Transaction Date
147	Cryptogram Amount
148	Cryptogram Currency Code

In addition:

- Card Sequence Number (field P-23) must be supported.
- Cryptogram Information Data (field P-60.4) must be supported.

7.2 EMV TOKENS

EMV data processed by the BASE24 LINK EMV module is passed within BASE24 in the form of data tokens appended to the STM. The following data tokens are used, dependent on the message type:

- **EMV Status Token (Token Id = 'B4')**. This token contains data generated by the transaction acquirer and the acquiring interface. This data is used in authorisation processing for both EMV and non-EMV transactions, initiated at EMV-capable terminals.
- **EMV Request Token (Token Id = 'B2')**. This token contains EMV data stored on the chip and generated by the terminal. This data is mandatory for EMV transactions, and is used in EMV transaction processing.
- **EMV Discretionary Token (Token Id = 'B3')**. This token contains EMV data stored on the chip and generated by the terminal. This data is optional for EMV transactions, and is not used in EMV transaction processing.
- **EMV Response Token (Token Id = 'B5')**. This token contains EMV data generated by the card issuer. The EMV card uses this data to authenticate the card issuer.
- **EMV Script Token (Token Id = 'B6')**. This token contains EMV data generated by the card issuer. The card issuer uses this data to re-configure the EMV card.
- **EMV Issuer Script Results Token (Token Id = 'BJ')**. This token contains information about the processing of EMV Script data. The card issuer uses this information to determine the success or otherwise of an EMV Script sent to an EMV card.

7.3 BASE24 ACQUIRER PROCESSING

Where BASE24 is acting as the transaction acquirer (i.e., the card is issued by another LINK member), the LINK EMV Module performs the processing described below.

7.3.1 Identifying EMV Transactions

There are, essentially, four levels of transaction support provided by the EMV product, based on the terminal capability, how the card details were read, and whether chip data elements are included in the transaction message.

On a request sent to the LINK Interchange, the four levels can be identified as described in the first three columns of the table below. Each level results in different processing requirements.

Terminal Capability	Card Details Entry	Chip Data Elements
EMV Status Token (indicating EMV-capable terminal) is not present (i.e., not EMV-capable).	Not applicable	Not applicable
EMV Status Token (indicating EMV-capable terminal) is present (i.e., EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is not set to '05x' (i.e., not chip-read).	Not applicable
EMV Status Token (indicating EMV-capable terminal) is present (i.e., EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is set to '05x' (i.e., chip-read).	EMV Request Token (indicating an EMV transaction) is not present (i.e., not an EMV transaction).
EMV Status Token (indicating EMV-capable terminal) is present (i.e., EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is set to '05x' (i.e., chip-read).	EMV Request Token (indicating an EMV transaction) is present (i.e., an EMV transaction).

7.3.2 Data Mapping

On a request sent to the LINK Interchange (an outbound message) and a response received from the LINK Interchange (an inbound message), EMV data is mapped as described below.

7.3.2.1 Outbound Messages

Where the transaction does not originate from an EMV-capable terminal (i.e., the first level of transaction support defined in the table in section 7.3.1), no additional EMV processing is performed.

Where the transaction does originate from an EMV-capable terminal, but the card details have not been read from the chip (i.e., the second level of transaction support defined in the table in section 7.3.1), data is mapped to the LIS5 message as shown in the table below.

SEM Field Name	Field Value
P-25 PSEM.PT-TRAN-SPCL-CDE	Move “55” into the SEM field, if the SEM field currently contains “54”. Move “57” into the SEM field, if the SEM field currently contains “56”.
P-60.3 PSEM.PRI-RSRVD1-PRVT.LINK. LAST-CHIP-STATUS	If the LAST-EMV-STAT field in the EMV Status Token contains “1” (i.e., an EMV card), then move “2” (i.e., a “fallback” transaction) into the SEM field. Otherwise, do not add P-60.3 to the SEM.
P-60.4 PSEM.PRI-RSRVD1-PRVT.LINK. CRYPTO-INFO-DATA	If this is a “fallback” transaction, set the SEM field to “C0” indicating “No Cryptogram Information Data Supplied”. Otherwise, do not add P-60.4 to the SEM.

Where the transaction does originate from an EMV-capable terminal and the card details have been read from the chip, but EMV-specific data is not included in the authorisation message (i.e., the third level of transaction support defined in the table in section 7.3.1), the transaction will be declined. The LINK Interchange does not permit this level of transaction support from a LIS5 acquirer – if the card details are read from the chip, then the EMV-specific data must be included.

Where the transaction contains full EMV data (i.e., the fourth level of transaction support defined in the table in section 7.3.1), data is mapped to the LIS5 message as shown in the table below.

SEM Field Name	Field Value
P-22 PSEM.ENTRY-MDE	When the first 2 bytes of PT-SRV-ENTRY-MDE field in the EMV Status Token are “05” and the DATA-SUSPECT field in the EMV Status Token is “1” then set the first 2 bytes of the SEM field to “95”. When the first 2 bytes of PT-SRV-ENTRY-MDE field in the EMV Status Token are “05” and the DATA-SUSPECT field in the EMV Status Token is “0” then set the first 2 bytes of the SEM field to “05”. Move the third byte of the PT-SRV-ENTRY-MDE into the third byte of the PSEM.ENTRY-MDE.
P-23 PSEM.MBR-NUM	Move “0” followed by the APPL-PAN-SEQ-NUM field in the EMV Status Token into the SEM field.
P-25 PSEM.PT-TRAN-SPCL-CDE	Move “55” into the SEM field, if the SEM field currently contains “54”. Move “57” into the SEM field, if the SEM field currently contains “56”.
P-60.3 PSEM.PRI-RSRVD1-PRVT.LINK. LAST-CHIP-STATUS	Move “1” (i.e., an EMV card) into the SEM field.
P-60.4 PSEM.PRI-RSRVD1-PRVT.LINK. CRYPTO-INFO-DATA	If position 2 of the request token bit map is set, convert the CRYPTO-INFO-DATA field from binary to ASCII before placing in the SEM field; otherwise set the SEM field to “80”.
T-130 TSEM.EMV-TERM-CAP	If position 2 of the discretionary token bit map is set, convert ICC-TERM-CAP from binary to ASCII. Move the left-hand 6 bytes into the SEM field, and set the tertiary bit map for Field 130 to true.
T-131 TSEM.ICC-TVR	If position 3 of the request token bit map is set, convert the TVR field from binary to ASCII before placing in the SEM field, and set the tertiary bit map for Field 131 to true.
T-132 TSEM.ICC-UNPREDICT-NUM	If position 13 of the request token bit map is set, convert the UNPREDICT-NUM field from binary to ASCII before placing in the SEM field, and set the tertiary bit map for Field 132 to true.
T-133 TSEM.ICC-TERM-SERIAL-NO	If position 1 of the discretionary token bit map is set, move TERM-SERL-NUM into the SEM field, and set the tertiary bit map for Field 133 to true.
T-134 TSEM.ICC-ISS-APP.LEN TSEM.ICC-ISS-APP.DATA	If position 16 of the request token bit map is set: Multiply the ISS-APPL-DATA-LGTH field by 2 and then convert from binary to ASCII before placing in the SEM field. Convert the ISS-APPL-DATA from binary to ASCII before placing in the SEM field. Set the tertiary bit map for Field 134 to true.

SEM Field Name	Field Value
T-136 TSEM.ICC-CRYPTOGRAM	If position 4 of the request token bit map is set, convert the ARQC field from binary to ASCII before placing in the SEM field, and set the tertiary bit map for Field 136 to true.
T-137 TSEM.ICC-ATC	If position 8 of the request token bit map is set, convert the ATC field from binary to ASCII before placing in the SEM field, and set the tertiary bit map for Field 137 to true.
T-138 TSEM.ICC-AIP	If position 7 of the request token bit map is set, convert the AIP field from binary to ASCII before placing in the SEM field, and set the tertiary bit map for Field 138 to true.
T-144 TSEM.ICC-TRAN-TYPE	If position 12 of the request token bit map is set, convert the TRAN-TYPE field from binary to ASCII before placing in the SEM field, and set the tertiary bit map for Field 144 to true.
T-145 TSEM.ICC-TERM-CNTRY-CDE	If position 9 of the request token bit map is set, convert the TERM-CNTRY-CDE field from binary to ASCII. The result from the conversion should be 4 bytes and the 3 right-hand side bytes should be placed in the SEM field. Set the tertiary bit map for Field 145 to true.
T-146 TSEM.ICC-TRAN-DAT	If position 11 of the request token bit map is set, convert the TRAN-DAT field from binary to ASCII before placing in the SEM field, and set the tertiary bit map for Field 146 to true.
T-147 TSEM.ICC-AMT-AUTH	If position 5 of the request token bit map is set, convert the AMT-AUTH field from binary to ASCII before placing in the SEM field, and set the tertiary bit map for Field 147 to true.
T-148 TSEM.ICC-AMT-CRNCY-CDE	If position 10 of the request token bit map is set convert the TRAN-CRNCY-CDE field from binary to ASCII. The result from the conversion should be 4 bytes and the 3 right-hand side bytes should be placed in the SEM field. Set the tertiary bit map for Field 148 to true.

Issuer Script Results are returned in PIN Management failure advices to inform the issuer of the cause of the script processing failure on the IC card. This data is mapped to the LIS5 message as shown in the table below.

SEM Field Name	Field Value
S-123.14 SSEM.SECNDRY-RSRVD2- PRVT.LINK. ISS-SCRIPT-RSLT	Default the Issuer Script Result to Script processing unknown (value '09'), Script sequence not specified (value '00') and the Issuer Script Identifier to spaces. If a script is found in the EMV-ISS-SCRIPT-RSLTS-TKN that has not succeeded (ISS-SCRIPT-PROC-RESULT is script not performed '0', script processing failed '1' or script processing unknown '9'), set the Script Processing part of the Issuer Script Result to '0' followed by ISS-SCRIPT-PROC-RESULT. If ISS-SCRIPT-SEQ is a valid hexadecimal number, set the Script Sequence part of the Issuer Script Result to the ASCII 2 digit decimal representation of the number e.g., if ISS-SCRIPT-SEQ is 'A', Script Sequence should be set to '10'. Set the Issuer Script Identifier part of the Issuer Script Result to ISS-SCRIPT-ID.

7.3.2.2 Inbound Messages

Where the transaction response includes Issuer Authentication Data (field T-139), the EMV Response Token is formatted as shown in the table below.

Response Token Field Name	Field Value
ISS-AUTH-DATA-LGTH	Convert TSEM.ICC-ARPC.LEN from ASCII to binary and divide by 2 before placing in the token field.
ISS-AUTH-DATA.ARPC	Convert TSEM.ICC-ARPC.DATA from ASCII to binary before placing in the token field. If more than 16 bytes of data have been returned from the issuer (resulting in more than 8 bytes of binary data in the token), the additional data will continue into ISS-AUTH-DATA.ADDL-DATA.
SEND-CRD-BLK	N - Do not send Card Block Script
SEND-PUT-DATA	N - Do not send PUT DATA Script

Where the transaction response includes Issuer Script Data (field T-142), the EMV Script Token is formatted as shown in the table below.

Script Token Field Name	Field Value
ISS-SCRIPT-DATA-LGTH	Convert TSEM.ICC-ISS-SCRIPT.LEN from ASCII to binary and divide by 2 before placing in the token field.
ISS-SCRIPT-DATA	Convert TSEM.ICC-ISS-SCRIPT.DATA from ASCII to binary before placing in the token field.

7.4 BASE24 ISSUER PROCESSING

Where BASE24 is acting as the card issuer (i.e., the transaction is acquired by another LINK member), the LINK EMV Module performs the processing described below.

7.4.1 Identifying EMV Transactions

There are, essentially, four levels of transaction support provided by the EMV product, based on the terminal capability, how the card details were read, and whether chip data elements are included in the transaction message.

On a request received from the LINK Interchange, the four levels can be identified as described in the first three columns of the table below. Each level results in different processing requirements.

Terminal Capability	Card Details Entry	Chip Data Elements
Field P-25 (POS Condition Code) is not '27', '55' or '57' (i.e., not EMV capable).	Not applicable	Not applicable
Field P-25 (POS Condition Code) is '27', '55' or '57' (i.e., EMV capable).	Field P-22 (POS Entry Mode) is not '05x' or '95x' (i.e., not read from chip)	Not applicable
Field P-25 (POS Condition Code) is '27', '55' or '57' (i.e., EMV capable).	Field P-22 (POS Entry Mode) is '05x' or '95x' (i.e., read from chip)	Field S-65 (Tertiary Bit Map) is not present (i.e., no tertiary data)
Field P-25 (POS Condition Code) is '27', '55' or '57' (i.e., EMV capable).	Field P-22 (POS Entry Mode) is '05x' or '95x' (i.e., read from chip)	Field S-65 (Tertiary Bit Map) is present (i.e., tertiary data)

7.4.2 Data Mapping

On a request received from the LINK Interchange (an inbound message) and a response sent to the LINK Interchange (an outbound message), EMV data is mapped as described below.

7.4.2.1 Inbound Messages

Where the transaction does not originate from an EMV-capable terminal (i.e., the first level of transaction support defined in the table in section 7.4.1), no additional EMV processing is performed.

Where the transaction does originate from an EMV-capable terminal, but the card details have not been read from the chip (i.e., the second level of transaction support defined in the table in section 7.4.1), the EMV Status Token is formatted as shown in the table below.

Status Token Field Name	Field Value
PT-SRV-ENTRY-MDE	If the first 2 characters of PSEM.ENTRY-MDE are not “90” then use this value. If the first 2 bytes of PSEM.ENTRY-MDE are “90” then set the first 2 bytes of the token field to “02”. Set the third byte of the token field to the third byte of the PSEM-ENTRY-MDE field.
TERM-ENTRY-CAP	“5” (ICC read capability)
LAST-EMV-STAT	If PSEM.PRI-RSRVD1-PRVT.LINK.LAST-CHIP-STATUS is “2” or “1” then set the token field to “1” (chip card); otherwise set it to “0” (not a chip card).
DATA-SUSPECT	Not used
APPL-PAN-SEQ-NUM	Not used
DEV-INFO	Not used
RSN-ON-LINE-CDE	Not used
ARQC-VERIFY	“0” (ARQC not verified)
USER-FLD1	Not used

Where the transaction does originate from an EMV-capable terminal and the card details have been read from the chip, but EMV-specific data is not included in the authorisation message (i.e., the third level of transaction support defined in the table in section 7.4.1), the EMV Status Token is formatted as shown in the table below.

Status Token Field Name	Field Value
PT-SRV-ENTRY-MDE	Set the first 2 bytes of the token field to “05”. Set the third byte of the token field to the third byte of the PSEM-ENTRY-MDE field.
TERM-ENTRY-CAP	“5” (ICC read capability)
LAST-EMV-STAT	“1” (Chip card).
DATA-SUSPECT	If PSEM.ENTRY-MDE is “95” then set the token field to “1” (CAM Data is unreliable); otherwise set it to “0” (CAM data is assumed correct).
APPL-PAN-SEQ-NUM	PSEM.MBR-NUM.BYTE[1] for 2 bytes
DEV-INFO	Not used
RSN-ON-LINE-CDE	Not used
ARQC-VERIFY	“0” (ARQC not verified)
USER-FLD1	Not used

Where the transaction contains full EMV data (i.e., the fourth level of transaction support defined in the table in section 7.4.1), the EMV Request Token and EMV Discretionary Token are formatted as shown in the tables below (in addition to the EMV Status Token formatting specified in the table above).

Request Token Field Name	Field Value
BIT-MAP	This is set according to the content of the subsequent fields. It follows ISO 8583 convention in that the highest order bit represents the first field in the token (following the BIT-MAP). This field is formatted according to the content of the following fields.
CRYPTO-INFO-DATA	From SEM field P-60.4 If this field is present in the SEM convert PSEM.PRI-RSRVD1-PRVT.LINK. CRYPTO-INFO-DATA from ASCII to binary before placing in the token field. If not present set the field to binary "80". Set bit map position 2 to true.
TVR	TSEM.ICC-TVR Convert this field from ASCII to binary before placing in the token field. Set bit map position 3 to true.
ARQC	TSEM.ICC-CRYPTOGRAM Convert this field from ASCII to binary before placing in the token field. Set bit map position 4 to true.
AMT-AUTH	TSEM.ICC-AMT-AUTH Convert this field from ASCII to binary before placing in the token field. Set bit map position 5 to true.
AMT-OTHER	Set to binary zeros. Set bit map position 6 to true.
AIP	TSEM.ICC-AIP Convert this field from ASCII to binary before placing in the token field. Set bit map position 7 to true.
ATC	TSEM.ICC-ATC Convert this field from ASCII to binary before placing in the token field. Set bit map position 8 to true.
TERM-CNTRY-CDE	TSEM.ICC-TERM-CNTRY-CDE Pad the left-hand side with a zero and then convert the 4 byte ASCII field to binary before placing in the token field. Set bit map position 9 to true.

Request Token Field Name	Field Value
TRAN-CRNCY-CDE	TSEM.ICC-AMT-CRNCY-CDE Pad the left-hand side with a zero and then convert the 4 byte ASCII field to binary before placing in the token field. Set bit map position 10 to true.
TRAN-DAT	TSEM.ICC-TRAN-DAT Convert this field from ASCII to binary before placing in the token field. Set bit map position 11 to true.
TRAN-TYPE	TSEM.ICC-TRAN-TYPE Convert this field from ASCII to binary before placing in the token field. Set bit map position 12 to true.
UNPREDICT-NUM	TSEM.ICC-UNPREDICT-NUM Convert this field from ASCII to binary before placing in the token field. Set bit map position 13 to true.
ISS-APPL-DATA-LGTH	TSEM.ICC-ISS-APP.LEN Convert from ASCII to binary and divide by 2 before placing in the token field.
ISS-APPL-DATA	TSEM.ICC-ISS-APP.DATA This field is used provided it is not zeros or spaces. Convert it from ASCII to binary before placing in the token field. If used, set bit map position 16 to true.

Discretionary Token Field Name	Field Value
BIT-MAP	This is set according to the content of the subsequent fields. It follows ISO 8583 convention in that the highest order bit represents the first field in the token (following the BIT-MAP). This field is formatted according to the content of the following fields.
TERM-SERL-NUM	TSEM.ICC-TERM-SERIAL-NO if present If present, set bit map position 1 to true.
EMV-TERM-CAP	TSEM.ICC-TERM-CAP if present. Pad the right-hand side with two binary zeros and then convert the 8 byte ASCII field to binary before placing in the token field. If present, set bit map position 2 to true.
EMV-TERM-TYPE	Not set
APPL-VER-NUM	Not set
CVM-RSLTS	Not set
DF-NAME-LGTH	Not set
DF-NAME	Not set

Issuer Script Results are returned in PIN Management failure advices to inform the issuer of the cause of the script processing failure on the IC card. This data is added to the EMV Issuer Script Results token, as shown in the table below.

Issuer Script Results Token Field Name	Field Value
NUM-ISS-SCRIPT-RSLTS	'1'
ISS-SCRIPT-PROC-RSLT	The second character of SSEM.SECNDRY-RSRVD2-PRVT.LINK.ISS-SCRIPT-RSLT, e.g., if the external message from LINK indicates that script processing failed ('01'), the ISS-SCRIPT-PROC-RSLT will be '1'.
ISS-SCRIPT-SEQ	The hexadecimal equivalent of the decimal value SSEM.SECNDRY-RSRVD2-PRVT.LINK.ISS-SCRIPT-SEQ
ISS-SCRIPT-ID	SSEM.SECNDRY-RSRVD2-PRVT.LINK.ISS-SCRIPT-ID

7.4.2.2 Outbound Messages

Where the transaction response includes the EMV Response Token, data is mapped to the Issuer Authentication Data (field T-139) as shown in the table below.

SEM Field Name	Field Value
TSEM.ICC-ARPC.LEN	Multiply ISS-AUTH-DATA-LGTH by 2 and then convert from binary to ASCII before placing in the SEM field.
TSEM.ICC-ARPC.DATA	Convert ISS-AUTH-DATA.ARPC and if necessary ISS-AUTH-DATA.ADDL-DATA (depending on the value of ISS-AUTH-DATA-LGTH) from binary to ASCII before placing in the SEM field. Set the tertiary bit map for Field 139 to true.

Where the transaction response includes the EMV Script Token, data is mapped to the Issuer Script Data (field T-142) as shown in the table below.

SEM Field Name	Field Value
ISOTSEM.ICC-ISS-SCRIPT.LEN	Multiply ISS-SCRIPT-DATA-LGTH by 2 and then convert from binary to ASCII before placing in the SEM field.
ISOTSEM.ICC-ISS-SCRIPT.DATA	Convert the ISS-SCRIPT-DATA from binary to ASCII before placing in the SEM field. Set the tertiary bit map for Field 142 to true.

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A. RESPONSE CODE TRANSLATION

A.1 BASE24-ATM TO LINK (LIS5) RESPONSE CODE

BASE24-atm RESPONSE CODE		LINK (LIS5) RESPONSE CODE	
00	Approved with balances	00	Approved
01	Approved - no balances	00	Approved
50	Unauthorised usage	62	Restricted card
51	Expired card	54	Expired card
52	Invalid card	14	Invalid card number
53	Invalid PIN	55	Incorrect PIN
54	Database problem	96	System malfunction
55	Ineligible transaction	12	Invalid transaction
56	Ineligible account	05	Do not honour
57	Transaction not supported	12	Invalid transaction
58	Insufficient funds no amount 3	51	Not sufficient funds
59	Insufficient funds - amount 3	51	Not sufficient funds
60	Uses limit exceeded	65	Exceeds withdrawal frequency limit
61	Withdrawal limit would be exceeded	61	Exceeds withdrawal amount limit
62	PIN tries exceeded	75	Allowable number of PIN tries exceeded
63	Withdrawal limit already reached	61	Exceeds withdrawal amount limit
64	Invalid credit card cash advance amount	13	Invalid amount
65	No statement information for the account	12	Invalid transaction
66	Statement information not available	12	Invalid transaction
67	Invalid cash back amount	13	Invalid amount
68	External decline	05	Do not honour
69	No sharing between the card issuer and terminal owner	05	Do not honour
70	System error	96	System malfunction
71	Contact card issuer	01	Refer to card issuer
72	Destination not available	91	Switch or issuer inoperative
73	Routing lookup problem	05	Do not honour
74	Message edit error	30	Format error
80	Signature check failed	05	Do not honour
81	HSM parameter error	05	Do not honour

BASE24-atm RESPONSE CODE		LINK (LIS5) RESPONSE CODE	
82	HSM failure	05	Do not honour
83	KEYI not found	05	Do not honour
84	ATC check failure	05	Do not honour
85	CVR check failure	62	Restricted card
86	TVR check failure	62	Restricted card
87	ARQC verification failure	Q1	Card authentication failure
88	Fallback transaction	Q2	Fallback not allowed

If the card disposition is 1 (Capture) and the response code is 50, the external response code is 43.

If the card disposition is 1 (Capture) and the response code is 51, the external response code is 33.

If the card disposition is 1 (Capture) and the response code is 62, the external response code is 38.

If the card disposition is 1 (Capture) and the response code is any other value, the external response code is 04.

If the response code is 70 and the error flag in the ATM 5.0 token indicates a sanity error, the external response code is 76.

If the transaction has been declined and the card verify flag in the ATM 5.0 token indicates that card verification failed, the external response code is either N2 (if the card disposition is 1) or N1 (otherwise).

Any BASE24-atm codes not listed are translated to 05 (do not honour).

A.2 LINK (LIS5) TO BASE24-ATM RESPONSE CODE

LINK (LIS5) RESPONSE CODE		BASE24-atm RESPONSE CODE	
00	Approved	01	Approved – no balances
01	Refer to card issuer	71	Contact card issuer
02	Refer to card issuer special conditions	71	Contact card issuer
03	Merchant invalid	68	External decline
04	Pick-up	68	External decline
05	Do not honour	68	External decline
07	Pick-up card special conditions	68	External decline
12	Invalid transaction	55	Ineligible transaction
13	Invalid amount	74	Message edit error
14	Invalid card number	52	Invalid card
15	No such issuer	68	External decline
19	Re-enter transaction	68	External decline
30	Format error	74	Message edit error
33	Expired card, pick-up	51	Expired card
36	Restricted card, pick-up	50	Unauthorised usage
38	Allowable PIN tries exceeded, pick-up	62	PIN tries exceeded
41	Lost card, pick up	50	Unauthorised usage
43	Stolen card, pick up	50	Unauthorised usage
51	Not sufficient funds	58	Insufficient funds no amount 3
52	No checking account	56	Ineligible account
53	No savings account	56	Ineligible account
54	Expired card	51	Expired card
55	Incorrect PIN	53	Invalid PIN
57	Transaction not permitted to cardholder	50	Unauthorised usage
61	Exceeds withdrawal amount limit	63	Withdrawal limit already reached
62	Restricted card	50	Unauthorised usage
65	Exceeds withdrawal frequency limit	60	Uses limit exceeded
66	Card acceptor - call acquirer security	68	External decline
67	Hard capture, pick up	68	External decline
75	Allowable number of PIN tries exceeded	62	PIN tries exceeded
76	Key synchronisation error	68	External decline
80	Invalid date	68	External decline
85	No reason to decline	01	Approved – no balances
91	Switch or issuer inoperative	72	Destination not available
93	Law violation	68	External decline

LINK (LIS5) RESPONSE CODE		BASE24-atm RESPONSE CODE	
96	System malfunction	70	System error
A0	Reserved for Pre-pay responses	68	External decline
A1	Reserved for Pre-pay responses	68	External decline
A2	Reserved for Pre-pay responses	68	External decline
A3	Reserved for Pre-pay responses	68	External decline
A4	Reserved for Pre-pay responses	68	External decline
A5	Reserved for Pre-pay responses	68	External decline
A6	Reserved for Pre-pay responses	68	External decline
A7	Reserved for Pre-pay responses	68	External decline
A8	Reserved for Pre-pay responses	68	External decline
A9	Reserved for Pre-pay responses	68	External decline
Q1	Card authentication failure	87	ARQC verification failure
Q2	Fallback not allowed	88	Fallback not allowed

For codes where the LINK description indicates “pick-up” (i.e., 04, 07, 33, 36, 38, 41, 43, 67), the card disposition is set to 1 (Capture) to ensure that the ATM retains the card.

If the transaction is a balance enquiry and has been approved, then the BASE24 response code is set to 00 (approved with balances).

Any codes not listed are translated to the BASE24 response code 68 (external decline).

B. FILE FORMATS

B.1 INTERCHANGE CONFIGURATION FILE (ICF)

This section gives a description of the fields included in the LINK (LIS5) Interchange Interface specific segment of the ICF.

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	ACQ-STAND-IN This field defines whether stand-in authorization is supported, where BASE24 is acting as the acquirer. Valid values are: 0 = not supported (default) 1 = supported	PIC 9
02	ISS-STAND-IN This field defines whether stand-in authorization is supported, where BASE24 is acting as the issuer. Valid values are: 0 = not supported (default) 1 = supported	PIC 9
02	CTVR-STAT This field defines the party responsible for initiating cutover processing. Valid values are: 0 = equal (default) 1 = master 2 = slave	PIC 9
02	FIXED-LGTH-FRMT Not used by the LINK (LIS5) Interchange Interface.	PIC 9
02	VERSION-NUM This field defines the release level of the BASE24 system. Valid values are: 00 = Release 4.0 (default) 01 = Release 5.0 or above	PIC 99
02	DATA-PREFIX-CHARS Not used by the LINK (LIS5) Interchange Interface.	PIC X(18)
02	SETL Not used by the LINK (LIS5) Interchange Interface.	TYPE BINARY 16
02	ITF-UPDATE-INTERVAL Not used by the LINK (LIS5) Interchange Interface.	TYPE BINARY 16
02	ITF-UPDATE-MAX-TRANS Not used by the LINK (LIS5) Interchange Interface.	TYPE BINARY 16

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	PROTO-TYP This field indicates the type of message header processing to be performed by the LINK (LIS5) Interchange Interface. Valid values are: 00 = None (default). 01 = SNA/CICS Protocol 02 = Bisync Multipoint Protocol	PIC 99
02	PROD-IND Not used by the LINK (LIS5) Interchange Interface.	PIC X(16)
02	STA The LINK (LIS5) Interchange Interface supports up to 20 stations (the two stations specified in the base segment of the ICF record are ignored).	OCCURS 20 TIMES
04	NAM	PIC X(16)
04	TYP	PIC X
02	REL-IND Not used by the LINK (LIS5) Interchange Interface.	PIC X(2)
02	BUY-RATE Not used by the LINK (LIS5) Interchange Interface.	TYPE BINARY 32
02	SELL-RATE Not used by the LINK (LIS5) Interchange Interface.	TYPE BINARY 32

B.2 LINK VERSION OF GENERIC BASE24 SWITCH TOKEN

The following LIS5 message fields are included in the token definition:

- Processing Code (field P-3)
- Systems Trace Audit Number (field P-11)
- Point of Service Entry Mode (field P-22)
- Point of Service Condition Code (field P-25)
- Forwarding Institution Identification Code (field P-33)
- Response Code (field P-39)
- Reversal Reason Code (field P-60)
- Point of Service Data (field P-61)
- Issuer Trace Id (field S-126)

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	FIID The FIID is hard-coded as 'LINK'	PIC X(4)
02	VER-ID The version number of the token, currently "04"	PIC X(2)
02	PROC-CDE The processing code	PIC 9(6)
02	STAN The Systems Trace Audit Number	PIC 9(6)
02	POS-ENTRY-MDE The point of service entry mode	PIC 9(3)
02	POS-COND-CDE The point of service condition code	PIC 9(2)
02	FWD-INST-ID-CDE The forwarding institution ID number	PIC X(11)
02	RESP-CDE The response code	PIC X(2)
02	RVSL-RSN The reversal reason code	PIC X(2)
02	CRD-DATA-INPUT-CAP The card data input capability	PIC X(1)

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	CRDHLDR-AUTH-CAP The cardholder authentication capability	PIC 9(1)
02	CRD-CAPTR-CAP The card capture capability	PIC 9(1)
02	OPERATING-ENVMT The operating environment	PIC 9(1)
02	CRDHLDR-PRSN The cardholder present indicator	PIC 9(1)
02	CRD-PRSN The card present indicator	PIC 9(1)
02	CRD-DATA-INPUT-MDE The card data input mode	PIC 9(1)
02	CRDHLDR-AUTH-METHOD The cardholder authentication method	PIC 9(1)
02	CRDHLDR-AUTH-ENTY The cardholder authentication entity	PIC 9(1)
02	CRD-DATA-OUTPUT-CAP The card data output capability	PIC 9(1)
02	TERM-OUTPUT-CAP The terminal output capability	PIC 9(1)
02	PIN-CAPTR-CAP The pin capture capability	PIC X(1)
02	SOCL-DEPRIVATN-AREA The cardholder authentication entity	PIC 9(1)
02	RESERVED-1	PIC X(7)
02	POST-CDE The card acceptor's postal code	PIC X(10)
02	RESERVED-2	PIC X(10)
02	ISS-TRACE-ID The issuer trace ID	PIC X(6)

C. LOG MESSAGES

Log messages provide information ranging from simple process statuses to problems of a serious nature requiring operator intervention. They are generated by the LINK (LIS5) Interchange Interface, based on conditions encountered, and are sent to the BASE24 EMS log.

Log messages also provide notification to operators of actions and conditions occurring in the system and, as such, provide an audit trail of these actions and conditions.

C.1 FORMAT OF MESSAGES

Log messages are generated to report an error condition or information. Error messages may require corrective action before the appropriate process can function properly.

The standard format for BASE24 log messages is:

YY/MM/DD HH:MM:SS X PROCESS ID MMMM TEXT

where: 'YY/MM/DD' specifies the date the message was sent.

'HH:MM:SS' specifies the time the message was sent.

'X' specifies the severity code where:

- (0) or (I) Status (Information)
- (1) or (W) Warning
- (2) or (E) Error
- (3) or (C) Critical

'PROCESS ID' specifies the symbolic process name.

'MMMM' specifies the message number.

'TEXT' specifies the text of the message.

For example:

07/05/10 09:09:48 (1) P1A^LINK^01 2300

Transaction Security Service will be used, Triple/Single DES support will be disabled and triple_to_single_g will be set to 0.

C.2 LIST OF MESSAGES

The LINKLOGM file on the SW60LINK subvolume documents the log messages output by the LINK (LIS5) Interchange Interface.

For each log message, additional information is given:

- The cause or reason the message is displayed.
- The remedy or solution for the message, if relevant.

Also, the following support/service personnel may need to be contacted:

- Operations Personnel

In-house operations staff who are responsible for initiating fault action for a particular problem.

- Service Personnel

In-house staff who are responsible for maintaining user data on BASE24.

- HELP24

ACI personnel who are responsible for solving customer queries and problems.

D. GLOSSARY

AWK	Acquirer Working Key
HSM	Hardware Security Module
IWK	Issuer Working Key
KCV	Key Check Value
KEK	Key Exchange Key
KPE	PIN Block Encryption Key
LIS5	LINK Interchange Standard, Release 5
LMK	Local Master Key
MFK	Master File Key
NCD	Non-Currency Dispense
NMM	Network Management Message
PCI	Payment Card Industry
TSS	Transaction Security Service
ZMK	Zone Master Key

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E. TSS MIGRATION

The BASE24 LINK (LIS5) Interchange Interface requires additional configuration steps in addition to those specified in the ***BASE24 Transaction Security Services Processing Guide***. These steps are required because the Triple DES “Interim” Solution uses the MAC exchange key to decrypt the Acquirer Working Key (AWK) received from LINK.

1. For each LINK (LIS5) Interchange Interface process, set MAC KEY LENGTH to ‘2’ in the KEY6 Screen 1
2. Run the conversion programs as documented in the ***BASE24 Transaction Security Services Processing Guide*** and start the GUI and TSS components
3. Logon to the ACI Desktop GUI as a user with access to Interface data
4. From the menu, select Configure>Transaction Security>Interface
5. For each LIS5 interface, do the following:

Change the PIN Keys Option to "Use separate keys"

Copy the PIN Import Key to the PIN Export Key

Copy the MAC Import Key to the PIN Import Key

Change the MAC Keys Option to "No keys exchanged"

Save the record.

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